



MSS Reference Architecture

VERSION 2.0



Mobile Satellite Services Association

Advancing the Future of
Global Mobile Connectivity

www.mss-association.org

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WHAT'S NEW IN VERSION 2.0?

This document has been produced by the Reference Architecture working group within MSSA's Technical Committee. This release is **Version 2.0** of the Reference Architecture document.

While Version 1.0 focused only on **transparent payloads**, Version 2.0 includes **regenerative payloads** and covers the challenges associated with concurrent **5G NR** and **NB-IoT** services. Upcoming versions will go further on topics of **TN-NTN integration**, coordination and interworking, **network selection** and **roaming**, and will provide deeper dives on selected areas introduced in earlier releases.

Version 2.0: Focus on Regenerative Payloads

The following presents a synthesis of the primary findings and recommendations arising from the updated content on regenerative satellite payloads in the second version (2.0) of the MSS Reference Architecture document. It evaluates the integration of regenerative satellite payloads and analyzes their strategic significance, technical impacts, and the associated implementation considerations for non-terrestrial mobile networks (NTN).

The initial MSS Reference Architecture released in November 2025 by MSSA focused exclusively on **transparent (bent-pipe) satellite architectures**, which can limit network performance by tightly coupling service and feeder links and requiring continuous ground connectivity. Regenerative payloads represent a major architectural evolution by embedding portions of the radio access network (RAN) — and in some cases an entire 5G gNB — directly onboard the satellite. This enables on-orbit processing such as demodulation, decoding, scheduling, routing, and local control-plane functions, transforming satellites into active elements of the mobile network.

Key drivers for regenerative architectures include the need for greater system autonomy and resilience, improved use of inter-satellite links (ISLs), and support for advanced 3GPP Release-19 capabilities such as store-and-forward services and direct UE-to-UE routing. By decoupling service, feeder, and inter-satellite links, regenerative architectures allow independent optimization of modulation and coding, significantly improving spectral efficiency and reducing feeder-link bandwidth requirements as traffic scales. Control-plane latency is also reduced by placing network intelligence closer to users, improving call setup, mobility signaling, and overall service quality.

The new content identifies substantial technical and operational challenges associated with regenerative payload deployment. Mobility management in LEO and MEO constellations is complex, as satellites, beams, and portions of the RAN are constantly in motion, requiring frequent handovers and reestablishment of network interfaces. Power and processing constraints onboard satellites further complicate implementation, as spacecraft must balance limited solar and battery energy against highly variable traffic demand. Efficient payload design and intelligent constellation-wide power and traffic management are therefore critical.

Standards alignment and interoperability represent additional risks. Although organizations such as 3GPP, the O-RAN ALLIANCE, and the Small Cell Forum have defined reference models, harmonizing regenerative architectures across vendors and operators remains challenging.

The MSS Reference Architecture evaluates three principal options: RU-only payloads, RU+DU payloads, and fully regenerative RU+DU+CU payloads. These options present clear trade-offs among onboard power consumption, feeder-link bandwidth, system complexity, mobility handling, and support for advanced features such as ISL-based handovers.

Overall, regenerative payload offers significant benefits in efficiency, autonomy, scalability, and performance for NTN systems, but introduces higher payload complexity, cost, power demand, and integration risk. No single architecture suits all use cases; operators must align architectural choices with business objectives, regulatory constraints, and deployment strategies. The 2026 revision of the MSS Reference Architecture is positioned as a critical industry reference to promote a common understanding of these trade-offs and enable interoperable, commercially viable regenerative NTN deployments.

Version 2.0: Focus on NR and NB-IoT

The following presents a synthesis of the primary findings and recommendations arising from the updated content on the coexistence and interworking of multiple radio access technologies in the MSS Reference Architecture 2.0 document. The new content focuses on the challenges associated with Radio Unit (RU) sharing and highlights the high-level architectural implications for system design, efficiency, and scalability.

Next-generation NTN systems are expected to support both 5G New Radio (NR) services and NB-IoT services using common satellite infrastructure. Supporting these two radio access technologies simultaneously introduces fundamental challenges because NR and NB-IoT have very different operational characteristics. NR typically operates with higher bandwidth and can tolerate lower duty cycles enabled through beam hopping, while NB-IoT requires narrow bandwidth but often demands higher duty cycles and more continuous coverage. These differences place additional stress on satellite payloads already constrained by limited power, onboard processing capability, and a finite number of simultaneously active beams.

Radio Unit sharing refers to the ability of a satellite payload to support multiple radio access technologies using a common set of RF chains, antenna elements, and beamforming resources. From a system perspective, RU sharing is essential to avoid duplicating hardware for each technology, which would significantly increase payload complexity, mass, power consumption, and cost. However, sharing resources across NR and NB-IoT requires careful architectural choices to balance flexibility against the inherent physical limitations of the payload.

The MSS Reference Architecture identifies two primary high-level approaches for RU sharing. One approach prioritizes NB-IoT requirements by reserving a larger fraction of RU resources — such as transmit and receive chains or simultaneously active beams — to support its higher duty-cycle needs. An alternative approach relies on wider beams or different beamforming strategies for NB-IoT, reducing the number of dedicated transmission chains at the expense of spatial efficiency and potentially link performance. Each option represents a clear trade-off between hardware cost, power efficiency, coverage characteristics, and service adaptability.

These RU sharing decisions have broader implications for overall system behavior. They directly affect beam hopping strategies, scheduler design, traffic prioritization, and how dynamically resources can be reallocated between NR and NB-IoT services as demand changes. Architectures optimized for NR efficiency may constrain NB-IoT scalability, while NB-IoT-optimized designs can reduce aggregate NR capacity. No single solution is universally optimal across all constellations or service mixes.

In conclusion, Radio Unit sharing between NR and NB-IoT is a critical enabling capability for multi-service NTN satellite systems, but it introduces unavoidable architectural trade-offs. MSS Reference Architecture 2.0 provides a framework to evaluate these trade offs at a system level, allowing operators and vendors to align RU designs with their service priorities, payload constraints, and long-term evolution strategy while maintaining overall efficiency and flexibility.



EXECUTIVE SUMMARY

This document serves as a preliminary guide to establish a baseline reference architecture, recommendations, and guidelines for the design, deployment, and realization of Non-Terrestrial Networks (NTN), based on 3GPP Release 17, primarily targeted at Mobile Satellite Services (MSS) and Direct-to-device (D2D) cellular services in FR1-NTN sub-3GHz bands. The primary intent is to offer a common architectural reference and clear system-level guidelines that industry stakeholders can adopt to improve the efficiency, reliability, scalability, and interoperability of NTN systems. These recommendations are not meant to serve as strict specifications but rather as flexible guidelines that can adapt to the evolving landscape of NTN technologies. Although technical in nature, the document is intended for a broad audience, from engineers and product managers to business and strategy leaders across equipment manufacturers, satellite network operators, and mobile device OEM organizations.

This document underscores the critical role of applying technological advancements and recommendations through system-level design to ensure consistent mobile device behavior and reliable performance. By prioritizing system-level design, industry stakeholders can position their NTN systems to meet the growing demand for high-speed, dependable connectivity. The guidelines extend across the entire service delivery value chain, spanning satellite constellations, ground infrastructure, and user devices.

The recommendations in this document are intended to guide and support specification development efforts by leading organizations such as the 3rd Generation Partnership Project (3GPP) and the GSM Association (GSMA). By adopting these recommendations, industry stakeholders can help advance a cohesive, interoperable, and sustainable NTN ecosystem.

Technological Advancements

A central focus of this document is the identification of recommendations for developing advanced technologies that enhance NTN performance, interoperability, and integration with terrestrial networks. Rapid advancements in satellite technology are creating new opportunities for NTN deployment, offering benefits such as broader coverage and improved connectivity in remote and underserved areas. Concurrently, satellite communications face inherent constraints that must be carefully addressed in the design of NTN-compliant constellations and ground infrastructure.

Interoperability and Mass Market Adoption

Interoperability is a critical determinant of NTN system success. Seamless integration with terrestrial networks is essential for delivering a consistent and reliable user experience and to ensure service continuity across NTN networks. This document emphasizes the importance of enabling mass-market adoption of NTN technologies, which depends on compatibility with a broad range of devices and applications. By prioritizing interoperability, the industry can accelerate widespread adoption and integration of NTN technologies into everyday use.

In addition to 3GPP specifications, this document analyzes NTN-specific use cases, including system resource management, mandatory signal scheduling, network selection in multi-orbits and cross-border scenarios, and approaches to network sharing.

EXECUTIVE SUMMARY CONTINUED

The recommendations provide guidelines on designing and implementing interoperable systems, addressing key aspects such as network architecture, protocols, and standards. By applying these guidelines, industry stakeholders can work together to optimize NTN systems and their sub-components to deliver consistent end-user services and elevate overall connectivity and user experience.

PRIORITIZED USE CASES FOR GLOBAL INTEROPERABILITY

This document prioritizes several use cases to ensure global integration and interoperability. These use cases include:

Resources Management: Efficient resource management is critical to the performance of NTN systems, particularly given the limited onboard power of satellites. This use case highlights strategies for allocating (e.g., beam hopping), utilizing, and optimizing resources to ensure NTN systems meet diverse application and service demands while remaining fully compliant with 3GPP specifications. Effective resource management drives network efficiency, lowers operational costs, and delivers superior user experience.

Network Selection and Network ID Management: This use case tackles the complexities of choosing the right network and managing identifiers in multi-network environments. It outlines methods for network selection, network ID management, and seamless inter-network handovers. Appropriate network selection and ID management ensure uninterrupted connectivity, optimized network performance, and full compliance with regulatory requirements.

Network Selection and Roaming Considerations: Seamless connectivity and uninterrupted service during roaming are vital for NTN system performance. This use case provides recommendations for roaming policies, including strategies to optimize network performance and maintain regulatory compliance. Effective roaming considerations enhance user satisfaction and extend the global reach of NTN services.

Network Sharing: Network sharing enhances the efficiency and cost-effectiveness of NTN deployments. This use case examines models such as infrastructure sharing, spectrum sharing, and service-level agreements to maximize resources. Network sharing reduces deployment costs, expands network coverage, and improves service availability.

Protocols Co-Deployment: With the high cost of launching a constellation, operators are looking to maximize their returns by generating multiple revenue streams across the widest range of services. A use case that is attracting a lot of interest at the moment is satellites that deliver both 5G NR as well as Narrowband Internet of Things (NB-IoT) services. Designing a payload and end-to-end system that can concurrently serve the different needs of these two protocols presents some unique challenges.

EXECUTIVE SUMMARY CONTINUED

END USER SERVICE TYPES

Building on the identified use cases, this section offers insights into which are the key services targeted by NTN and how services can be delivered using NTN technologies. It examines service models and architectures that enable effective deployment and delivery of NTN services. These include:

Messaging Services: NTN systems support a broad range of messaging capabilities, including SMS, MMS, emergency messaging, and instant messaging. These services enable the exchange of text, multimedia, and real-time communication, strengthening communication and connectivity in remote and underserved areas.

IoT Services: NTN systems deliver energy-efficient IoT connectivity with NB-IoT standards for monitoring and tracking applications, etc.

Voice/Telephony Services: NTN systems deliver reliable voice and telephony services, allowing users to make and receive calls anywhere. These services are critical for emergency response, remote work, and personal communication, ensuring continuous connectivity regardless of their location.

Data Services: NTN systems provide high-speed data access for internet access, video streaming, and online gaming. These services expand access to information, entertainment, and applications, supporting diverse use cases and elevating the overall user experience.

Conclusion

In conclusion, this document serves as a comprehensive resource for the NTN industry, providing recommendations across key areas including technological advancements, interoperability, system-level use cases, and service models. These adaptable guidelines support the evolving landscape of NTN technologies, supporting the growth and development of the industry.

By adopting these recommendations, industry stakeholders can accelerate the development of a cohesive and interoperable NTN ecosystem while shaping standardization efforts by prominent bodies such as the 3GPP and GSMA. Following these guidelines ensure NTN systems can meet rising demands for high-speed, reliable connectivity, while addressing critical requirements for security and interoperability, and ultimately fostering a robust and resilient NTN ecosystem.

1 INTRODUCTION

1.1 MSSA Technical Committee Main Objectives

The main objectives that are supported by the Reference Architecture document are:

- Unlocking interoperable architectures and standards for use in multi-orbit, multi-band, satellite systems, ground infrastructure, and end-user equipment.
- Enhancing seamless global interaction between terrestrial and multi-orbit, multi-band, satellite networks through the development of recommended specifications.
- Providing guidance to industry and network operators on network architectures and features necessary to achieve interoperability between various non-terrestrial networks and terrestrial networks.

1.2 Purpose of the Architecture Document

The vision of Non-Terrestrial Networks (NTN) for Mobile Satellite Services (MSS) is to provide service ubiquity while maximizing the use of assets (existing and future) for mass market adoption. To achieve these objectives, several aspects of an NTN MSS architecture need to be considered and further defined, including:

- **End-to-End System Operation** – Accounting for all the components and segments of the system, both 3GPP- and non-3GPP-defined, their functional and non-functional requirements, demarcation points and interfaces, interoperability and deployment scenarios.
- **Space Segment** – Accounting for payload technology and other space segment constraints and challenges, cost-effectiveness, service and system implications, and future-proofness.
- **Ground Segment** – Including both 3GPP and non-3GPP elements, Access Layer (RAN, Gateways, etc.), Backhaul and Transport, Core Network and Interconnect and their interfaces.
- **User Segment** – Considering key mandatory features and capabilities, different target services and markets, UE classes and form factors.
- **Resource Management** – Including spectrum management, dynamic resource allocation, network capacity optimization and load balancing, alignment with 3GPP network procedures such as initial access, mobility and paging, addressing different service requirements, KPIs and objectives, as well as system and regulatory constraints such as spectrum availability, different spectrum bandwidth and gateway capacity, regulatory requirements, cross-border spectrum, emission and network identity management, etc.
- **Inter-Network Integration and Coordination** – Addressing interoperability schemes that ensure service transparency and continuity for customers, whilst in a range of different scenarios ranging from cross-border operations, TN-NTN and NTN-NTN network selection, roaming and mobility, including multi-orbit, multi-constellation coexistence, interoperability, mobility and cooperation schemes, etc.
- **Network Infrastructure Sharing** – Enabling space, ground and network infrastructure cooperation, spectrum asset sharing and pooling, neutral host, sharing and slicing, establishing requirements and operating principles to share NTN MSS infrastructure and definition of the different types of sharing architectures, demarcation points and interoperability interface specifications.

- **NTN MSS Service Definition, Delivery, and Evolution** – Analyzing various services and their specific requirements, impact on the system, and their evolution over time as the networks, markets, and ecosystems mature and grow.
- **NTN MSS 3GPP Compliance** – Ensuring alignment with 3GPP standards and how this affects operations utilizing existing and upcoming satellite systems.

This document, throughout its evolution over time, aims to analyze all these topics, identify the expected challenges and areas requiring further consideration, and provide a common industry view on the possible solutions and implementation approaches.

The ultimate goal of the MSS Reference Architecture document is to eventually provide guidelines to the industry for NTN MSS system design and interoperability, to help design and build systems that can easily interoperate and enable new shared NTN MSS infrastructure models, whilst meeting service and user experience objectives and achieving maximum acceptance by the NTN ecosystem/industry participants, as well as compliance with international standards and regulations.

1.3 Document Structure

The document is structured to provide a comprehensive understanding of NTN technologies and their applications.

Section 2 (System Overview) provides an overview of the high-level components of NTN systems and, in particular, Mobile Satellite Systems (MSS), including architectures, key features, as well as the key challenges and issues faced by the NTN industry, establishing the foundational knowledge necessary for subsequent discussions and setting the context for recommendations by highlighting the areas requiring attention.

Section 3 then delves into critical aspects of NTN technologies, identifies the key challenges to achieve some of the key use case objectives, covering technological advancements, interoperability, security, and use cases, offering in-depth analysis and insights into potential solutions and strategies.

Finally, Section 4 collects some initial key recommendations for the industry to take into consideration.

1.4 Scope and Demarcations

This section defines the scope of the document and highlights the critical demarcations and interfaces between network elements. While some elements are tailored to individual satellite networks, others are shaped by 3GPP standards and their integration with MNO networks and user devices. This document focuses on NTN-specific components, emphasizing satellite resource management and the distinct performance characteristics of geostationary (GEO), medium-Earth orbit (MEO), and low-Earth orbit (LEO) satellites.

- The 3GPP-defined functions in this document refer to services and subsystems based on 3GPP Release-17 to Release-19, predominantly focusing on NTN NB-IoT and NTN NR protocols for GEO, MEO, and LEO scenarios, parameters and use cases, and for NTN bands definition in FR1-NTN (e.g., L-bands, S-bands). Systems using IMT/terrestrial spectrum, including Supplemental Coverage from Space (SCS) bands as defined by the FCC in the U.S., are not addressed in this document, though it might be addressed in future work.

- This document emphasizes systems targeting Mobile Satellite Services (MSS) in corresponding FR1-NTN bands below 3 GHz, such as L- and S-bands.
- The end-to-end architecture will support different types of services (voice, messaging, data, emergency communication, etc.) expected in such systems.
- This document covers both transparent and regenerative payload architectures.
- A Generic System Description, the important Demarcations and Interfaces of interest (shaded yellow), is shown in Figure 1 below. It illustrates the end-to-end architecture with emphasis (shaded areas) on functionality enablement between 3GPP-defined and non-3GPP elements and functions.

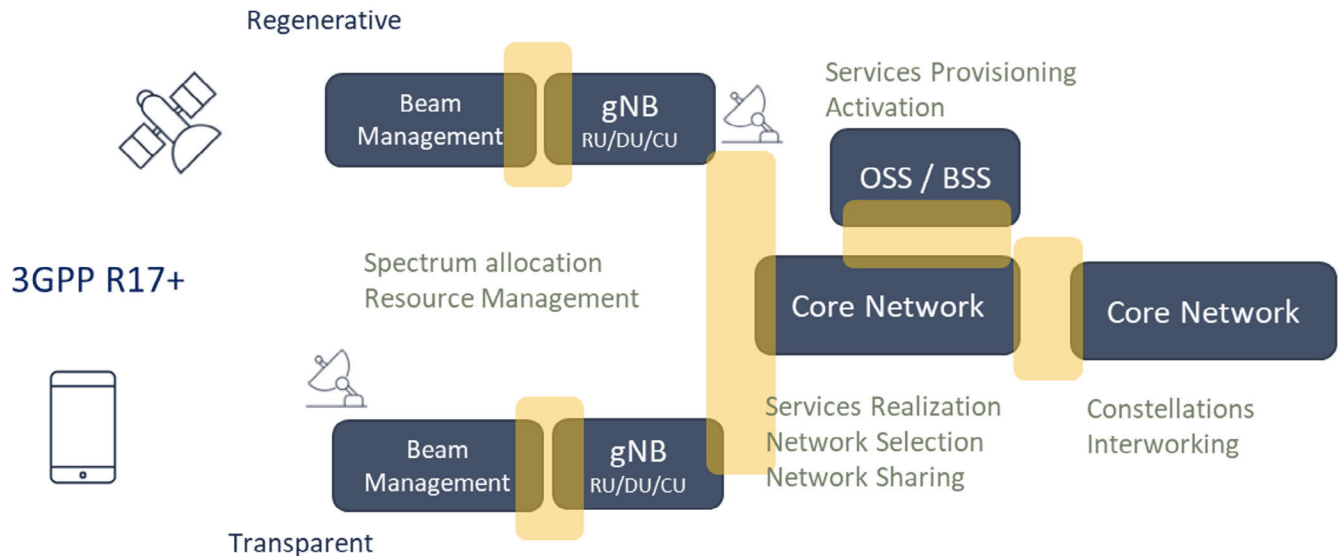


Figure 1: Demarcation Points and Interfaces of Interest

1.5 Target Audience

Ecosystem players such as satellite network operators, mobile network operators, network element providers, chipset manufacturers, device manufacturers, satellite manufacturers, standards body participants, regulators, and system integrators are the primary audience for this document.

2 SYSTEM OVERVIEW

2.1 Satellite Systems

As 5G terrestrial networks (TNs) continue to be steadily deployed around the world to support increasing mobile services demand, satellite-based communications are becoming an important part of this paradigm, as they can provide extended coverage and emergency services where TN coverage and services are not available.

Mobile Satellite Services (MSS) constellations have been and are being deployed across Geostationary Earth Orbit (GEO), Medium Earth Orbit (MEO), and Low Earth Orbit (LEO) orbits, each optimized to balance coverage, capacity, service level performance, cost, and latency. GEO satellites have long been the backbone of satellite communications, whilst LEO satellite systems support emerging technologies like global internet access, direct-to-device services, and Earth monitoring. MEO systems are utilized for both communications and radio navigation applications. Together, these satellite architectures enable a broad spectrum of technological capabilities.

2.1.1 Geostationary Earth Orbit (GEO)

Satellites in geostationary Earth orbit remain stationary relative to the Earth's surface, providing continuous coverage to a specific region or country. While ideal for services requiring constant communication links, higher latency is introduced, which leads to greater path loss due to the greater distance signals must travel. The greater path loss results in significant challenges for MSS services, as user devices typically have exceptionally low gain omnidirectional antennas and limited transmitter power due to battery and radiation safety concerns. Therefore, GEO satellites for MSS services typically have exceptionally large reflectors (9-meter to 22-meter diameters) to enable high gain satellite performance to close the links with disadvantaged mobile devices such as handheld units.

Characteristics:

- Altitude: ~35,786 km (22,236 miles) above the Earth's equator
- Orbital Period: Matches Earth's rotation (24 hours)
- Positioning: Appears stationary relative to the Earth's surface
- Fixed position over one spot on Earth
- Large coverage area (up to 42% of Earth's surface per satellite)
- Ideal for continuous communication with specific regions

Applications:

- Telecommunications: Mobile Satellite Services offering voice, data, and IoT; Fixed Satellite Services offering internet and long-haul data
- Broadcasting: Television and radio broadcasting
- Weather Monitoring: Satellites like GOES (Geostationary Operational Environmental Satellites)
- Global Navigation: Augmentation systems like WAAS (Wide Area Augmentation System)

Advantages:

- Continuous coverage of specific areas
- Simplified ground station tracking (no need to track moving satellites)
- Lowest satellite cost for regional coverage

Disadvantages:

- High latency (signal delay due to altitude)
- High transmission path loss
- Expensive to launch and maintain
- Cannot cover/low elevation angles for high latitudes
- Higher radiation environment increasing satellite cost

2.1.2 Medium Earth Orbit (MEO)

Satellites in MEO provide a balance between coverage and latency. Fewer satellites are needed compared to LEO, and they can cover larger areas with each satellite pass.

Characteristics:

- Altitude: ~2,000 to 20,000 km (1,243 to 12,427 miles)
- Orbital Period: 2-12 hours, depending on altitude
- Positioning: Moves relative to the Earth's surface
- Each satellite covers a larger area than LEO but smaller than GEO
- Requires fewer satellites for global coverage compared to LEO

Applications:

- Navigation: GPS, GLONASS, Galileo, and BeiDou systems
- Communication: Certain communication satellite constellations (e.g., O3b, Omnispace)

Advantages:

- Moderate latency (less than GEO)
- Fewer satellites are required than LEO constellations
- Balance between coverage area and altitude

Disadvantages:

- Satellites move relative to the ground, requiring tracking systems for directional ground antennas
- Individual satellite cost is higher than LEO satellites
- Higher radiation environment increasing satellite cost

2.1.3 Low Earth Orbit (LEO)

Satellites in LEO offer low-latency communication but require large numbers of satellites to ensure global coverage due to their fast movement relative to the Earth's surface. Examples include the Iridium and Globalstar systems.

Characteristics:

- Altitude: ~160 to 2,000 km (99 to 1,243 miles)
- Orbital Period: 90-120 minutes
- Positioning: Rapidly moves across the Earth's surface
- Smaller coverage area per satellite
- Requires constellations (multiple satellites) for continuous regional/global coverage

Applications:

- Earth Observation: Imaging and remote sensing (e.g., Foresight, Landsat, Sentinel)
- Communication: Constellations like Starlink, OneWeb, and Iridium
- Science and Research: International Space Station (ISS) and experimental missions

Advantages:

- Low latency (faster signal transmission)
- Shorter path loss
- Closer proximity enables high-resolution imaging and data collection
- Less expensive to launch due to lower altitudes
- Lower radiation environment, reducing satellite cost

Disadvantages:

- Shorter satellite lifespans due to atmospheric drag
- Complex/large constellations needed for consistent coverage
- Frequent replacements and potential for space debris

The size of a satellite constellation is determined by coverage, capacity, and redundancy. LEO constellations often comprise dozens to hundreds of satellites, whereas GEO systems may rely on just a few strategically placed satellites to cover large areas.

To maintain seamless service, satellites must be precisely spaced and phased to deliver continuous coverage and reduce handover interruptions as users transition between footprints. End users should typically have visibility to multiple satellites at elevation levels higher than 20 degrees.

2.2 Mobile Satellite System (MSS) Network Components

2.2.1 MSS Architecture Overview

Figure 2 provides a general D2D satellite communication services delivery components. It includes the interconnection with MNOs/MVNOs or private networks.

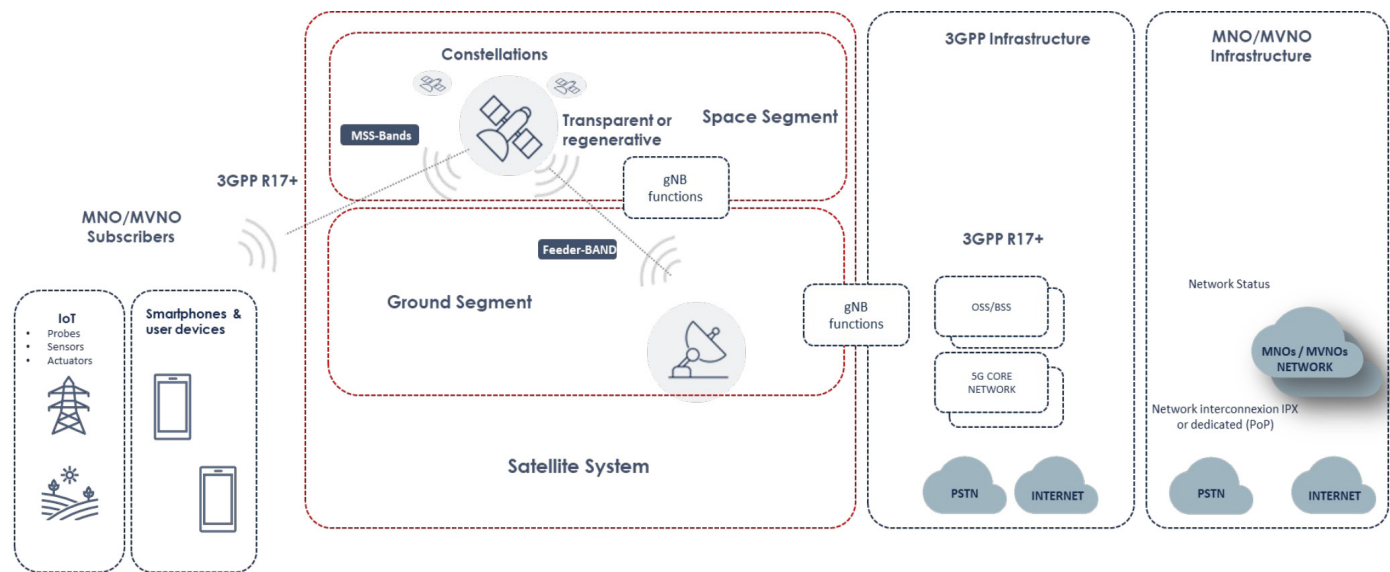


Figure 2 : Generic Satellite Communication Services Delivery Components

This is intended as a generic architectural representation to highlight the key high-level segments and domains that will be examined in further detail in this and the following sections.

2.2.2 Payload Architectures

One of the key architecture options in satellite communications, and in particular in mobile satellite systems, is the payload architecture, which in turn drives significant differences in the ground segment as well as a number of other architectural components and operational aspects.

Satellite payloads are often simply grouped into either “Transparent” or “Regenerative,” but in practice there are a range of possible top-level architecture options for the payload and the overall Satellite System, which place key baseband and processing components of the satellite base station (e.g., gNB or eNB) on the ground or in space.

The most common options are summarized in Figure 3, some of which were studied by 3GPP throughout Releases 16 to 19.

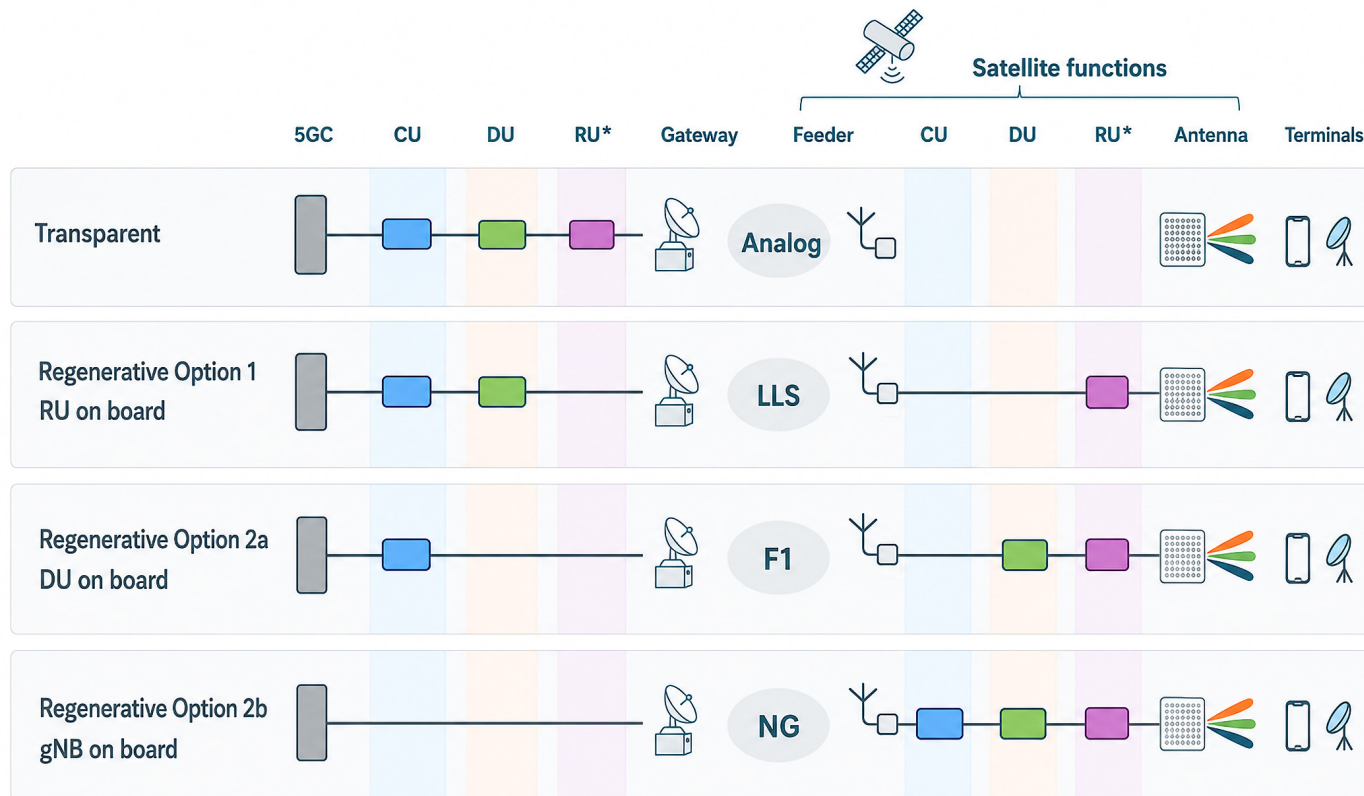


Figure 3 : Most Common Options Between Transparent and Regenerative Architectures

Each option has benefits and trade-offs, and operators will choose one based on a number of factors driving the design of their constellation. Currently, 3GPP standards are only considering the full Transparent option and Option 2b for regenerative (full gNB/eNB on board); however, the standards do not exclude the implementation of other options.

In certain scenarios, it is possible for some core network functions to be hosted directly on satellites. This approach was further studied and specified by 3GPP from Release 19 and is typically considered for specific use cases where on-orbit processing or network management offers distinct operational advantages. Examples of this include the 3GPP-defined concept of UE-satellite-UE, in which user plane data is exchanged between UEs connected to the same satellite constellation (or even the same individual satellite), without passing through any ground-based infrastructure. The 3GPP-defined concept of store-and-forward (S&F) is another example, in which satellites with intermittent connectivity can store user plane data until connectivity to ground-based infrastructure or user terminals is reestablished. Despite its relevance, it should be noted that the hosting of core network functions on satellites falls outside the scope of this document.

2.2.3 Drivers impacting top-level architecture choice

Early deployments have focused on Transparent architectures, motivated primarily by their apparent simplicity and commonality with existing satellite communications systems. Advances in space-based processing have made it possible to incorporate regenerative options in satellite payloads and create opportunities for improvements not possible with transparent architectures. Drivers for regenerative architectures include:

2.2.3.1 The need for more autonomy and resilience in satellite connectivity.

In a transparent architecture, the satellite must have continuous connection with all the UEs and a ground station for the link to function. With transparent architectures, it is generally complex to make use of inter-satellite links (ISLs) to remove this dependency on the feeder link. Regenerative solutions greatly simplify the implementation of ISLs. Another consequence of transparent architecture is modulation; coding and error correction schemes cannot be optimized separately for the different characteristics of the service, ISL, and feeder links, reducing both performance and resilience.

2.2.3.2 Use cases benefiting from on-orbit processing, including UE-to-UE satellite links and store-and-forward.

3GPP Release 19 introduces standards for store-and-forward supporting use cases (e.g., as IoT devices) where the satellite and a suitable ground station are not simultaneously in view of the satellite.

Release 19 introduces direct UE-to-UE routing using Inter-Satellite Links (ISLs) to provide a space-based backhaul network, reducing load on service links, reducing load on terrestrial networks, and potentially increasing security and resilience.

Both of these features require regenerative architectures with the complete gNB in space.

2.2.3.3 Enhancing system efficiency by reducing feeder-link load and through optimization of individual link modulation and coding.

Service link bandwidth is typically more constrained than feeder link bandwidth; as traffic scales, in a transparent architecture, the required feeder link bandwidth becomes very large. Because the feeder link and service link are closely coupled, many techniques that could increase the spectral efficiency of the feeder link, such as higher modulation orders to exploit the improved link data throughput, cannot be used. By separating the service and feeder links, each can be optimized independently, improving overall system performance.

2.2.3.4 Increasing demand for lower control plane latency and improved service performance.

A transparent architecture introduces approximately double the control plane propagation delay between the user equipment (UE) and the base station (gNB). The latency introduced causes significant impact on signaling such as connection and mobility management. The control plane latency will cause delays in call set-up, beam-to-beam transition, and other mobility functions.

2.2.4 Space Segment

The space segment of a Mobile Satellite Services (MSS) system primarily refers to the constellation of satellites in orbit, which are responsible for relaying communication signals between users and the ground segment (e.g., gateway stations, network control centers). This segment is critical to the functionality of the MSS system, providing coverage, capacity, and reliability for mobile communications across vast areas, including remote and underserved regions. Below is an expanded view of the key components and considerations within the space segment for an MSS satellite constellation.

2.2.4.1 Satellites:

Satellites forming the space segment are broadly composed of the two following blocks:

- **Bus** – Provides the physical platform hosting all the satellite's functions, including power, propulsion, processing, control, as well as the **mission Payload(s)**
- **Payload(s)** – Provides the main mission functions of the satellite. In an MSS system, the main payload is typically a **Communications Payload**

Satellite Payload Specifics (transponders, antennas):

- **Communication Payload:**
The payload is the core of each satellite's functionality, comprising transponders, antennas, and other equipment that facilitate the transmission and reception of signals. For MSS, the payload typically operates in specific frequency bands (such as L-band or S-band) that are allocated for mobile satellite communications.

There are two types of satellite payloads:

- a. Transparent (including digital transparent)
- b. Regenerative

Transparent payloads perform frequency translation, signal amplification, and beam transmission. Digital Transparent payloads receive the I-Q data signals.

Regenerative payloads implement regeneration of the signals (conversion to digital and demodulation) from Earth, which is equivalent to a base station. More details on satellite payload architectures are available in Section 3.2.1.10.

2.2.4.2 Radio Access Network (RAN) Platform - Regenerative Architectures

In regenerative architecture, the RAN Platform is part of the space segment.

RAN platforms are based on 5G or 4G Terrestrial Network (TN) technology. Satellite adaptations – e.g., timing advance – are required to accommodate the 2-way satellite delay between the gateways and UEs.

The gNB is the central logical node of the 5G NG-RAN, including the 5G NR NTN RAN.

The architecture of the NG-RAN is described in 3GPP TS 38.401 and follows the model below (Figure 4):

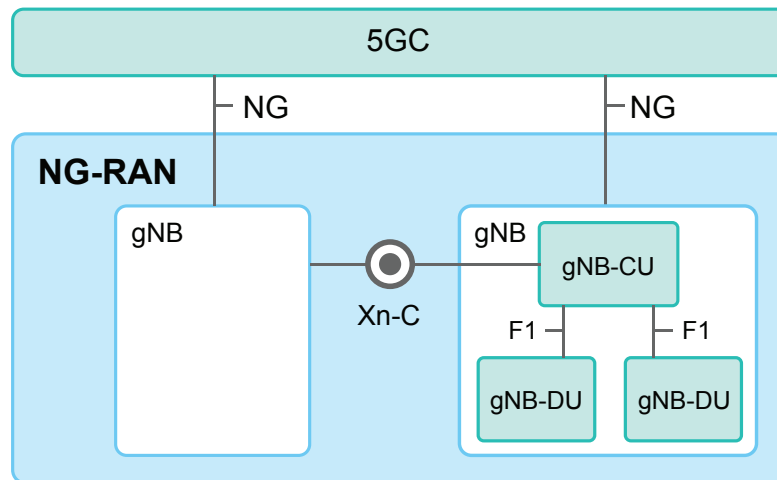


Figure 4: 5G RAN Reference Architecture in 3GPP



In the 3GPP architecture, the gNB (Figure 5) could be considered as a monolithic unit or can comprise a gNB-CU and one or more gNB-DUs connected via F1 interface. Under the 3GPP architecture, the radio unit (RU) is part of the Distributed Unit (DU), and the interface between them is out of the scope of 3GPP. The O-RAN ALLIANCE, however, defines an open fronthaul interface, the O-RAN Open Fronthaul (OFH), to standardize the Low Layer Split (LLS) between a DU and a RU. Hence, here we depict the gNB to have 3 different entities, RU, DU, and CU, which can be described as follows:

- The CU hosts RRC, PDCP, and SDAP protocols and is responsible for handling the signaling/control-plane and non-real-time functions.
- The DU hosts L2 (RLC, MAC) and can host part of the physical layer. It is also responsible for scheduling the UEs and managing power control, link adaptation, and timing alignment.
- The RU generates the signal sent to the UE (over the Uu-interface) and receives the signal from the UE; it hosts (part) of the physical layer and handles UL/DL radio frame timing.

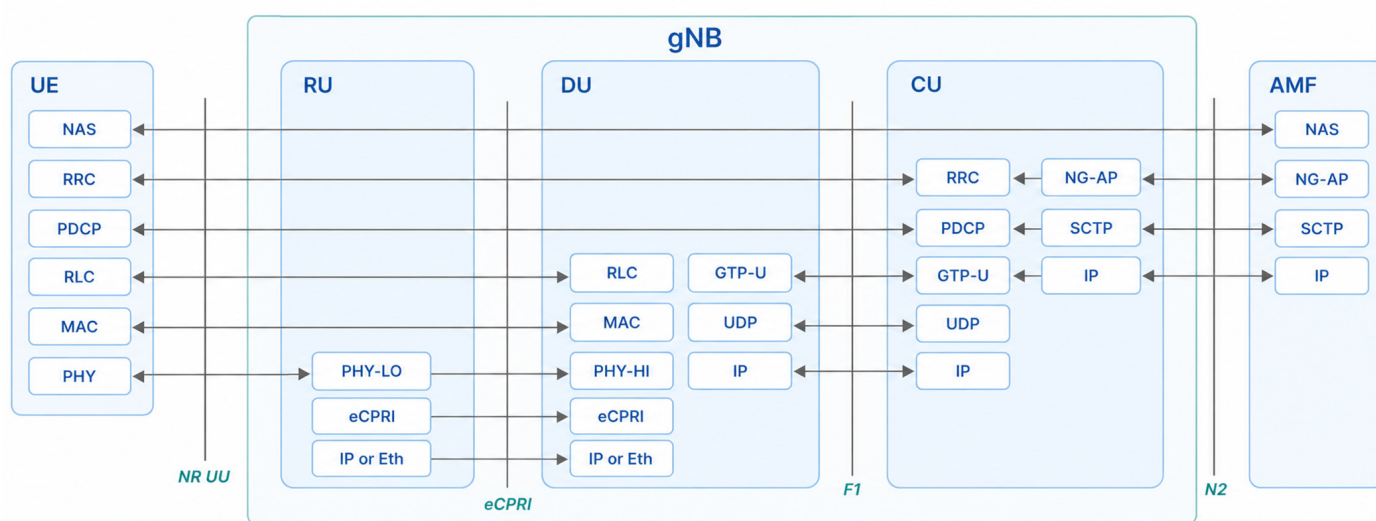


Figure 5: gNB Entities and Interfaces for User Plane and Control Plane Protocols

It is up to the RAN vendor to split up or combine the gNB entities in their products. Typical splits used in terrestrial networks are described in section 2.2.3, however in a regenerative architecture, it is expected that different solutions will be chosen to satisfy the different constraints of a satellite payload.

There is ongoing work in the O-RAN community to review architecture splits for NTN applications. A white paper, Deployments of O-RAN-based Non-Terrestrial Networks¹ was published in May 2025 which examines the advantages and disadvantages of some of the options.

¹ Deployments of O-RAN-based Non-Terrestrial Networks, ORAN Alliance, May 9, 2025

Option 1 RU on board the satellite, DU and CU on the ground

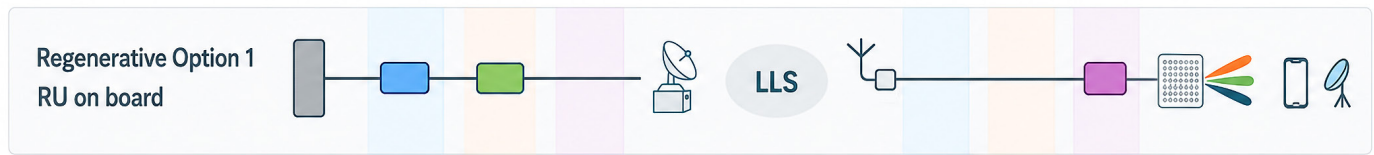


Figure 6 : RU On Board the Satellite

This architecture (Figure 6) could be referred to as a split 7.2 architecture. The feeder link is used to convey the O-RAN Fronthaul LLS interface, which includes a representation of the transmitted and received RF waveform as IQ samples requiring a higher bandwidth than the waveform itself.

Advantages

- Relatively low complexity payload
- The RU is stateless, so handover between satellites in LEO and MEO is relatively simple
- No DU handover between CUs

Disadvantages

- Very high feeder link bandwidth requirements – approximately 10 times higher than other regenerative architectures
- O-RAN LLS has very tight timing and delay requirements that will be challenging to meet, particularly if ISLs are used

Option 2a RU and DU on board the satellite; CU on the ground

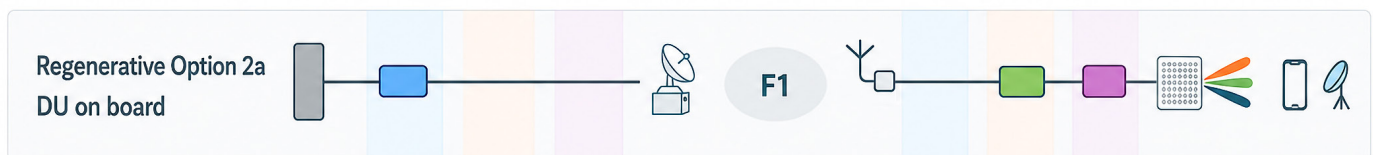


Figure 7 : RU and DU On Board the Satellite

This architecture (Figure 7) could be referred to as a split 2 architecture. The feeder link is used to convey 3GPP F1 interface, which is an IP interface.

Advantages

- Relatively efficient use of feeder links is comparable with most other regenerative architectures
- Less stringent timing and latency requirements than split 7.2, which also enables multiple-hop ISLs

Disadvantages

- Relatively high payload complexity (compared with option 1)
- Higher payload power requirements (compared with option 1)
- Large number of handovers between DUs and CUs, particularly in LEO constellations
- F1 interface is not an open interface in terrestrial networks, so it is not currently possible to mix DU and CU components from different vendors
- In most implementations, there is a requirement for high-bandwidth, low-latency connectivity between some of the processes in the DU and CU; these must be adapted to accommodate the feeder link

Option 2b RU, DU, and CU on board the satellite

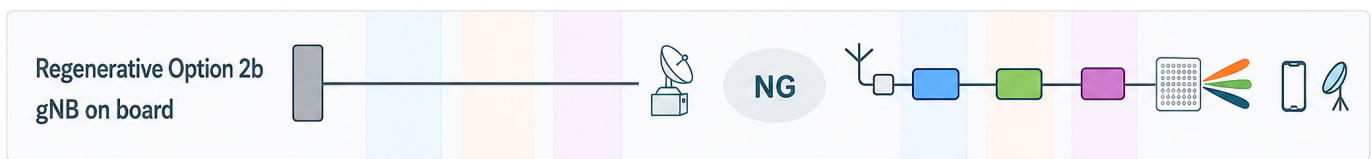


Figure 8 : RU and DU and CU On Board the Satellite

This architecture (Figure 8) could be referred to as a split 0. The feeder link is used to convey the 3GPP N2 interface between the CU and the core network. Feeder link bandwidth requirement is like Option 2a.

Advantages

- No DU to CU handovers
- All gNB functions are co-located with no need for adaptations for limited feeder link bandwidth and high latency
- Xn handovers between NTN cells are possible where ISLs are used to carry the Xn interface

Disadvantages

- Highest payload complexity (compared with option 1 or 2a)
- Highest payload power requirements (compared with option 1 or 2a)
- Multiple handovers of UEs and all state information, particularly in LEO constellations, mitigated by the availability of Xn links over ISLs to avoid feeder link traffic

2.2.4.3 Architecture option trade-offs

Feeder link capacity considerations

Options 2x provides distinct benefits when it comes to feeder link bandwidth requirements, especially when compared with Option 1. The reduction in necessary feeder link bandwidth comes at the expense of increased payload complexity and higher power requirements. However, the overall power consumption may be balanced out by the energy savings resulting from the lower bandwidth demands on the feeder link, which can offset the need for additional processing power onboard the satellite.

For constellations characterized by low throughput, it is possible that sufficient feeder link bandwidth to support Option 1 is readily available. In such cases, the simplicity of the payload in Option 1 can present a significant advantage, as it reduces onboard complexity.

Mobility considerations

In a LEO constellation, each satellite is only in view of a given location on the Earth for a matter of minutes. If a satellite were to form fixed beams that moved across the Earth's surface, UEs would need to hand over to a new beam every few seconds. The concept of "Earth fixed" beams is used where the satellite steers the beam to cover a fixed area on Earth for as long as possible. Using these earth fixed beams reduces handovers from every few seconds to every few minutes.

3GPP networks were traditionally designed on the assumption that the UEs move, and the gNB is stationary. Links between the elements of the gNB and the core network were assumed to be static. In NTN networks, either part of, or the complete gNB moves, and the link between it and the rest of the network must be broken and reestablished.

Within the 3GPP framework for Distributed Unit (DU) and Centralized Unit (CU) functionality, the Radio Resource Control (RRC) function is divided between the DU and CU, and each platform maintains per-user equipment (UE) state information. In Medium Earth Orbit (MEO) and Low Earth Orbit (LEO) satellite constellations, the mobility of the satellites means that serving satellites change regularly due to their orbital motion. As a result, state information for many UEs must be synchronized between the DU and CU during handovers between satellites triggered by satellite movement. Because of this operational requirement, 3GPP is currently considering only Option 2b for implementation in such environments.

Power considerations

A satellite is limited to the power it generates with solar power whilst it is in sunlight and when it is operating on energy stored in its batteries. Alignment between the times that a satellite can generate electricity and times of peak demand will vary with every orbit. Peak generation may occur when the spacecraft is over empty ocean, and peak traffic may be over land while the satellite is in eclipse.

Transparent architectures typically have the lowest payload power requirements as they require the least onboard processing capability; however, this is offset by the requirement for more transmit power for the service and feeder links, due to their reduced efficiency compared to regenerative architectures.

Comparing regenerative options 1 and 2b, a similar trade-off exists. Option 1 has the lowest power requirement for onboard processing but **has a much higher feeder link bandwidth, requiring higher feeder link transmitter power**. The balance of these two factors will depend on many factors.

Power must be considered at a constellation level to ensure that all factors are considered.

Overall system complexity and management

Architecture options not only affect payload complexity and feeder link requirements but also play a significant role in resource management strategies. The selection of functional split points, such as between options 2a and 2b, determines where centralized scheduling and coordination are implemented within the system. This placement directly influences the complexity of efficiently managing resources like spectrum and power across the constellation. For a more comprehensive analysis of these impacts, refer to the section on Resource Management.

Regulatory considerations

In regenerative option 2, it may be the case that processing of sensitive data that is covered by regulations is carried out on board the satellite. The implications of this must be considered; however, further discussion is currently outside the scope of this document.

2.2.4.4 Antennas

Satellites are equipped with highly directional antennas that focus signals to and from the Earth's surface.

Satellite antenna design is crucial for ensuring efficient power usage and maintaining signal strength across vast distances.

A multibeam antenna is designed to produce beams that cover distinct areas on the Earth's surface (like terrestrial cellular systems). Each beam can be independently controlled for gain, beam size/shape, frequency, and polarization. Two technologies are typically used to create beams. Table 1 provides a comparison of the technologies:

- **DRA (Direct Radiating Array)** is composed of radiating elements directly pointed and creating spot beams on Earth's surface thanks to amplitude and phase modulation management by onboard beamformers. DRAs can be implemented using analog beamformer systems, digital beamformers, or hybrid analog/digital beamforming technology.

	Multi Feed/Array fed antenna	DRA
Size and compactness	Large size and volume required in addition to a classic reflector; Mechanically steered	Compact, suitable for space-constrained applications and is electronically steered
Performances (G/T and EIRP)	High level of performance on limited bandwidth; High gain and directivity	Good level of performance on large bandwidth
Performance Field of Regard	Limited Field of Regard (10-15 degrees)	Wide Field of Regard (40-60 degrees)
Flexibility	Limited flexibility since beams are defined by the antenna geometry	High-level of flexibility since fully reconfigurable
Complexity	Complex deployment mechanisms are less reliable than electronically steered systems	Electronically steerable without antenna deployment mechanisms; Complex beamforming systems

Table 1: Reflector-Based Antenna and DRA Characteristics

- Multi-feed/array fed antennas with large unfurlable reflectors have been standard on GEO MSS satellites offering more than 200 beams over large areas (e.g., Thuraya, Inmarsat).
- DRA antennas have been adopted primarily for MEO and LEO satellites, offering a larger field of regard and a high-level of flexibility to configure and manage beams, like those offered in terrestrial cellular networks.

The performance of DRA antennas for 5G NTN satellite applications is driven by a combination of several factors. Orbit altitude, number of satellites, satellite field of regard, and required antenna performance to close the link budget are factors considered in the design of the satellites and their antennas.

The optimization involved in determining cost-effective solutions is not trivial. It would seem logical to minimize the number of satellites in a constellation, which would involve higher orbits and/or lower user elevation angles, to serve wider areas per satellite (via larger FoV). This solution would require larger antennas and more power to compensate for the higher free-space losses, as well as increasing the number of beams being created and managed by each satellite. It would also require larger antennas to receive signals from the ground. This can soon reach the limits of the satellite technology available on reasonable time scales, as well as the size and the cost of the satellites, and the number of launch vehicles required, resulting eventually in an unattractive commercial solution. On the other hand, moving toward lower orbits demands more satellites in the constellation, more launches, more ground infrastructure, and overall results in a more complex system. The cost trade versus satellite constellation performance and cost favors lower altitudes for non-terrestrial mobile satellite systems, leveraging economies of scale.

As an example, a satellite at 600km attitude with 30deg minimum user elevation angle will require the number of beams to serve 5G NTN users across the entire FoV would be around 1000. Since the number of satellites is large and multiple satellites have overlapping fields of view, each satellite will be required to service between 200 and 500 beams at this altitude, depending on various factors, such as constellation design, operating latitude of a given satellite, link budget constraints, capacity demand, as well as power

and processing constraints limiting the number of beams that can be illuminated simultaneously. Beam hopping techniques are a widely adopted solution to resolve some of these limitations. This, however, complicates the resource management of the satellite network and reduces the effective capacity available over a given Earth fixed reference cell. A complex analysis is required to understand the effective capacity of an individual beam/cell, individual satellite, and for the entire constellation.

2.2.5 Ground Segment

2.2.5.1 Earth stations (Gateways)

Satellite gateways (GW) are ground stations that house antennas, RF and communication equipment enabling communications between the satellite and the terrestrial network.

In this document version, the focus is on the open reference system based on a transparent architecture (no CU/DU on board). A future target for next versions would be to address the integration of regenerative payloads into the open reference system.

A GW site comprises several of these systems and services multiple satellites. Multiple antenna systems include the following:

- **Radio Frequency Systems (RFS):** Antenna, RF electronics, pointing and tracking mechanisms, etc.
- Telemetry, Tracking, and Control modems (for spacecraft operations), in-band and payload command and telemetry modems (for payload configuration).
- **Provided a transparent payload on board, the RAN receives and transmits modulated signals, only RF electronics in the GW** (other splits to be detailed in a subsequent version of this document).
- Monitoring and control mechanisms (M&C) (including calibration).
- Frequency and timing subsystem: including GPS receivers, frequency generators, feeder link, Doppler compensation, etc.
- Transport facilities for interconnection.

The GWs have the main functionality of receiving signals from the satellite and transmitting signals to the satellite. A GW interfaces with the feeder link on the radiation side (towards and from the satellite at the feeder link uplink and downlink frequencies), and with the Radio Access Network domain on a digital or analog conducted interface (e.g., towards and from the ground network via an intermediate frequency (IF) network).

- **Analog feeder link interface:** The signals received by the RAN would be the modulated OFDM carriers, as defined by 3GPP 5G NTN standards, to be uplinked to the satellite and directly transmitted to the UTs in a transparent way **(only frequency conversion and amplification are performed on board the satellites)**.
- Feeder link frequencies (Ka/Q/V).

For a GEO satellite, only one GW is required to enable space-to-ground continuous connectivity. This is because the GEO satellite and the GW site line of sight (LoS) can be considered constant/fixed. More than one GW can be implemented depending on the link availability requirement and feeder link capacity requirements.

For other satellite orbits (NGSO), several GWs are required to enable continuous space-to-ground connectivity (Figure 9). This is because the satellites move with respect to the GW site, and LoS is lost when a satellite moves below the horizon. Handovers or other satellite mobility mechanisms are required to keep a continuous space-to-ground communication link, and the satellite will always have at least one GW with direct LoS.

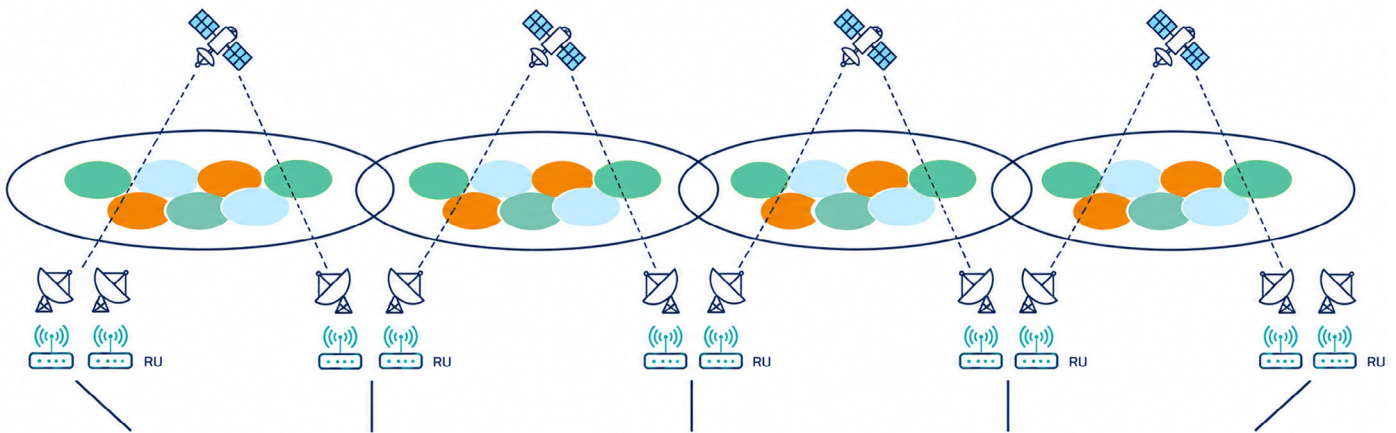


Figure 9 : GW Requirements for Continuous Connectivity

There are several mobility considerations for 5G NTN constellations:

- **Feeder Link switchovers:** relates to procedures involving a satellite change of feeder link from a source gateway to a target gateway from the point of view of a given satellite. It can be triggered by satellite mobility (satellite losing LoS with one GW, so switching to another visible GW), link availability such as rain events (satellite switching to a redundant GW with the purpose of keeping service availability commitments, QoS), network load/resources optimization (satellite switching gateways because of a higher-level optimization algorithm), or other events such as GW maintenance, etc. 3GPP distinguishes between a hard and soft feeder link switch, depending on the capabilities of the satellite².
- **Soft Feeder Link Switch:** When a payload connection can cope with more than one gateway during a given period, i.e., a temporary overlap that can be ensured during the transition between the feeder links. This requires the capability within the satellite to maintain at least two feeder link connections in parallel. Higher layers of network architecture are required to tackle duplication and packet loss.
- **Hard Feeder Link Switch:** When a payload connection is established to one gateway at any given time, i.e., a radio link interruption may occur during the transition period between the feeder links. The feeder link switch causes an interruption in the connection.

² 5G NTN Takes Flight: Technical Overview Of 5G Non-Terrestrial Networks, Reiner Stuhlfauth

- **Satellite Switchover:** Relates to a procedure involving a UE change of its user link from a source satellite to a target satellite. This can be triggered by the deterministic mobility of satellites across the coverage, by the mobility of users across the coverage, or by network load/resources optimization. The system shall manage continuous mobility procedures to allow UEs to switch the satellite it is being served by. This switch can be performed either by a handover (in which case the source and target satellites are serving different cell IDs) or by a satellite switch with resync (in which case the source and target satellites are serving the same cell ID, and only a physical layer resynchronization is needed). The complexity of the satellite switch option depends on the architecture.

2.2.5.2 Network Operation Center (NOC) and Satellite Control Center (SCC)

Network Operation Centers (NOCs) are centralized locations where operators can monitor and control a network. A satellite NOC supports the monitoring and operations of the satellite as well as the gateways. These centers play a crucial role in network management, ensuring smooth communication and optimal performance. The NOC allows for quick detection and resolution of issues, minimizing service interruptions.

The Satellite Control Center (SCC) is responsible for managing satellites, ensuring their proper functioning and alignment with network objectives. Together, the NOC and SCC provide comprehensive monitoring and control of network operations.

2.2.5.3 Radio Access Network (RAN) Platform - Transparent Architectures

In a transparent architecture, the RAN Platform is part of the ground segment. RAN platforms are based on 5G or 4G Terrestrial Network (TN) technology. Satellite adaptations – e.g., timing advance - are required to accommodate the 2-way satellite delay between the gateways and UEs. The gNB is the central logical node of the 5G NG-RAN, including the 5G NR NTN RAN. The architecture of the NG-RAN is described in 3GPP TS 38.401 and follows the model below (Figure 10):

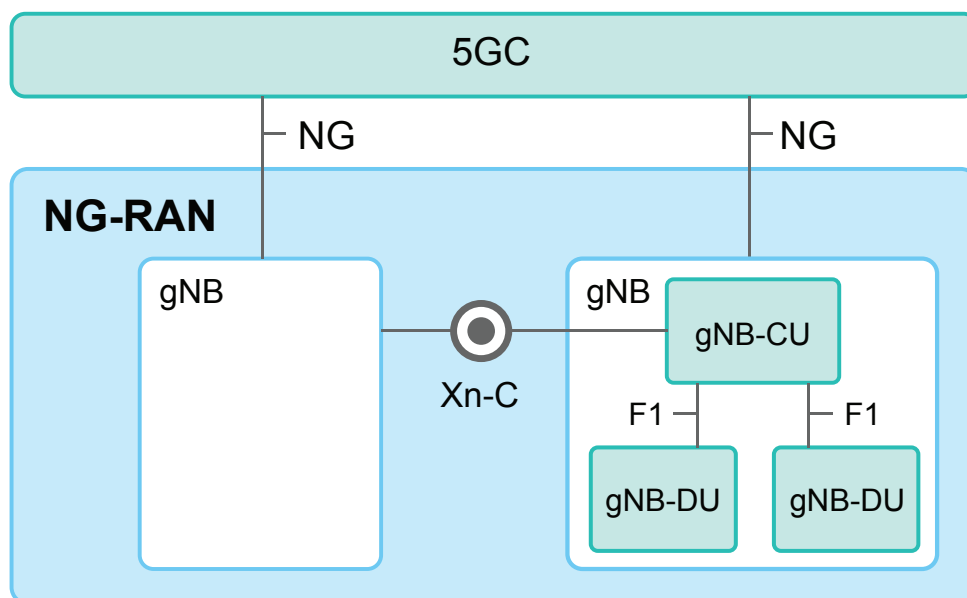


Figure 10: 5G RAN Reference Architecture in 3GPP

In the 3GPP architecture, the gNB (Figure 11) could be considered as a monolithic unit or can comprise a gNB-CU and one or more gNB-DUs connected via F1 interface. Under the 3GPP architecture, the radio unit (RU) is part of the Distributed Unit (DU), and the interface between them is out of the scope of 3GPP. The O-RAN ALLIANCE, however, defines an open fronthaul interface, the O-RAN Open Fronthaul (OFH), to standardize the Low Layer Split (LLS) between a DU and a RU. Hence, here on we depict the gNB to have 3 different entities, RU, DU, and CU, which can be described as follows:

- The CU hosts RRC, PDCP, and SDAP protocols and is responsible for handling the signaling/control-plane and non-real-time functions.
- The DU hosts L2 (RLC, MAC) and can host part of the physical layer. It is also responsible for scheduling the UEs and managing power control, link adaptation, and timing alignment.
- The RU generates the signal sent to the UE (over the Uu-interface) and receives the signal from the UE; it hosts (part) of the physical layer and handles UL/DL radio frame timing.

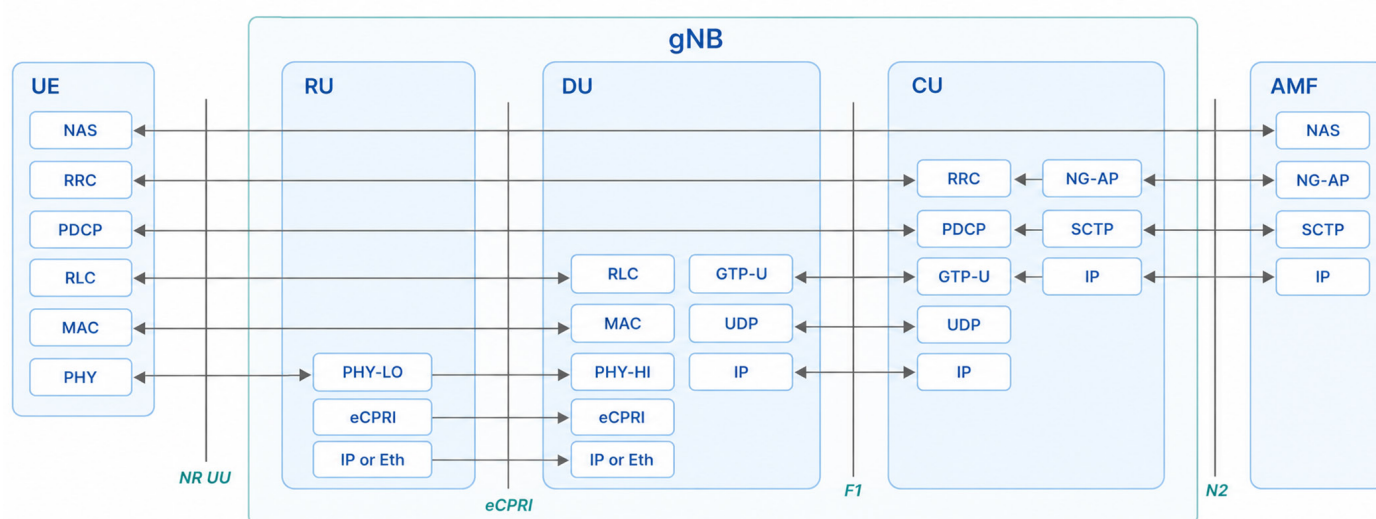


Figure 11 : gNB Entities and Interfaces for User Plane and Control Plane Protocols

It is up to the RAN vendor to split up or combine the gNB entities in their products. In the RAN product domain, the split between products is traditionally identified with the split between the radio (RU) and the RAN processor (a product typically including both CU/DU), also called “baseband unit” (BBU). The most supported split by RAN vendors is the RU-DU split, also known as Low-Layer Split (LLS), with most vendors using a proprietary CPRI- and/or eCPRI-based interface implementation connecting the radios with the RAN processors. Currently, in the product space, for terrestrial networks, there is a strong movement toward implementing a standardized open fronthaul interface, O-RAN OFH, to allow for a multivendor mix of radios and RAN processors.

Despite the existence of the CU-DU split via F1 in the 3GPP specifications, the 3GPP-defined F1 interface between CU and DU has so far failed to gain traction and become a verified open multi-vendor interface. F1 or a similar proprietary protocol is, however, often present in the RAN processor products from various RAN vendors. This allows RAN processors to run the DU function and another processor to run the CU in a centralized location. But since the F1 has failed to become a multi-vendor interface, the RAN processors need always to be from the same manufacturer.

Latency introduced by the Fronthaul Transport Network adds to the overall propagation delay observed at the DU and at the UE, thereby increasing the total HARQ Round Trip Time (HARQ RTT). If HARQ RTT exceeds the number of HARQ processes (32), HARQ would need to be disabled, resulting in a loss of the HARQ combining gain. It is therefore recommended to maintain a low fronthaul latency by deploying the DU close to the RU.

A centralized CU (Figure 12; at a country or region level) could benefit the system by facilitating PCI unchanged GW handovers while eliminating the need for CU/UE context exchange between dedicated CUs per GW.

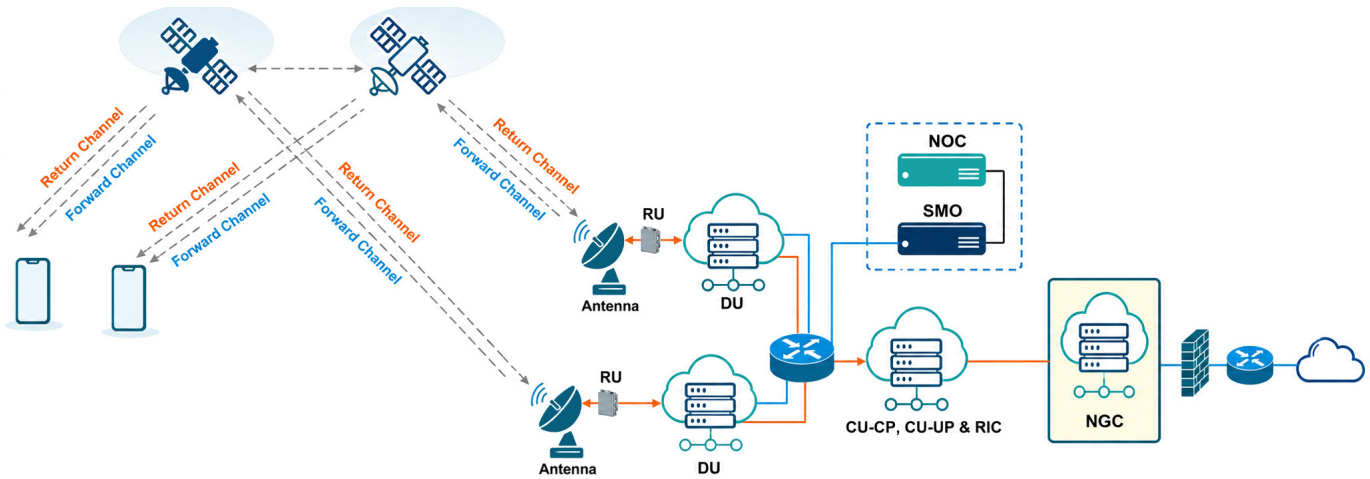


Figure 12: Distributed CU/DU Deployment Transparent Option Example

2.2.5.4 Core Network (CN)

The core network (CN) is a central piece of a cellular network; it handles session management, mobility management, authentication/authorization, policy control, subscriber data management, exposure, routing of user data and being the point of presence where the cellular network interface the internet. To handle all these tasks, the CN has several functions that are divided into a control plane and a user plane. Figure 13 depicts the nodes of three parallel 5GC's serving users in PLMN 1 and PLMN X based on different architectural principles (gNB in home network, gNB in visited network via RAN sharing and roaming).

In a transparent architecture, the core network is based entirely within the ground segment. In transparent architectures, it may be desirable to place some functions of the core on the satellite payload. Some examples of why this might be required are to support advanced features in 3GPP Release 19, such as UE-satellite-UE where traffic is routed entirely within the constellation, and store-and-forward for IoT where data is stored until a suitable feeder link becomes available.

Both 4G- (EPC) and 5G- (5GC) based core networks will be used in NTN. EPC is the model used for NB-IoT NTN, while NR NTN makes use of the 5GC. 5GC, being the newer architecture, supports more services; one notable difference is the support of network slicing, which is only available in 5G. The following bullets (and Figure 13) elaborate on CN architectures in an NTN service context.

- The core network (CN) architecture and required system interfaces depend on whether the subscription is provided by the satellite network operator (SNO) that owns the satellites/GW (i.e., the RAN) or by another operator, e.g., a mobile network operator (MNO), that has a business agreement with the SNO (e.g., via RAN sharing or roaming agreements).
- A roaming architecture (see 3.4 Roaming) allows a serving network (visited PLMN) to serve UEs that have subscriptions with another operator (e.g., with an MNO). This requires a roaming agreement between the SNO and the MNO. Typically, there is also an intermediate “roaming exchange” (IPX) that acts as a “hub” between operators. In the case of roaming, all data goes back to the home network before being sent to the data networks (e.g., the internet). In the case of roaming, the SNO needs to support a minimum CN interfacing the CN of the MNOs.
- Network sharing architecture (see section 3.2.4) allows multiple participating operators (MNOs) to share resources of a single shared RAN (provided by the SNO) according to agreed allocation schemes. The cellular systems support Multi Operator Core Network (MOCN) and Multi Operator Radio Access Network (MORAN). MOCN is deployed for the case that each connected PLMN shares the spectrum, which would be the most common case if using MSS frequencies. MORAN allows for the case when each PLMN uses dedicated spectrum. In the case of RAN sharing, the SNO does not need to have any CN functions.
- In case the SNO supports its own subscribers, the SNO needs to at least deploy the corresponding functionality to handle subscriptions, functionality to handle mobility and session management, as well as charging support. In addition, other functionality of an MNO such as policy control and functions for network slice selection may be required, depending on use cases and requirements
- It is possible to construct a CN architecture allowing a SNO to provide RAN sharing and roaming, including support for own SNO subscribers. Such architecture (Figure 13) is outlined below.

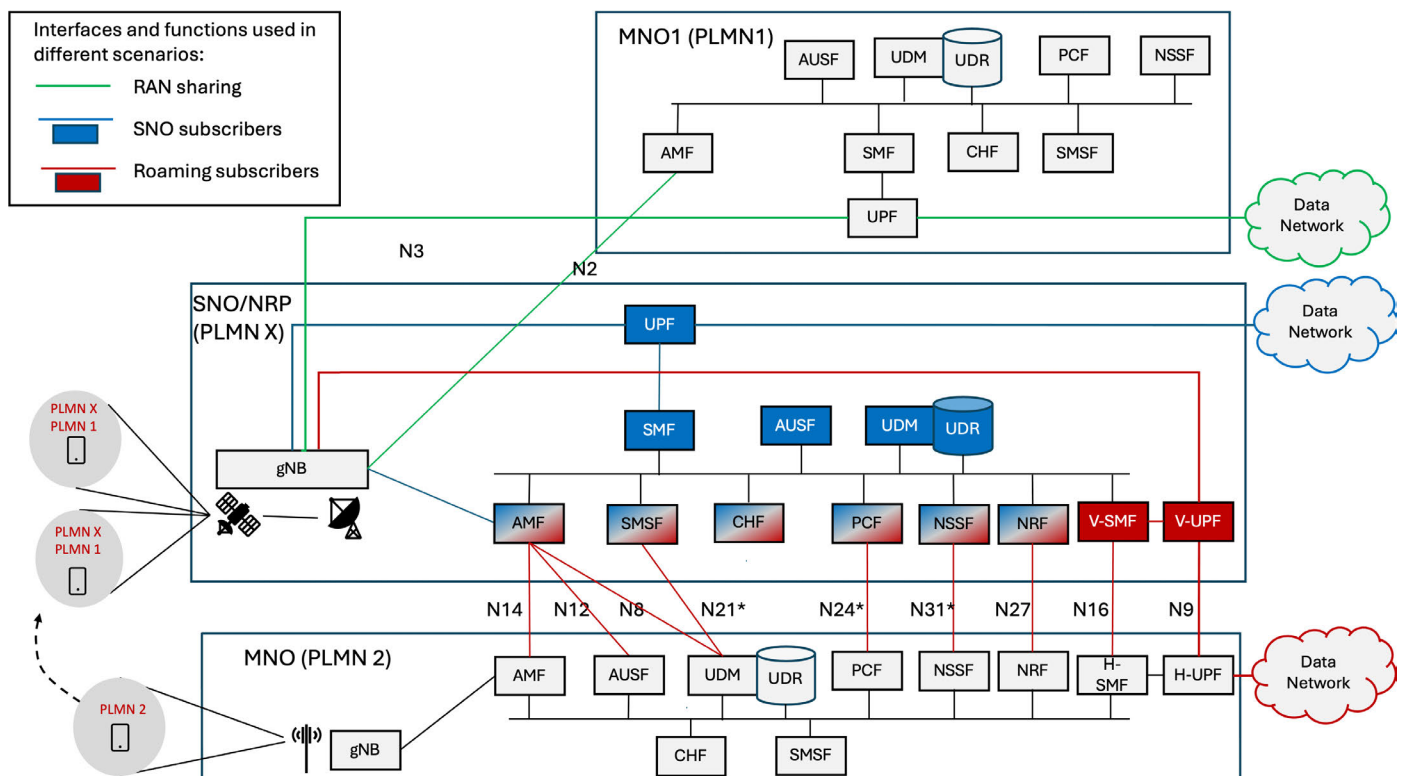


Figure 13: 5GC CN Architecture for RAN Sharing, Roaming and Own SNO Subscribers

NTN introduces some impact on core network functions. Examples of impacts are:

- A new access type (RAT type) used by the UE and network to, e.g., be able to have different policies/charging when over NTN compared to TN.
- The QoS framework is impacted, especially by the longer packet delay budget (PDB) longer/optimized protocol timers for session and mobility management (mainly for GEO deployments).
- Support for Store-and-Forward function.
- Enhancements for handling discontinuous coverage.
- The regenerative architecture will impact the CN when 3GPP Release 19 is completed.

Core Network – Deployment Perspective

The virtualization journey of core networks started much earlier than for RAN. The new 5GC products for standalone operation are, as a rule of thumb, built upon cloud-native service-based architectures. Older 4G core (EPC) products in production are a mix of PNFs, VM-based virtual network functions (VNFs), and products built upon cloud-native principles.

EPC and 5GC will co-exist for the near future; for instance, 4G IoT services are going to be supported for a long time.

The most typical coexistence mechanism is dual-mode (shown in Figure 14) operation, also known as “dual-stack,” allowing operators to shift to a cloud-native architecture for their core networks and continue to support the “4G only” devices/subscribers alongside 5G subscribers.

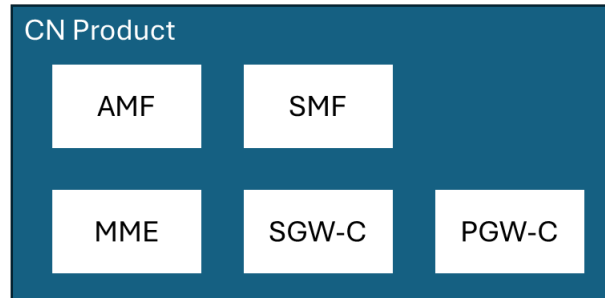


Figure 14 : Dual-Mode CN Product (Control Plane) Running as 4G CN or 5G CN or Both Simultaneously

Another choice that operators are facing is whether to implement the CN on an in-house Telco Cloud, the Public Cloud, or in a hybrid approach. Operators have so far been quite conservative, and even most cloud-native implementations have been done on private Telco Cloud. Still, public/hybrid deployments are of interest. To support current and future customers, Core Network vendors need to support several different deployment options.

A deployment approach for upcoming global NTN constellations might be a hybrid approach, leveraging Control-User Plane Separation (CUPS), to enable deployment of a control and management plane in regional public clouds, with high-throughput user plane deployed at the edge, in Edge clouds or on-premises Telco Cloud deployments.

2.2.6 Resource Management System

2.2.6.1 Role of the Resource Management System

Satellites are inherently limited in power and on-board processing capabilities, often driven by size, mass, and impact on manufacturing and launch costs. This affects the way network and resource management are performed, since their limited resources must be carefully managed and allocated to ensure an optimal balance between coverage, capacity and service performance, within the available limits and constraints. In a satellite or other non-terrestrial network, this function is typically fulfilled by the Resource Management System (RMS) or Segment.

In addition, network demand experiences continuous changes due to geographic areas, time of the day or year, ephemeral events, user mobility and, more fundamentally, the combination of multiple different service types and user groups. Due to the wide-scale or even global nature of Non-Terrestrial Networks (NTN), these effects can impact the network on a very large, if not global, scale and often more dynamically, in particular in the context of Non-Geostationary (NGSO) constellations, which also introduce the added complexity of platform motion into the equation.

To address these challenges, satellite and overall network and constellation resources must be allocated as much as possible in a dynamic fashion, whilst at the same time complying with system and network hardware and operational constraints, as well as, of course, the specified parameters of the 3GPP standards.

This dynamic allocation involves a number of different decisions and operations, the most obvious of which is the continuous adjustment of the distribution of power and beams based on real-time demand and traffic patterns, as well as constellation evolution and orbital motion, depending on the architecture. By doing so, the satellites can efficiently serve multiple regions and users, ensuring that high-priority areas receive the necessary coverage when needed. This approach must not only maximize the utilization of the space segment's limited power, but also ensure compliance with industry standards, thereby maintaining the quality and reliability of the service provided.

In a 3GPP NTN MSS system, the Resource Management System (RMS), often implemented as a logically centralized function, must interface and interact with the different components of the system end-to-end, including the Space Segment payloads, the Ground Segment Gateways, RAN eNBs or gNBs and other subsystems, and potentially more, in order to coordinate the dynamic resource allocation ensuring alignment to the 3GPP air interface and service configuration and requirements.

Figure 15 shows the resource management system, where centralized allocation interacts with local functions and eNB/gNB activities.

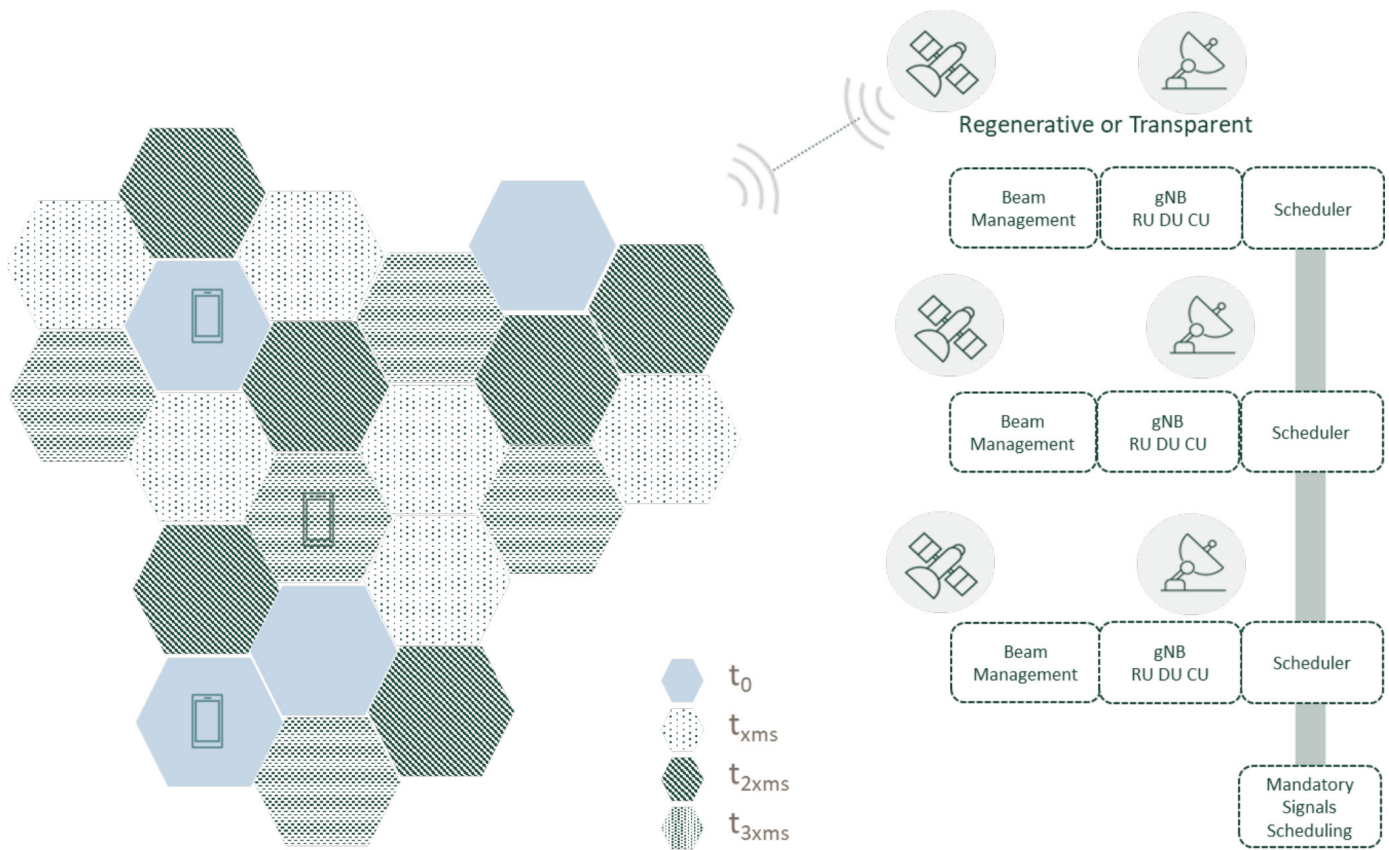


Figure 15 : High-level Resource Management

Some of the key space segment and system limitations and constraints that need to be addressed by the Resource Management System (RMS) include the following:

Total Power Limitation:

- A typical satellite deployment may be designed to reach a given Power Flux Density (PFD) level over its coverage area. The coverage area of a satellite will be subdivided into multiple beams. Due to limited solar array output and battery capacity, the satellites are expected to have an upper threshold of total aggregate payload power that they can radiate, often referred to as Aggregate Equivalent Isotropic Radiated Power (AEIRP). As such, it will not be possible to achieve the desired PFD level across all the possible beams in the satellite's coverage area simultaneously.

Beamformer Processing Limitation:

- In addition to the power limitation, a satellite may be limited by the number of independent signals it can simultaneously generate. Under an antenna array architecture with N antenna elements, the satellite may be able to only simultaneously generate K independent signals (each one corresponding to a beam), with typically $K \ll N$. Under a fully digital beamforming architecture, this limitation arises when the beamformer hardware can only perform a limited number of products per sample (having full beamforming flexibility would require N^2 complex products per sample).

Spectrum Availability Limitations:

- Spectrum is a very limited resource, especially in Mobile Satellite Systems (MSS), often driven by regulatory requirements as well as multiple users of the spectrum, both on a global and regional level, which often places region-specific limitations on spectrum availability and usage constraints, such as emissions, PFD, or field strength requirements.

Interference Management Limitation:

- In satellite communications, interference management is crucial to ensure optimal performance. One key limitation is that adjacent beams cannot use the exact same resources (e.g., time, frequency, code) without some form of precoding or resource orthogonalization, often addressed through various reuse patterns. This approach minimizes interference and maximizes the efficiency of the satellite system.

Bandwidth / Channel Sizes / Feeder Link Limitations:

- Furthermore, bandwidth and channel sizes are critical factors. Limited bandwidth restricts data transmission rates, while smaller channel sizes can lead to congestion. Feeder link limitations further complicate the situation, as they impact the satellite's ability to maintain continuous connectivity with ground stations, affecting overall system performance.



A consequence of these limitations is that not all beams or earth fixed cells can be illuminated simultaneously. For instance, a reference scenario capturing these limitations has been studied in 3GPP as follows (Table 2):

LEO600km Set1-1 FR1 (i.e., S-band)	
Maximum Bandwidth per beam	5 MHz
SCS	15 kHz
Beam size (Note 1)	50km
Satellite EIRP density /beam (dBW/MHz)	34
Payload Total DL power level (dBW)	31.24
Aggregated EIRP (Total) (dBW)	61.24*
Satellite Tx max Gain	30 dBi
Maximum EIRP per Satellite beam (dBW)	41
Total number of beam footprints***	1058
Total number of simultaneously active beams **	106
% simultaneously active beams**	10.02 %
*Note: EIRP limit is 61.24 dBm for the reference configuration. **Assuming 100 % Resource Block utilization within the same beam at max power. Absolute number of simultaneously active beams is up to 212 (due to limitation of RF). *** For a constellation design at 600km with low elevation angle with 30° and selected (i.e., Set 1 parameters) beam size. Note 1: At least this beam size is considered in this scenario, larger beam sizes evaluated and reported by companies.	

Table 2 : LEO 600km S-Band Example

Under this scenario, the satellite can illuminate ~10% of its beams at the desired PFD (for the full 5MHz) or illuminate up to ~20% of its beams at reduced power per beam (either at a reduced PFD or with reduced bandwidth).

Given the limitations, it is imperative that the communication system architecture be optimized to achieve the objectives of interoperability and seamless interaction between multiple satellites' communication systems.

2.2.6.2 Resource Management System: Role and Key Functions

The Resource Management System (RMS) plays a crucial role in satellite network operations by addressing satellite-specific operational constraints and working alongside the service-level orchestration. Together, these components ensure efficient, reliable, and coordinated service delivery and network resource utilization across terrestrial and non-terrestrial domains.

The Resource Management System's responsibilities can be summarized as follows:

Frequency Band Usage and Prioritization:

- Manages allocation and prioritization of frequency bands for satellite communications, ensuring compliance with regulatory frameworks, and minimizing interference.
- Addresses legacy constraints, such as pre-assigned frequency plans or coordination with terrestrial networks.
- Dynamically adapts frequency assignments in response to changing network conditions, satellite position, or user demand.

Payload Supervision and Control:

- Monitors and manages satellite payload operations, including power budgets and duty-cycle limits to maximize operational efficiency and extend satellite lifetime.
- Oversees implementation of operational constraints related to Optical Inter-Satellite Links (OISL), ensuring optimal link performance and resource utilization.
- Coordinates payload settings with real-time network demands and SMO policies.

Definition and Management of OISL Routing:

- Designs and manages routing schemes for Optical Inter-Satellite Links, enabling efficient data transmission between satellites within the constellation.
- Continuously optimizes inter-satellite link configurations based on network topology, traffic flows, and satellite positions.
- Integrates OISL routing decisions with ground network routing and overall constellation management.

Priority Handling Across Multiple RAN Domains:

- Resolves resource conflicts and manages priority among different Radio Access Network (RAN) domains, especially in scenarios where several SMOs may request access to the same satellite resources.
- Implements policies for fair and efficient resource sharing, supporting service level agreements (SLAs) and quality of service (QoS) guarantees.
- Facilitates coordinated multi-domain operations, ensuring seamless service delivery across terrestrial and satellite segments.

By structuring its functions in this way, the Resource Manager ensures that satellite network operations remain efficient, adaptable, and aligned with the broader RAN management objectives set by the SMO. This approach allows for the integration of satellite and terrestrial domains, addressing the complexities and dynamic nature of modern communication networks.

2.2.7 User Segment

User terminals are devices used by end users. A device could be classified as fixed or mobile. Fixed devices are usually larger devices with external antennas for higher data rate applications, such as routers. Examples of mobile devices include smartphones, tablets, and IoT sensors. Vehicular, maritime, and aviation terminals would also be classified as mobile devices. Fixed terminal devices are units that are in a permanent location or are portable terminals which typically do not move once they are active.

MSS, with the understanding that the satellite component is aimed at complementing, i.e., extending or filling outdoor coverage gaps of the terrestrial network with voice, text, and low speed data service, and is not providing the same 4G/5G performances as terrestrial networks. The main MSS use cases would be as follows:

- **Narrowband connectivity:** Support of low data rate connectivity using narrowband communications (e.g., NB-IoT 180kHz) to IOT or mobile devices. Services expected to be delivered over narrowband connectivity comprise sensor/metering data, telemetry, messaging services, emergency services and potentially low-rate voice. Data rates provided by this use case are in the order of a few kilobits per second.
- **Mobile broadband:** Support for larger data rates, including high-rate voice and data services. Mobile broadband services are expected to be directly delivered to mobile devices and provide a similar user experience to cellular networks (although with lower data rates). Data rates provided under these use cases are in the order of tens of kbps to several Mbps.
- **Fixed broadband:** Support of even higher data rates through larger EIRP / antenna gain at the UE (e.g., in the form of CPEs which may be mounted in vehicles). These larger terminals would act as concentrators from which the end user may use a separate technology to connect to (e.g., wired, Wi-Fi, or small cellular cells). Data rates for this use case are in the order of 10 Mbps.

Table 3 below summarizes the assumptions necessary to calculate the link budgets for each device class for both uplink and downlink performance.

Parameter	Units	Terminal Type				
		Handheld ¹	NB-IoT ²	Vehicular ³	Maritime/ Aviation ⁴	Fixed Terminal ⁴
Frequency Ranges (bands 255 and 256)	MHz	Uplink -- 1626 to 1660 and 1980 to 2010 Downlink -- 1525 to 1559 and 2170 to 2200				
DL Signal Bandwidth⁵	MHz	5,10	0.180	5, 10	5,10	5,10
DL Resource Blocks⁵	#	25,50	1	25,50	25,50	25,50
UL Signal Bandwidth Per resource block⁶	kHz	180	3.7, 15	180	180	180
Antenna Gain	dBi	-5.5	-5.5	1.0	8.0	8.0
Transmit Power	dBm	23	23	23	29	29
EIRP	dBm	17.5	17.5	24.0	37	37
Noise Figure	dB/K	7	7	7	7	7
G/T	dB/K	-36.1	-36.1	-29.7	-21	-21
Body Blockage	dB	3	0	0	0	0
Polarization Loss	dB	3	3	3	.5	.5

Table 3 : Device and Terminal Types Characteristics

Notes:

- Mobile devices such as smartphones can support voice, data, and location services.
- IoT devices such as sensors can only support data transmission. NB/IOT devices can extend performance with repeat transmissions.
- Vehicular antennas assume a patch antenna.
- Maritime/Aviation/Fixed terminals assume 10+ cm antennas.

2.3 MSS User Services

2.3.1 Services Overview

Services delivered by Mobile Satellite Systems can be broadly grouped into three segments, each with different characteristics and delivery models:

- Direct-to-Consumer Segment (or simply Consumer Segment)
- Enterprise Segment
- Government and Public Sector Segment

This version of the document focuses primarily on the first two.

2.3.2 Direct-to-Consumer Services

The Direct-to-Consumer Segment covers services that are directly targeted at the commercial consumer mass market, providing NTN mobile satellite connectivity to consumer-oriented device categories, such as smartphones (the predominant category), wearables, consumer drones, portable CPEs, etc., and for which the end customer and source of revenue of the service is generally the individual consumer.

This segment includes services generally provided through agreements with Mobile Network Operators (MNOs) as an extension of cellular network coverage.

MSS services offer two primary business models for direct-to-consumer applications:

- Business-to-MNO-to-Customer (B2MNO2C)
- Business-to-Customer (B2C)

The first model, **B2MNO2C**, extends cellular network coverage through agreements with Mobile Network Operators (MNOs). This model ensures that consumers can access services such as text messaging, data messaging, emergency services, location-based services, voice services, and data services including video streaming, even when they are outside the reach of terrestrial networks. Whilst not always the case, these services are primarily accessible when a customer is not served by the MNO's terrestrial network.

The second model, **B2C**, involves satellite service overlays where subscribers directly access satellite services. This model provides similar services but operates independently of terrestrial networks, offering seamless connectivity in remote or underserved areas.

Below is an overview list of the services falling into the Consumer category:

- Text Messaging: SMS, MMS
- Data Messaging: Chat, emails
- Emergency Services:
 - SOS/Rescue Services (including both messaging and voice-based services)
 - Location-based services/navigation
 - Natural hazards (broadcast, including Public Warning System - PWS, terrestrial network outage, others)
- Voice Services:
 - IMSI/E.164 number-based (either native or application-based)
 - VoIP service
 - 2-way calls
 - Push-to-Talk
 - Video calls
 - Data services including video streaming
- Connected Objects for Consumers
 - Wearables
 - Home automation
- Data Service and Internet-Based Application
 - Throughput ranges: kbps to Mbps

2.3.3 Enterprise Services

In the enterprise segment, NTN services cater in large part to connected objects and/or private networks. These services are crucial for industries such as agriculture, automotive, industrial, smart building, tracking, maritime, aviation (including unmanned), and utilities.

NTN services enable reliable data transmission for IoT devices, ensuring continuous monitoring and control of connected objects. Additionally, private networks benefit from NTN services by maintaining secure and efficient communication channels for national security, drones, shipping, aviation safety, and maritime applications.

The enterprise segment leverages NTN's ability to provide robust connectivity solutions in environments where traditional networks may fall short.

IoT Services / Verticals

Agriculture:

IoT services in agriculture enable precision farming, which involves monitoring soil moisture, weather conditions, and irrigation systems to optimize crop yields. For example, IoT sensors can be placed in fields to measure soil moisture levels and send real-time data to farmers, allowing them to adjust irrigation schedules accordingly.

Automotive:

In the automotive industry, IoT services enhance vehicle connectivity and safety. IoT-enabled vehicles can communicate with each other and with infrastructure to improve traffic flow and reduce accidents. For instance, IoT sensors can be used for fleet tracking and route optimization in logistics.

Industrial:

IoT services in industrial settings improve efficiency and productivity. IoT sensors can monitor equipment performance and detect anomalies, allowing for predictive maintenance and reducing downtime.

Smart Building:

IoT services in smart buildings enhance energy efficiency and occupant comfort. IoT sensors can monitor and control lighting, heating, ventilation, and air conditioning (HVAC) systems based on occupancy and environmental conditions.

Asset Tracking:

IoT services for tracking are crucial in various industries, including logistics and transportation. IoT-enabled tracking devices can monitor the location and condition of assets in real-time.

Utilities:

IoT services in utilities improve the management of energy and water resources. IoT sensors can monitor the performance of power grids, renewable energy farms, and water management systems.

Other Enterprise / Private Networks Services

Non-Terrestrial Networks (NTN) services are revolutionizing private network solutions by offering seamless connectivity in critical areas. Examples include:

Public Safety:

NTN ensures reliable communication during emergencies, enhancing response times and coordination.

Drones:

Equipped with NTN, they can perform surveillance and monitoring tasks efficiently, providing real-time data in remote or hazardous locations.

Mining Industry:

NTN supports private networks, enabling robust communication and data transfer in areas with limited terrestrial coverage.

Shipping and Maritime Transportation:

Benefits from NTN by maintaining continuous connectivity, ensuring safety and operational efficiency even in the most remote oceanic regions.

3 USE CASES AND ARCHITECTURE CHOICES

3.1 Background and Main Use Cases

Native support for Non-Terrestrial Networks (NTN) in the 3GPP system is achieved by implementing a set of standardized capabilities in 3GPP-compliant Radio Access Networks (RAN), User Equipment (UE), and Core Networks (CN), coupled with the suitable Space and Ground Segment architectures. Using these capabilities, in conjunction with the appropriate Space Segment, Ground Segment and Resource Management considerations, different end-user services can be realized.

As outlined in Section 2, the choice of the system architecture and orchestration among the different components needs to account for some critical service and network management aspects that require specific considerations for NTN, such as:

- Beam and Cell Management
- Access and Load Management
- Control Signaling
- Location Management
- Mobility Management
- Network Selection
- Roaming, Lawful Intercept
- Multi-RAT Support

The following section explores a set of specific use cases with corresponding architecture choices, potential implementation solutions, and corresponding recommendations in relation to these topics. Seven main use cases have been identified to be addressed as a priority for this second version of the Reference Architecture document. Subsequent revisions will address additional areas of interest:

- Network Resource Management
- Network Selection and Network ID Management
- Roaming
- Network Sharing
- NB-IoT and NR Co-deployment
- Use of Existing Space Segment
- MSS User Services

3.2 Network Resource Management

3.2.1 Overview

As detailed in section 2.2.4, satellites are inherently limited in power and on-board processing capabilities, which affect the way resource management is performed. Given the limitations, it is imperative that the communication system architecture be optimized to achieve the objectives of interoperability and seamless interaction between multiple satellites' communication systems.

The following sections provide the foundation for such objectives.

3.2.2 Resource Management System Domains

Resource management functions can be divided into three main domains: non-real-time, near-real-time and real-time, with each category handling various aspects of network control and optimization.

Whilst the following grouping should not be considered “strict,” with some aspects depending on architecture and even implementation, it should provide a broad example of what functions tend to fall into each of the three groups.

Non-Real-Time

- Satellite orbit tracking and prediction
 - First-level traffic resource orchestration and scheduler
 - Grid definition
 - Frequency allocation (spectrum) and interference management
 - Feeder link allocation (contact plan)
 - Gateway allocation (contact plan)
 - Processing traffic (regenerative and digital OISL)
- Beam allocation (Cell to Sat allocation)
 - Cell allocation/satellite (CU/DU allocation)
 - QoS and SLA management
 - Ground network routing (PoP, GW sites ...)
- Routing feeder links and OISL traffic
- Status of the constellation (Via Satellite Control Center interface)
 - Payload monitoring and control including OISL
- Status of the ground network (routers and GWs)
- Payload crypto management

Near-Real-Time

- Fine-tuning resource orchestration interfacing with the RAN
 - Traffic load per cell/slices
 - UE association/cell
 - Mandatory cell signal scheduling
 - Beam arbitration and beam hopping schedule generation
- Cross-satellite and multi-RAN cluster scheduling
- Resource arbitration in case of traffic priority:
 - Multiple RAN
 - NB IoT vs 5G NR
- Space segment payload orchestration
 - Beam layout and shaping

Real-Time

- User plane traffic processing
- RAN L3 RRC/PDCP functions
 - RRC signaling and message encoding/decoding
 - QoS mapping to PDCP sessions
 - UE context and state machines
- RAN L2 MAC scheduler functions
 - PRB/Slots per UE allocation
 - Control/Data channel scheduling/multiplexing
 - Reference Signal scheduling
 - Transport Block selection
- RAN L1 PHY baseband functions
 - Resource mapping
 - Channel Coding (encoding, decoding)
 - Modulation and Coding Scheme (MCS) selection
 - Modulation
- Payload physical layer and RF functions
 - Beamforming

It should be noted that, generally, the Real-Time domain of the Resource Management System is actually implemented by the RAN nodes themselves, due to the very tight timing requirements involved.

Once the resource allocation is defined, the resource management will transfer the new configuration to all components, such as the satellite payloads, RAN nodes, Gateways and their sub-systems, etc.

The following sections describe some of these functions and operational concepts in further detail.

3.2.3 Satellite Tracker Function

The integration allowing for satellite system information to be fed into the 5G systems initially designed for terrestrial networks is crucial for ensuring a working communication solution with as good a performance as possible. This process involves several key functions and techniques that bridge the gap between the Satellite Operational Domain and the Radio Access Network (RAN) Operational Domain. See Figure 16 below.

First, the system bridges the gap between the Satellite Operational Domain and the RAN Operational Domain by ensuring that satellite-specific behaviors are isolated from the RAN to a maximum extent possible. This isolation is essential to maximize the reuse of the 5G RAN solutions developed for terrestrial system but still be able to leverage satellite information to realize the satellite services.

The system proposed tracks current satellite positions and activates/deactivates satellites for RAN coverage. This dynamic tracking and activation process ensures that the network can adapt to changing satellite positions and maintain optimal coverage.

Additionally, the system maps Cells/EFAs to active satellites and generates SIB19 information. This mapping process is crucial for providing accurate and timely information to the network, enabling efficient resource allocation and management.

To ensure seamless connectivity, the system establishes and tears down fronthaul connectivity between gNBs (DUs) and satellites in a timely manner. This process is vital for maintaining high-quality communication links and minimizing latency.

Moreover, the system keeps satellite network operations running in case of software upgrades, outages, or failures in the Satellite Operation Center (SOC) or the Network Operations Center (NOC/OSS). This resilience is essential for ensuring continuous service availability and reliability.

One of the significant challenges in this integration is to create isolation between cells. This isolation is necessary to prevent interference and ensure efficient communication. The solution to this challenge involves efficient frequency reuse and beam hopping techniques. These techniques optimize the use of available frequencies and dynamically allocate resources to minimize interference and maximize coverage.

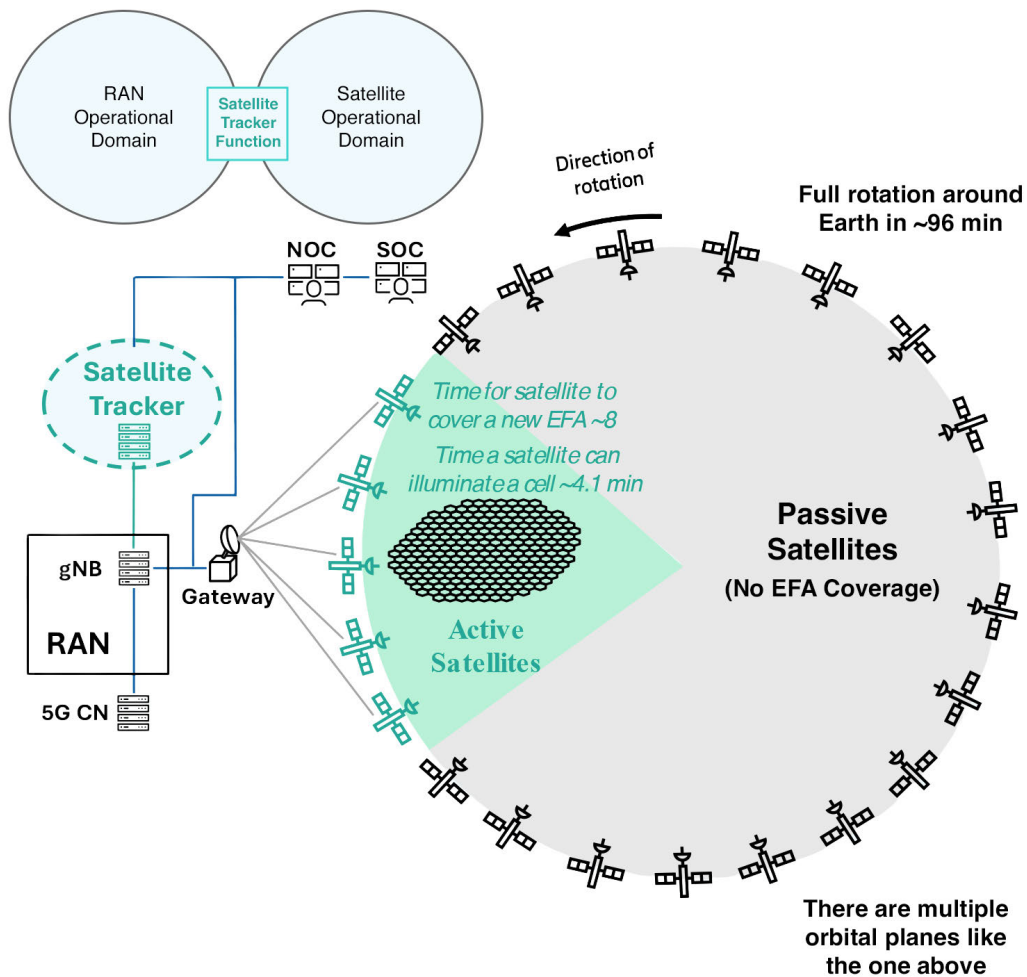


Figure 16 : Satellite Tracker Functions

3.2.4 Beam Hopping with TR and FR

3.2.4.1 Beam Hopping with TR and FR

The satellite can direct N simultaneous beams (usually limited by the available output power of the satellite and the number of RF chains) towards M cells in the satellite footprint, as the satellite moves between the minimum elevation angles (Figure 17).

The number of simultaneous full band beams that a satellite can provide varies depending on satellite design. The number of beams can be increased by allocating a smaller bandwidth and power to each of the beams, for instance with frequency reuse (FR) (e.g., three times with FR3 but can be limited by the number of RF chains).

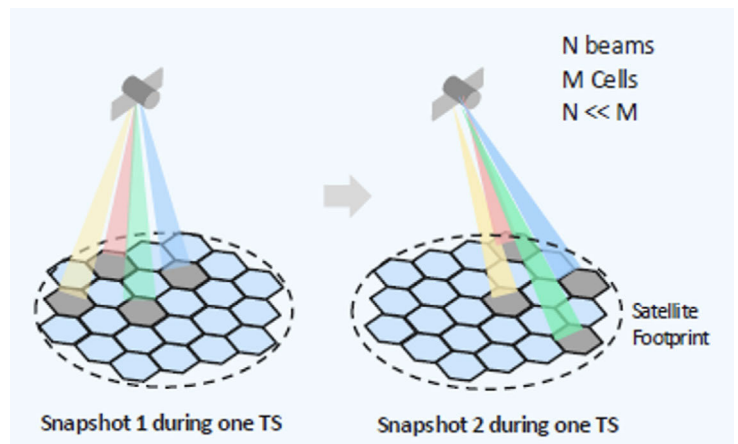


Figure 17 : Time and Frequency Hopping

Time, frequency, and power resources are allocated to beams that shall illuminate each cell in the satellite footprint (Figure 18).

- A pre-scheduled beam hopping (BH) plan that takes traffic demands into consideration.
- Traffic demand-driven BH plan (transmits packets to each UE as they arrive, where the illumination plan is not constrained by a regular repetitive pattern).
- A combination of pre-scheduled and traffic-driven BH plans.

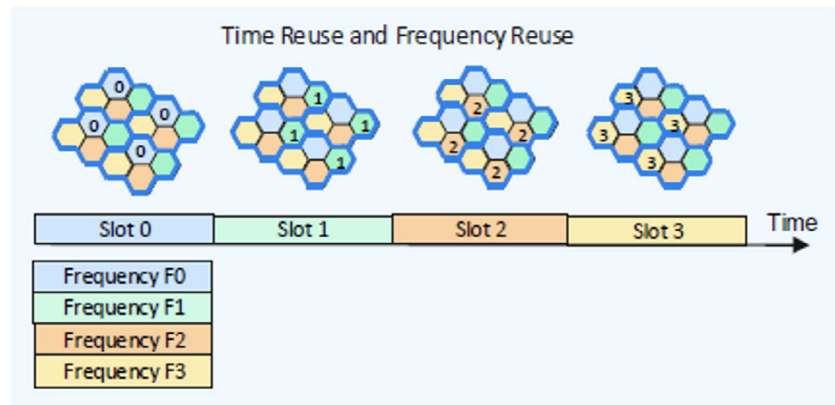


Figure 18 : Time Reuse and Frequency Reuse

In general, the BH plan(s) cannot be updated in real time: although the BH plan may have a granularity of a single slot (i.e., different cells may be illuminated in different consecutive slots) or smaller, the sequence of cells to be illuminated may need to be decided well in advance (e.g., the BH plan may be programmed 100s of ms in advance).

Although moving beam hopping scheduling to the payload in a regenerative architecture reduces the latency and allows a faster response to real-time information, the need to schedule TR and FR between satellites means that a proportion of the processing will still need to be carried out on the ground, and the complexity of splitting the processing might outweigh any benefits.

Time Reuse (TR) and Frequency Reuse (FR) are essential techniques to mitigate interference in satellite communications. Frequency Reuse enables the creation of more simultaneous beams, enhancing capacity. Beam Hopping (BH) strategically allocates resources to beams and directs them to cells, optimizing coverage and performance by dynamically adjusting to traffic demands and minimizing interference. This approach ensures efficient utilization of satellite power and resources, allowing for better service quality and reliability. By combining these techniques, satellite systems can effectively manage interference and maximize their operational capabilities.

The main benefits with Beam Hopping Interference mitigation (by avoiding illuminating neighbor cells simultaneously with the same frequency or with use of FR) are:

- Increased coverage
- Improved link budget
- Load balancing by illuminating cells with high traffic load for a longer time
- Reduces the constellation size needed to provide global coverage with a given satellite size (i.e., payload power/mass budget)
- Allows support of spectrum-limited systems where frequency reuse is not possible or desirable

3.2.4.2 Mandatory Cell Signals

3.2.4.2.1 NR Mandatory Cell Signals

For a 5G NR NTN system to work, you need to comply with the mandatory cell signaling rules defined in 3GPP. BH creates some additional complexity when considering NR mandatory cell signals that could impact system capacity negatively if not managed in a proper way.

To achieve efficient operation of the system, the resources allocated by BH would need to be distributed according to the load of a cell. In the most extreme case, a cell may have no active traffic at a given point in time and would just need to transmit the minimum set of signals for a new user to establish a connection.

For a cell to be detectable by a UE (and for the UE to camp on it), the minimum set of channels that need to be periodically transmitted in the downlink is as follows:

- SSB, SIB1, SI messages

In the uplink, the beam should be illuminated in the random-access occasions (RO).

Although there are several other signals that may not be mandatorily transmitted, due to the delay in pre-programming the BH plan, we would need to account for potential messages that the network may need to transmit or receive depending on traffic. These include necessary signals to establish a connection and the paging channel.

Note that, in the case where the system is limited by power and not by RF chains, the beam hopping may be “overbooked” (e.g., the same slot for msg2 can be illuminated in several cells with the understanding that there is a low probability of having to send msg2 in many of these cells simultaneously). These channels are marked as “Always on = No.”



The following Table 4 includes the set of channels and the corresponding periodicities per 3GPP specifications.

	Channel	Always on	Periodicity	Min duty cycle
DL	SSB	Yes	Up to 160ms (Rel-19)	1/160
	SIB1	Yes	[Up to 160ms (Rel-19)]	1/160
	Other SIBs	Yes	From 80 to 5120ms	1/5120
	Paging	No	From 320 to 2560ms	1/2560
	RAR/msg2	No	Up to 160ms	1/160
	Msg4	No	Up to 160ms	1/160
UL	PRACH	Yes	Up to 160ms	1/160
	Msg3	No	Up to 160ms	1/160
	Msg4 HARQ-ACK	No	Up to 160ms	1/160

Table 4 : Channels and Periodicity

With these constraints, the duty cycle of an idle cell in the downlink would be equal to $0.0256 = 2.56\%$, and the duty cycle of an idle cell in the uplink would be equal to 1.87% .

3.2.4.3 Beam Hopping Scheduler Interaction

The primary objective of the Beam Hopping Architecture is to minimize changes to the legacy NR system design and architecture. This approach ensures that existing infrastructure and functionalities are preserved while integrating new capabilities.

In the legacy NR system, the L2 scheduler architecture is designed to operate on a per-cell basis. This means that each cell is managed independently, with the scheduler handling tasks such as resource allocation and traffic management within that cell. However, beam hopping introduces a new dimension to this architecture, requiring the scheduler to manage multiple cells simultaneously. This shift necessitates a more complex and integrated approach to scheduling.

Despite the challenges, the legacy L2 schedulers in NR baseband products are advanced and robust solutions. They have been optimized over time to deliver high performance and reliability. Changing these schedulers would be costly and time-consuming, potentially disrupting existing operations². Therefore, it is crucial to find a solution that leverages the strengths of the legacy schedulers while accommodating the new requirements of beam hopping.

From a system capacity perspective, scaling the L2 scheduler to handle many cells is not feasible. The scheduler is designed to operate within specific parameters, and expanding its scope to manage multiple cells would exceed its capabilities. This limitation underscores the need for a solution that maintains the integrity of the legacy scheduler while addressing the demands of beam hopping.

The proposed solution involves adding a beam arbitrator as a multi-cell scheduler function to the legacy L2 schedulers. The beam arbitrator focuses on distributing transmission slot grants for cells that meet beam hopping pre-conditions, such as reuse distance, number of satellite beams, and cell signal transmission needs⁴. This approach ensures minimal impact on the legacy L2 scheduler algorithm, allowing it to continue managing tasks within individual cells.

The beam hopping slot assignment adapts in a non-real-time manner, typically every 100ms to seconds, based on traffic cell-specific needs. A control interface between the beam arbitrator and the L2 scheduler is necessary to facilitate communication and coordination. This interface provides cell-specific traffic needs information and TX slot assignment signaling.

By computing the slot assignment in this non-real-time manner, there is sufficient time to carry out the computation on the ground and pass the assignments over the feeder link to a layer 2 (L2?) scheduler located on the satellite payload in option 2 regenerative architectures.

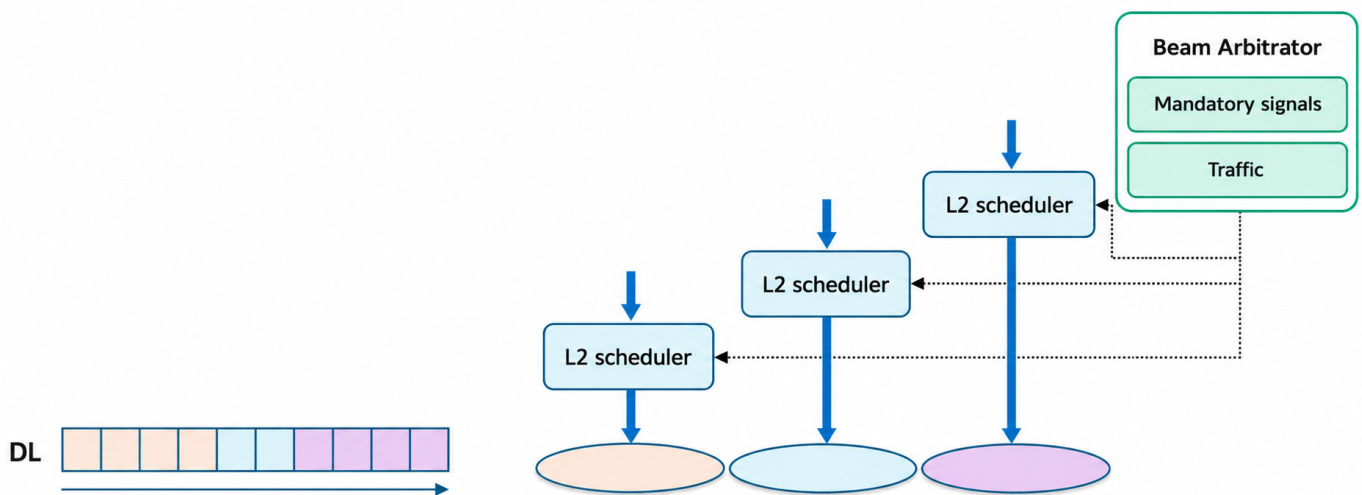


Figure 19 : Beam Arbitrator to L2 Scheduler Interaction

The solution is to add beam hopping as an added multi-cell scheduler function (beam arbitrator) to the legacy L2 schedulers (see Figure 19).

- Beam Arbitrator only focuses on distributing transmission slot grants for the cells fulfilling beam hopping pre-conditions (re-use distance, number of satellite beams, cell signal transmission needs, etc.). What then is transmitted in a slot remains the responsibility of the legacy L2 scheduler, i.e., minimal impact on the legacy L2 scheduler algorithm.
- Beam hopping slot assignment adapts in a non-real-time manner (e.g., every 100ms to seconds) to traffic cell-specific needs.
- A control interface between the Beam Arbitrator and the L2 scheduler is needed - cell-specific traffic needs information and TX slot assignment signaling.
- Arbitrator TX decision input relies on: Cell traffic load, mandatory TX, TX dependency schemes.
- The arbitrator also needs to handle dynamic variations such as cell-specific re-use distance variance; the available cells change due to satellite mobility.

Major challenges, as multiple aspects need to be considered in real-time:

- Mandatory transmissions: Not only a performance optimization aspect, but there is also a dependency on the Cell CCH and reference signal deployment design.
- DL and UL cell beam handling decisions cannot work independently; Chained TX dependency.
- Need to handle future TX needs, i.e., not only single time-instance choice but a mix of TX length (symbol, slot) and chained TX need.
- Need to consider the spatial interference aspect: Dependency to the Cell CCH and reference signal deployment design.
 - Especially challenging for preamble, and it has a long chain of UL/DL TX dependency.

3.2.4.4 Beam Arbitration Concept

Figure 20 shows the relationship between the mandatory signals for cell deployment, the Beam Arbitrator, as well as the L2 scheduler.

At **Cell deployment**, cells and their reference signals are configured.

- Deployment fulfilling RAN TX rules, cell reuse distance limits, and satellite beam capabilities.
- Need to be updated only due to cell deployment changes, e.g., New Satellites are activated, or inconsistencies have happened.

The **Beam Arbitrator** distributes satellite beams to selected cells.

- The beam arbitrator needs input on which cells are **activated-on** (=cells that should be allowed to transmit if there is a need).
- The beam arbitrator assigns beams needed for transmission of mandatory signals, via input from cell deployment, for activated-on cells.
- Remaining satellite beams (not consumed by transmission of mandatory cell signals) are assigned to cells based on traffic needs and priority, fulfilling re-use distance requirements.
- It should be noted that even though the cell status input is updated every slot from the respective cell-specific L2 scheduler, the beam hopping plan does not need to be updated at the same rate; it may be adapted at a rate of few 100ms to as many as few seconds.

The **L2 Scheduler** schedules data per slot per cell.

- Legacy L2 scheduler, mainly updated with per slot “transmission on/off” grant from the beam arbitrator.
- If the cell is assigned by the beam arbitrator to transmit in a slot, then the L2 scheduler determines what to transmit based on its internal needs and priorities. If not assigned a transmission capability for the slot, nothing is scheduled even if there is a need.
- The L2 scheduler performs an evaluation of every slot per cell having a “transmission on” grant.

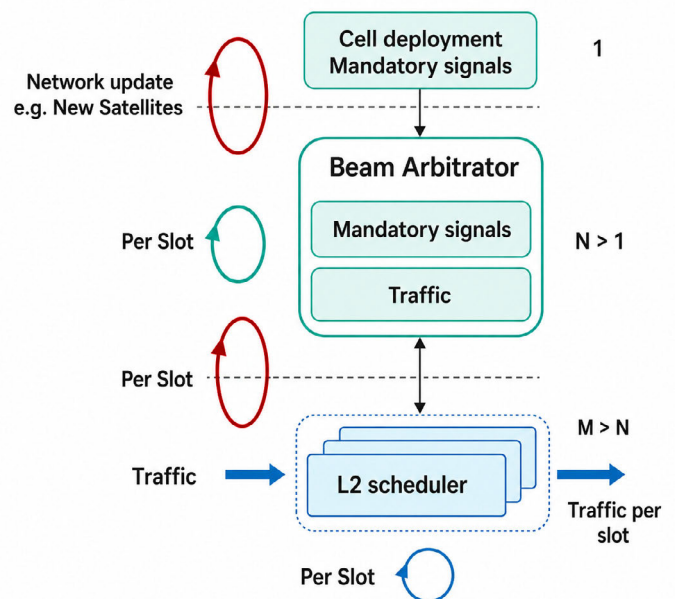


Figure 20 : Beam Arbitrator Architecture

Note that L2 schedulers need to have a low-latency interconnection to the beam arbitrators. Latency considerations between L2 scheduler and beam arbitrator need to be taken into consideration when deploying the system, but co-location is not required.

3.2.4.5 Cell Mandatory Signal Deployment – General Aspects

A cell coverage plan consists of a satellite footprint, consisting of several Earth Fixed Areas (EFA) of the same size (see Figure 21 below).

- Each EFA is covered by two cells (PCIs). If deploying Release 18 “satellite switch with resync,” each EFA can be covered by a single cell (PCI).
- There is a one-to-one mapping between the cells in the two layers and the EFA.
- The EFA and the cells have the same geographical position and radius.
- The cells and the EFA are modelled as regular hexagons.

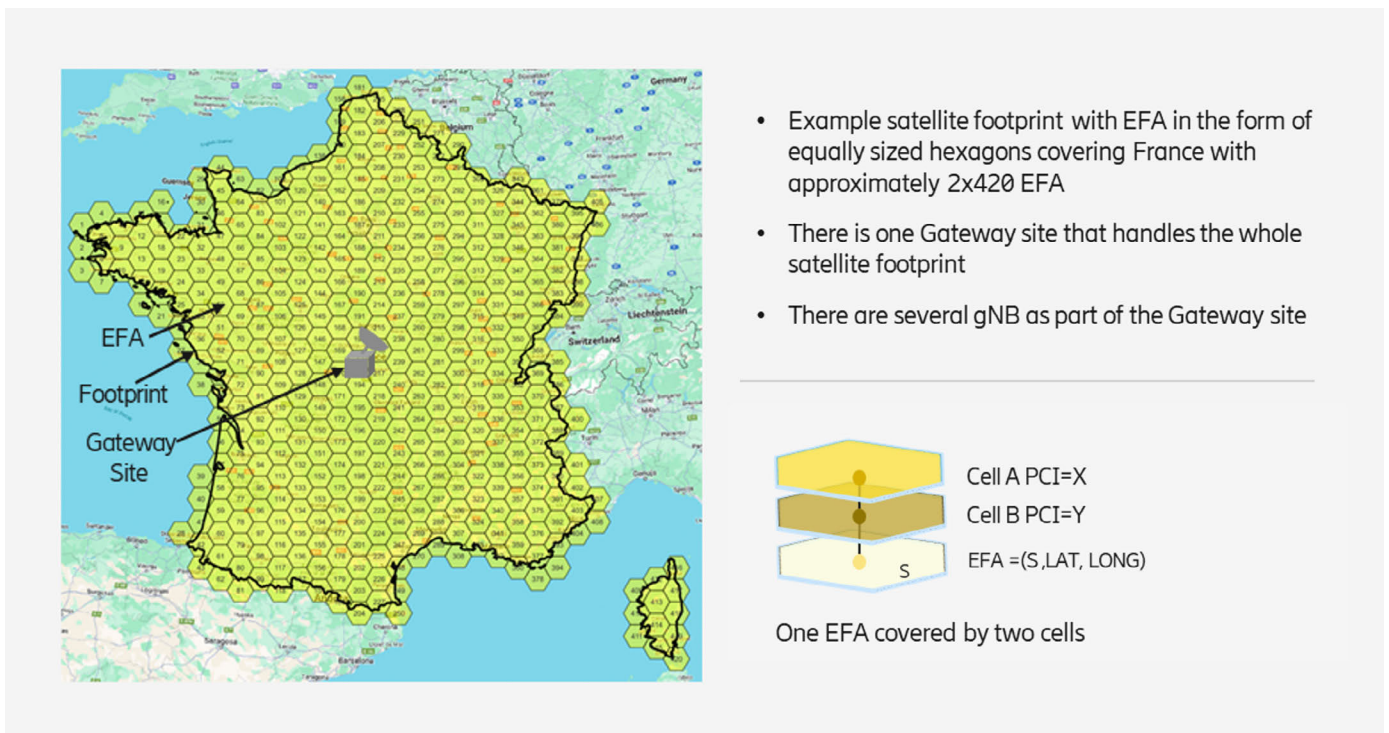


Figure 21 : Cell Coverage Plan Example

In this satellite footprint:

- One cell layer is active at a time, except for a short transition period during satellite handover (if used). If the assignment of EFA to satellite is performed with the same PCI, both satellites may be active simultaneously for a short period of time (when using a soft satellite switch), while for the case of a hard satellite switch, no overlap is needed.
- Different satellites will be responsible for a cell during consecutive periods when it is active.
- There is a satellite footprint area associated with at least one Gateway site, typically covering a large geographical area, sometimes a whole country.
- A footprint will have at least one Gateway site with at least one gNB, but in many cases several gNBs or even many GW sites.

The cells will be grouped into reuse groups (RG), where the number of cells in an RG corresponds to the reuse factor needed for interference mitigation (Figure 22).

- Reuse can be implemented with Time Reuse (TR) or Frequency Reuse (FR) or both.

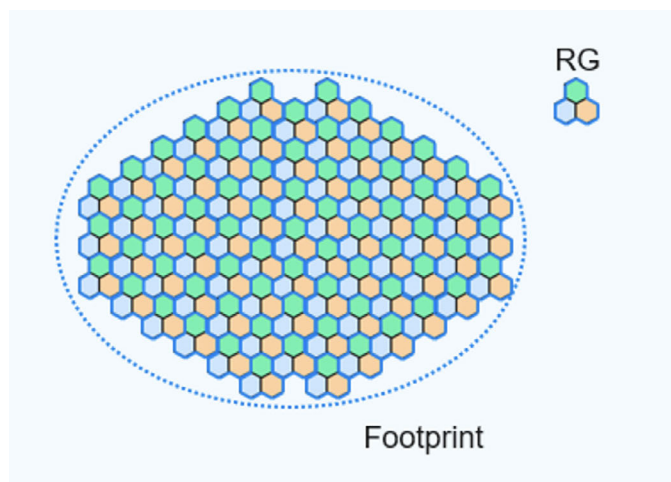


Figure 22 : Cell Reuse Group Pattern

Each satellite will then be assigned time-changing coverage snapshots (Figure 23).

- The duration and size of the coverage snapshots can vary. In the example below, the satellite first covers a smaller part of the footprint in the first coverage snapshot (at $T=N$) and then a larger part of a footprint in the coverage snapshot ($T=N+1$).

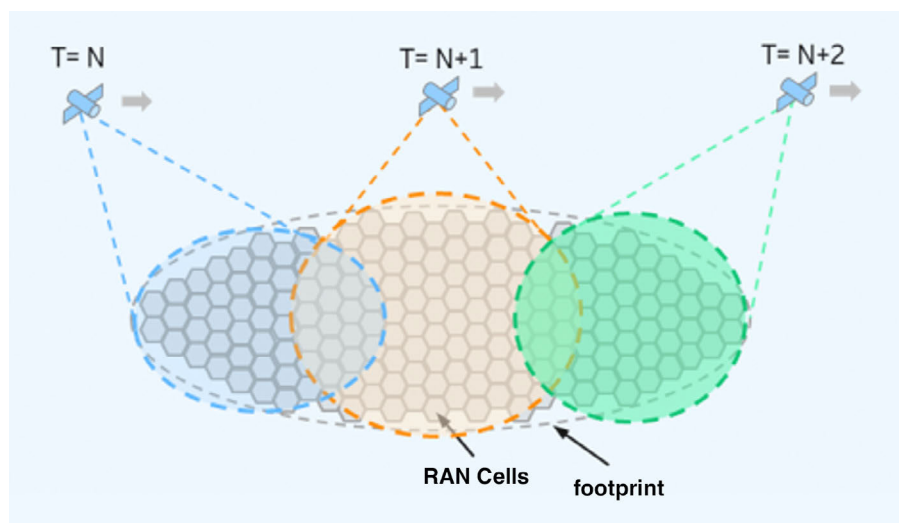


Figure 23 : Satellite Footprint Over Time

3.2.4.6 Mandatory Signal Deployment – Cell Signaling Scheme

The mandatory signaling scheme must be designed to fulfill the NR rules for mandatory signaling and that there is only one cell that has a transmission of signaling per slot.

One issue to consider is beam sharing, i.e., the number of cells that share a beam. The number of cells that share a beam depends on the number of cells in a coverage snapshot. If the number of cells is low, no beam sharing is needed, and the satellite can serve all cells. In other cases, with large coverage snapshots, a beam may need to serve, e.g., seven cells (beam sharing ratio 1:7).

According to the analysis in Section 3.2.1.4, the duty cycle for an idle cell (i.e., a cell not serving any active traffic) is 2.56% in the downlink, achieving a beam sharing ratio of 1:39.

Since there are multiple coverage snapshots, the satellite coverage snapshots are changing as the satellite moves over a footprint. For each snapshot, the following needs must to be understood:

- Number of cells to be covered
- Number of beams needed
- Minimum reuse distance
- Interference mitigations implemented
- Mandatory signaling scheme
- Constellation: Distance between satellites in the same orbital plane; are there satellites in several orbital planes that cover the footprint? Size of constellations.
- Satellite capacity: Number of simultaneous full-band beams/sub-band beams the satellite supports. Available bandwidth and total beam power.
- Inconsistencies that may occur lead to invalid snapshots (e.g., due to malfunction of a satellite leading to an insufficient number of beams or insufficient bandwidth).

Before a satellite enters the coverage of a footprint, the scheme of mandatory RAN cell signals and the BH scheme on the footprint must be determined using the information mentioned above. The deployment of mandatory RAN cell signals on a footprint must be determined/validated for each defined coverage snapshot over the footprint. This needs to be done using satellites info that may vary over time, and this needs to be done just before a satellite starts to cover the footprint. Activation of a new coverage snapshot is signaled by the Satellite Tracker function.

The beam hopping scheme is best calculated per activated coverage snapshot and applied on a per snapshot basis. In some cases, it may be fixed, i.e., calculated once, stored as a table.



Beam to cell sharing ratio 1:1



TRG	Cells	Beams	Slots
1.3		9	

- First snapshot is added
- BH starts

Beam to cell sharing ratio 1:1



TRG	Cells	Beams	Slots
1.3		9	

Beam to cell sharing ratio 1:2



TRG	Cells	Beams	Slots
1.8		24	
9.16			

Time →

Beam to cell sharing ratio 1:2



TRG	Cells	Beams	Slots
5.8		24	
9.16			

- More snapshots are added and BH is applied over the new snapshots

Beam to cell sharing ratio 1:3



TRG	Cells	Beams	Slots
5.11		30	
12.22			
22.31			

Beam to cell sharing ratio 1:4



TRG	Cells	Beams	Slots
1.18		30	
11.19			
26.28			
29.37			

- All cells in all TRG in the snapshot has been activated
- BH is done over the whole snapshot

Figure 24 : Beam Hopping/Mandatory Cell Signal Schemes Example

In the example above (Figure 24) time goes from right to left. In the first coverage snapshots (e.g., at lower elevation angles), the satellite only serves few cells and can have a beam sharing ratio of 1:1 and 1:2, leaving many slots open for user data. Later, the beam hopping scheme changes as the number of cells in a coverage snapshot increases, and the beam sharing ratio becomes 1:3 and 1:4.

3.2.4.7 Beam Hopping Recommendations

Beam hopping in combination with support of the mandatory cell signals in NR threatens to produce a high overhead, significantly reducing the communication capacity of the 5G NR NTN-based satellite system. Especially if the system supports only narrow bandwidths like 5 MHz and has a small number of satellites deployed. Care needs to be taken to optimize this.

In this section, a Beam Arbitrator function that works on slot level to aid the legacy L2 schedulers is proposed. The Beam Arbitrator tries to minimize overhead by optimizing the mandatory cell signalling by using satellite constellation/beam hopping information that it gets via the proposed Satellite Tracker function. Several implementation options for such Beam Arbitrator function exist, either as a SW tightly integrated with the RAN SW, or a more loose integration as a xApp.

It could also be recommended that Rel-19 devices are preferred, this since they have a longer SSB periodicity that is helpful in reducing the overhead.

Wider frequency bands and well-selected MCS are also beneficial. Future studies will come with more detailed recommendations.

3.2.5 RAN Related Functions, Deployment Perspective

The RAN function deployments depend on the NTN architecture chosen. There are four main architecture stereotypes (see Figure 25 below):

- Transparent having all RAN-related functions on ground
- Regenerative having all RAN-related functions on board the satellite
- Regenerative having RAN compute on ground and (parts of) the 3GPP radio unit functionality in the satellite
- Regenerative with split of CU-DU having the CU on ground

This section uses regenerative option 1 (sometimes referred to as semi-transparent) as an example. The characteristics of that deployment model are that it resembles the deployment model of terrestrial networks. Basically, the RAN computing function (CU/DU) is placed in another location than the radio (RU), and they are interconnected with a digital interface (e.g., O-RAN LLS over eCPRI).

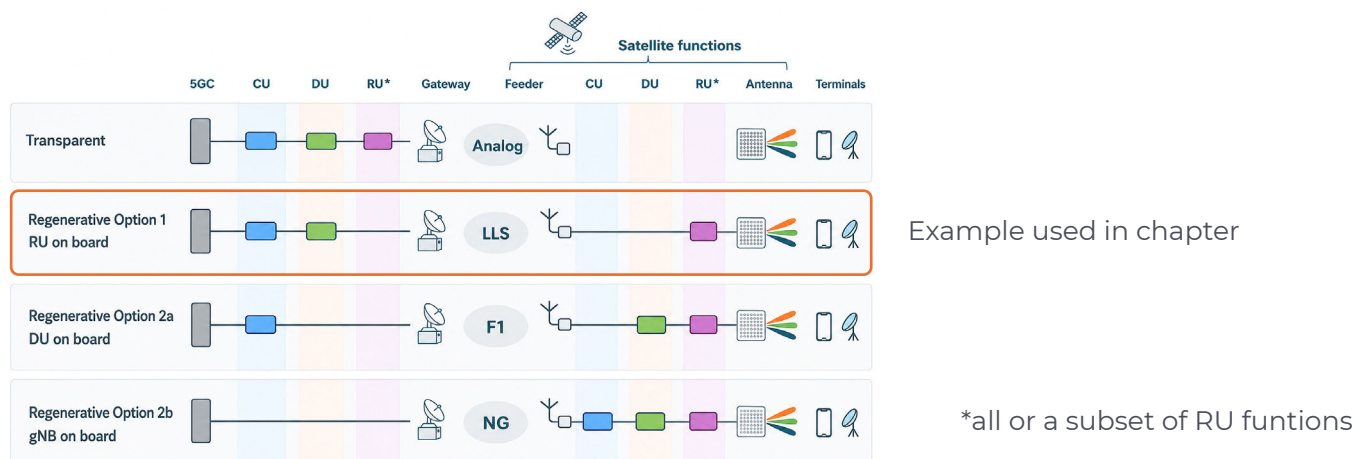


Figure 25 : NTN Architecture Types

In Figure 26 below, a deployment view of a regenerative option 1 NTN architecture is presented. Blue functions are 3GPP/O-RAN specified functions, while purple support functions are needed to help combine satellite constellation and RAN info for efficient NTN operations.

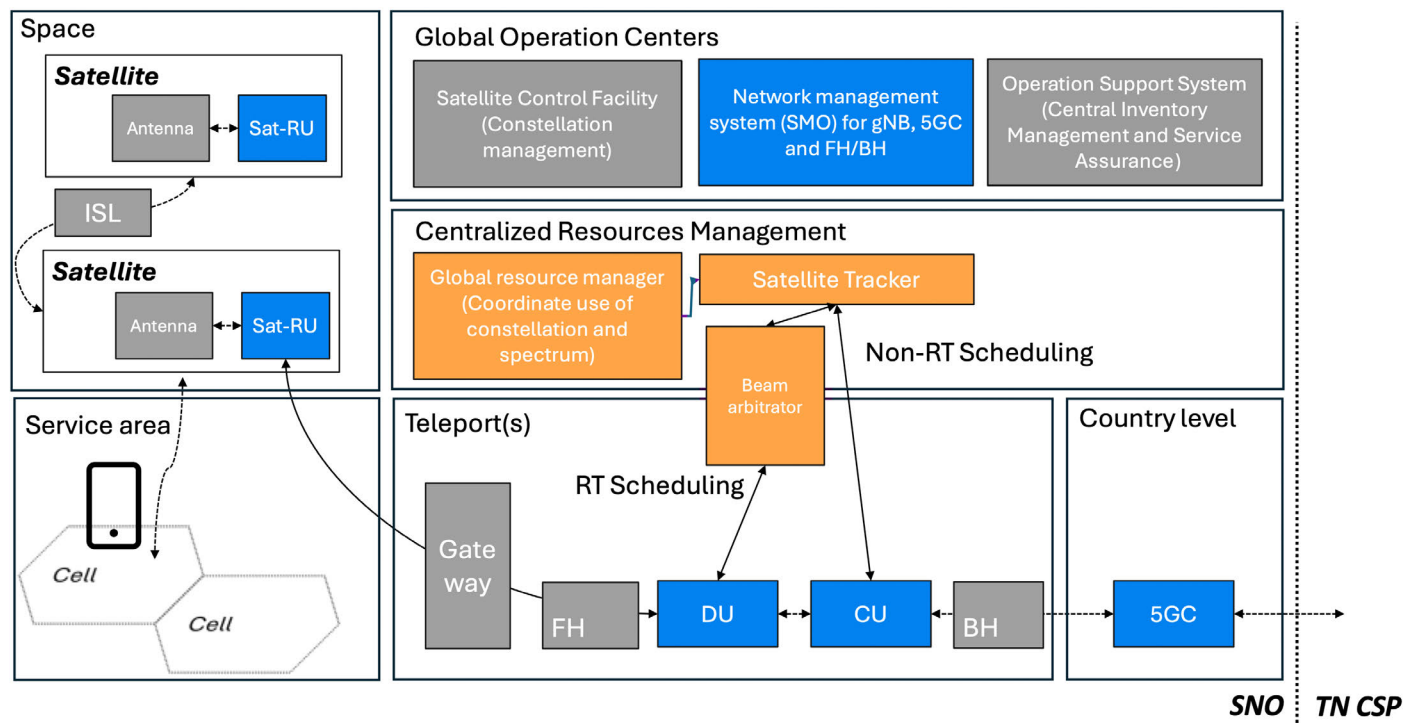


Figure 26 : Deployment View of a Regenerative Option 1 NTN Architecture

The Global Resource Manager could be deployed on one global centralized architecture or split with one master and multiple slaves on, e.g., country-level. The GRM needs to consider and allocate the following resources:

- Select the appropriate constellation based on specific criteria.
- Define contact plan with satellites interfacing with Satellite Control (constellation health/ephemeris).
- Define Feeder links and OISL contact plans.
- Define Onboard routing based on Ground/Board split (CU/DU/RU) and functional architecture.
- Define antenna beams allocation and constraints (power).
- Cell allocation to satellites.

Note: The satellite tracker function is presented in section 3.2.1.2 and the beam arbitrator in 3.2.1.5.

The satellite tracker function bridges the gap between Satellite Operational Domain and RAN Operational Domain. The intent with it is to minimize the impact of Satellite-specific behaviour on RAN implementation. The data sent between the satellite tracker and beam arbitrator does not need real-time characteristics and can thus be deployed separately.

The beam arbitrator distributes satellite beams to selected cells and needs to be deployed collocated with the DU due to low latency requirements between the beam arbitrator and the L2 scheduler in the DU.

The CU and DU can be deployed separately if a hierarchical system is preferred.

The current O-RAN RIC/SMO architecture is still lacking many other satellite-specific resource management and orchestration functions that would be required by a fully-fledged satellite resource management system. Some of these gaps are currently being discussed in the O-RAN ALLIANCE as part of a set of active Work Items (WI).

With this in mind, an NTN Resource Management System could be realized—at least in part—through the combined capabilities of the O-RAN Service Management and Orchestration (SMO) and RAN Intelligent Controller (RIC), both Non-RT and Near-RT, suitably adapted to cater for the specific requirements and functionalities needed to support NTN Resource Management.

The O-RAN ALLIANCE specifications define the Service Management and Orchestration (SMO) as responsible for RAN domain management.

While the O-RAN ALLIANCE defines the SMO as responsible for overall RAN domain management, the Resource Management System focuses on the unique requirements and challenges of satellite-based systems.

The SMO is an open automation platform allowing deployment and management of RAN network functions and applications through standardized and open interfaces. As part of and along with the SMO, the non-Real-Time RAN Intelligent Controller (non-RT RIC) allows running modular non-real-time RAN management functions in the form of applications, called “rApps,” via the open R1 API. Examples of functions that those apps can handle include:

- Network deployment
- Network evolution/capacity planning
- Network optimization/performance diagnostics
- Network healing

Note: Market consensus is that the SMO is the RAN automation platform of the future, but the SMO market has just started to pick up and is still quite limited in size, but a large growth is expected in coming years.

Further into the RAN stack, the O-RAN ALLIANCE also defines a near-real-time RAN Intelligent Controller (near-RT RIC), which opens up the RAN control plane by enabling fine-grained control and coordination of RAN control plane functions with a much more granular control loop, via the E2 interface and the deployment of modular control plane applications, the “xApps.” Example functions that those apps can enable include:

- Radio resource management policy control
- Handover and Reselection triggering and control
- QoS policies and slice policies
- Load balancing
- Dynamic cell re-configuration
- Network energy saving policies
- Measurement collection and interference management policies

The current O-RAN ALLIANCE specifications distinguish between the non-RT RIC and the near-RT RIC based on their control loop latency. The former, which integrates with the network orchestrator, operates on a time scale longer than 1 s, while the latter, which drives control loops with RAN nodes, operates on a time scale between 10 ms and 1 s.

The O-RAN ALLIANCE is currently in a study phase on the introduction and impact of support for Non-Terrestrial Networks in the O-RAN architecture and specifications, as part of the ongoing Working Group 1 (WG1) Use Case Task Group (UCTG) work item “RIC Enabling NTN Deployments (RIC4NTN).”

The latest release of this work is published as O-RAN.WG1.TR.RIC4NTN-R005-v02.00. The Work Item studies use cases applicable to NTN, their impact on the O-RAN architecture and specifications, identifying possible architectural gaps and enhancements, including possible architectures to support TN-NTN interworking and coordination at the RAN layer.

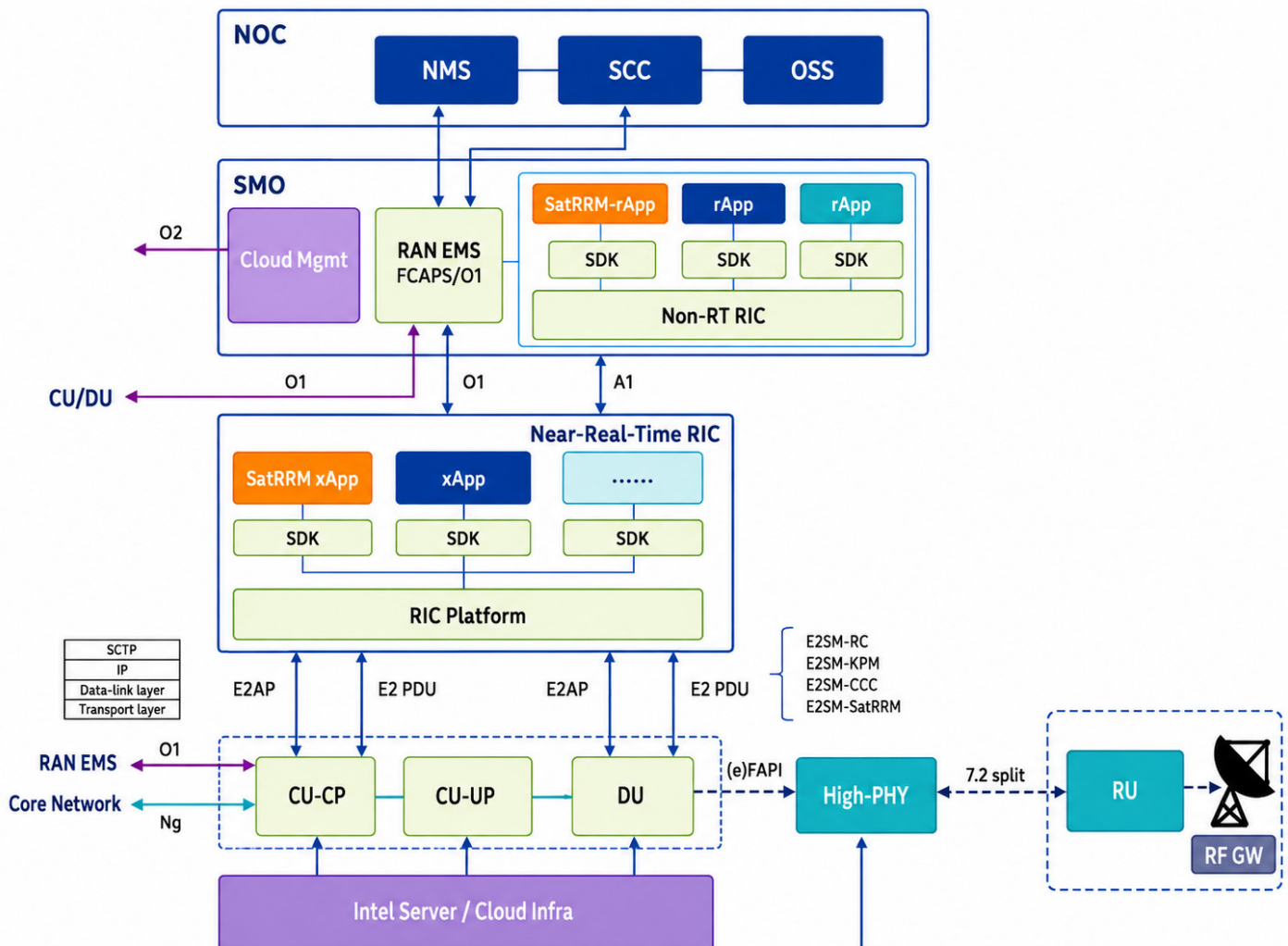


Figure 28 : Stay connected: Seamless Transitions Between Mobile Broadband Networks

A view of the role split between the different layers of the RAN control and management architecture, including the O-RAN RIC complex, is as follows (Figure 29):

- Non-Real-Time RIC coordinates beam/cell mapping schedule across coverage areas to avoid interference at coverage boundaries and overlap areas.
- Near Real-Time RIC balances PRB allocations to cells within a satellite's aggregate EIRP limit and the pre-assigned beam/cell mapping (from the Non-Real-Time RIC), coordinates scheduling allocations across the same Earth Fixed coverage area from multiple satellites, intra-satellite and inter-satellite mobility, close coordination of timing aspects across multiple cells, etc.
- The Legacy L2 scheduler handles PRB allocation among users of the same cell.

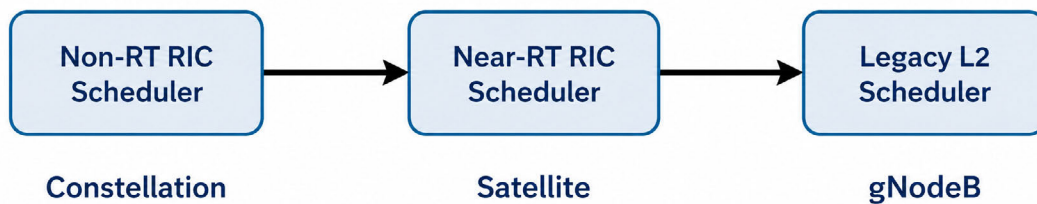


Figure 29 : Tiered RRM with O-RAN Architecture, Non-RT RIC, Near-RT RIC and gNB L2 Scheduler

3.2.7 Implementation considerations for transparent and regenerative architectures

As detailed above, there is a range of use cases that may be implemented by the resource management system. Each use case will have requirements for processing time windows and latency, and availability of timely information, which will influence the choice of implementation architecture, in some cases potentially requiring the need for further programmability control functions closer to the gNB or even the on-board payload.

Looking through the lens of the current O-RAN Architecture (as of Release 5), programmability is achieved via xApps deployed on the Near-RT RIC or rApps and SMO Services (SMOS) on the SMO and Non-RT RIC.

In addition, in the scope of NTN, while the SMO and Non-RT RIC are almost universally assumed to be deployed in a logically centralized architecture on the ground segment, there is some debate ongoing on the placement of the Near-RT RIC, especially considering the current O-RAN Architecture normative specs and the difference between Transparent and Regenerative payload architectures.

Some consider a need to place the Near-RT RIC closer to RAN clusters (e.g., in Transparent payload architectures), whilst others consider that the relaxed latency requirements of NTN, contrasted with the higher dynamicity requirements for NTN network configuration, instead warrant a logical centralization of at least some of the Near-RT RIC functions in tight cooperation with some of the traditionally more Non-RT RIC and SMO functions, which tend to need to operate at faster pace.

It is also possible that, for certain NTN use cases, programmability at the lower limit of the near-RT RIC control loop latency window (<1ms) may be required, requiring programmable functions to be co-located with the DU and/or CU, to achieve the latency required. This might be particularly relevant to regenerative architectures.

To this end, some research is ongoing on the introduction of so-called “dApps” or Distributed Apps at or near the gNB O-CU, O-DU, or even O-RU nodes, to provide close-to-real-time programmability in the gNB stack itself.

This has also led some research to consider the possibility of placing a near-RT RIC on board the payload, with the AI link passing over the feeder link.

3.3 Network Selection and Network ID Management

The inherent complexity of network selection arises when Earth fixed cells and Tracking Areas have different diameters. This challenge is compounded by 3GPP's focus on maintaining existing UE handling mechanisms as much as possible. The situation becomes even more complex when multi-orbits and multiple inclinations are considered, leading to the creation of Earth fixed cells of varying diameters.

This requires careful coordination and synchronization to ensure seamless connectivity and optimal performance. Addressing these complexities involves developing advanced algorithms and strategies to manage the dynamic nature of satellite networks while adhering to 3GPP standards and ensuring efficient resource allocation.

- References: 3GPP TR24.821 v17.0.0; TR23.737 v17.2.0; TR 22.822 V16.0.0

3.3.1 General Description

- For similar antenna array performance, satellites at different altitudes will address different amounts of Earth Fixed Areas (Figure 30) on the ground and/or different Earth Fixed Area sizes ($D_1 > d_2$).
- Satellites at different altitudes travel at different speeds; different inclinations create speed differentials from an Earth Fixed Areas viewpoint.
- Earth Fixed Areas may cross countries or frequency licenses boundaries.
- 3GPP-based systems are synchronized, requiring proper interworking architecture.

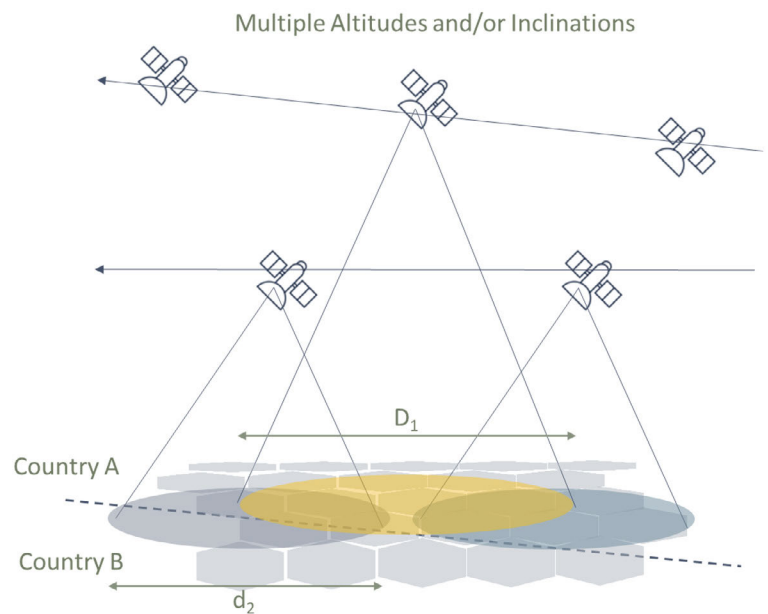


Figure 30 : Altitude/Inclination to EFA Relation

3.3.2 Frequency Bands and Channels Management

- Satellite systems must dynamically change frequency bands and channels at border conditions (Figure 31) to comply with regulatory requirements and license conditions.
- This applies to satellites from different orbits or inclinations.
- Borders can be between countries or license areas, necessitating careful coordination to ensure seamless connectivity and adherence to regulations.

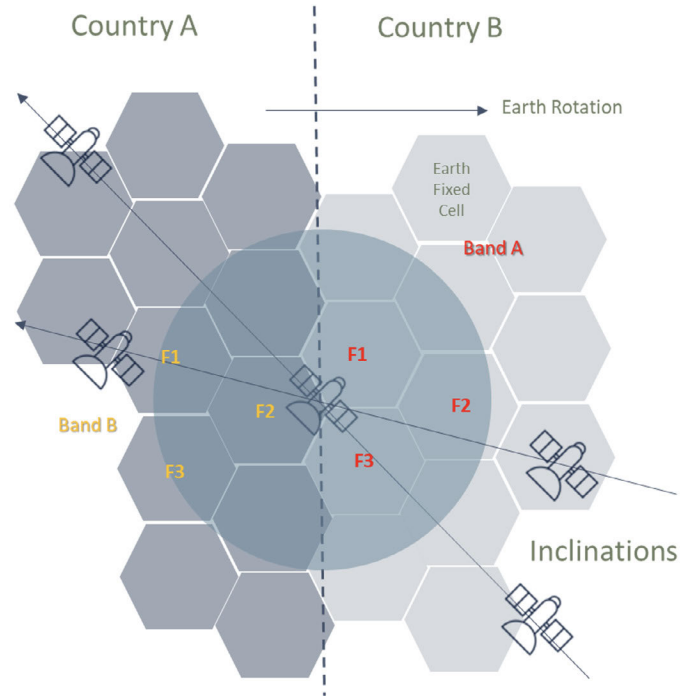


Figure 31 : Frequency Band Management

3.3.3 Network ID Management

- To support different business models and SNO to MNO relationships, satellite system must be able to broadcast proper PLMN IDs (Figure 32) for subscribers to attach to satellite resources.
- The system should be able to handle PLMN ID from MNOs or SNOs, or both.
- Proper PLMN IDs coordination between orbits is required for access priorities and traffic balancing between channels and bands.

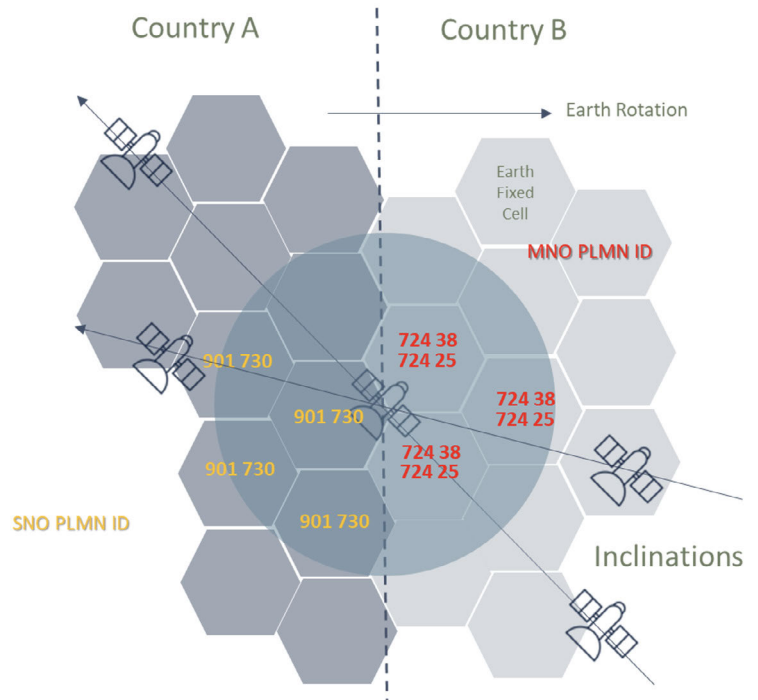


Figure 32 : Network ID Management

3.3.4 Recommendations

The following are to be supported:

- Earth fixed cells hierarchy concept to address Earth fixed area with various diameters from multi-orbits satellites to maintain 3GPP TA/RA logic
- Dynamic beam size/EFA diameters adjustment for overlapping constellations
- Support for Multiple PLMN ID broadcasts; dynamic allocation as a function of a shared EFA reference grid
- Multiple ephemeris information broadcast – SIB19
- Business agreement/rules-based network selection priorities; SNO to MNOs agreement
- Traffic-based/traffic-balancing network selection priorities

3.4 Roaming

Roaming services, both domestic and international, between Mobile Network Operators (MNOs) are well-documented and widely practiced. When Non-Terrestrial Network (NTN) services are provided domestically by a Satellite Network Operator (SNO) to MNOs as a footprint extension service for mobile subscribers, it is essential for the MNO and SNO to reach an agreement. This agreement should detail mobile subscriber account ownership, the type of network interconnection, service characteristics, and management, as well as billing and charging details.

When a subscriber moves outside their MNO's footprint and connects to an MNO roaming partner network, the roaming agreement between the MNOs specifies the supported services and rates. However, if the same subscriber moves beyond the roaming partner's footprint and connects to the SNO network, the services and rates may vary depending on the business rules and priorities.

Given that different business models exist between SNOs and MNOs, involving external clearing house services, it is crucial to identify and analyze the possible models to support national and international roaming for a shared satellite network. These models must address various aspects such as subscriber ownership, network interconnection, service management, and billing. By doing so, they can ensure seamless connectivity and optimal performance for mobile subscribers, regardless of their location.

The collaboration between SNOs and MNOs is vital for providing comprehensive roaming services. By establishing clear agreements and analyzing different business models, they can effectively support national and international roaming for a shared satellite network.

- References: GSMA IR88 EPS Roaming Guidelines, CLP.28 NB-IoT Deployment Guidelines, Guidelines for IPX Provider networks, International Roaming Explained; GSMA PRD NG.113; 3GPP TS 23.501

3.4.1 Background

Terrestrial Networks Roaming:

Figure 33 shows commercial and technical details for international mobile roaming. The diagram focuses on the international roaming wholesale and retail arrangements, for simplicity.

The mobile user (Mobile User A) has an international roaming service with their home operator (Home Operator) and is automatically connected to a visited network (Visited Operator A) while roaming. Mobile User A is automatically granted access to Visited Operator A's network when arriving in the visited country by an exchange of data between Home Operator and Visited Operator A, where Visited Operator A confirms Mobile User A is a roaming customer with Home Operator.

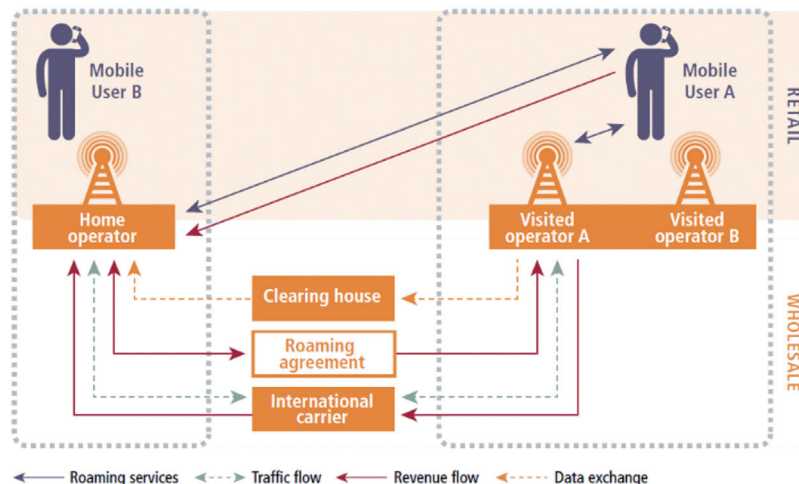


Figure 33: Roaming and Clearing House Concepts

As such, the wholesale roaming agreement between Visited Operator A and Home Operator specifies how this data is to be provided to the visited operator. Home Operator usually has wholesale roaming agreements with more than one operator in the same visited country, which in this case is Visited Operator A and a second network, Visited Operator B. As a result, Mobile User A can call home using either visited operator networks, both of which use international transit services to carry the call back to Mobile User A's home country. Mobile User A pays a retail price to Home Operator for the roaming service and does not pay Visited Operator A. Provided Mobile User B is not also roaming, they will not incur any extra charges to receive a call from, or to make calls to Mobile User A.

Visited Operator A sends transferred account procedure (TAP) files to a clearing house, which forwards them to the Home Operator. TAP files are used for billing of calls while roaming. Home Operator can then pay Visited Operator A the wholesale charges as per call volumes in the TAP file and rates in the wholesale roaming agreement.

Visited Operator A pays an international carrier (International Carrier) for carrying the call and handing over the call to Home Operator. International Carrier pays Home Operator a termination rate for terminating the call in the home country.³

³ Reference extract: GSMA International Roaming Explained

Shared Satellite System

- In the context of a shared satellite system offering global coverage (refer to Figure 34), local coverage is facilitated through local MSS licenses and their associated landing rights utilizing MSS spectrum.
- Within a given country, there may be one or more MSS licensees, each capable of providing satellite communication services to in-country MNOs.
- With licensees in each country sharing the satellite system and multiple MNOs present, there could be numerous MNO to SNO relationships.
- To ensure adequate service quality, a fair satellite system and spectrum sharing mechanisms are essential.

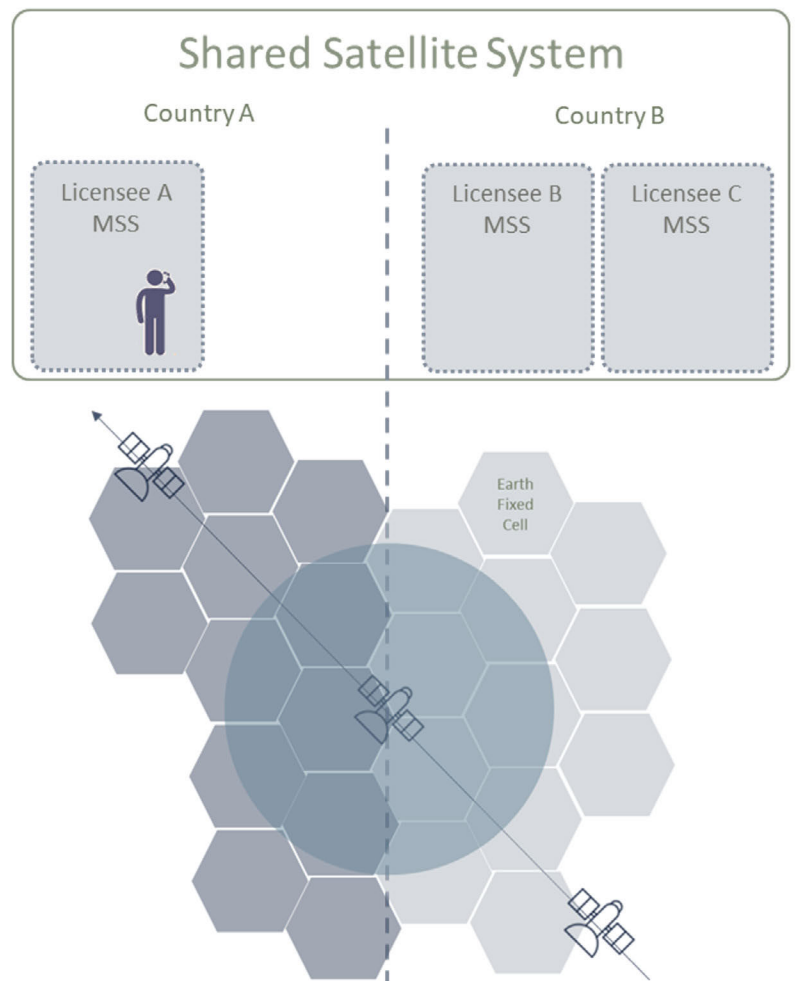


Figure 34 : Satellite System, MSS Licensee and Boundaries

3.4.2 SNO/MSS Licensee to MNO Relationships

Regional licensee(s) in a region can contractually agree with local MNO(s) to provide the MNOs' subscribers with network extension services using the shared satellite system.

Depending on the business model, the regional licensee can also provide MSS services directly to the MNOs' subscribers. In which case, the regional licensee (or the SNO) must have separate accounts with the subscribers.

The seamless subscriber session transition between a regional terrestrial network and a shared satellite system depends on the contractual agreement between the parties and the type of network interconnection.

International roaming use cases can also be supported. A mobile subscriber with an NTN-capable device can be serviced abroad by the regional licensee (or the SNO). Exact device behavior depends on the relationship between the MNO and their roaming partner abroad, and the relationship between the regional MNO and the local licensee.

Figure 35 depicts the generic relationship between the SNO, the MSS licensees, and the MNO for domestic and international roaming.

For international roaming, there exist multiple SNO/Licensees to MNO relationship scenarios:

- Domestic MNO to International MSS Licensee; the local MNO has an agreement with a foreign MSS service provider to provide international roaming services.
- Domestic MSS Licensee to International MSS Licensee; the local MSS has an agreement with a foreign MSS service provider to provide services to home MNO subscribers or to local MSS subscribers.
- Domestic MSS Licensee to Satellite Service (SNO); like the previous case but where the only SNO ID is broadcasted, so the SNO has the subscribers' ownership.
- MNO/MSS Domestic to foreign MNO/MSS; the local MNO has an agreement with the local MSS licensee, the foreign MNO also has an agreement with the local MSS licensee, and the MNO has a roaming agreement and desires to use this agreement to include MSS services.

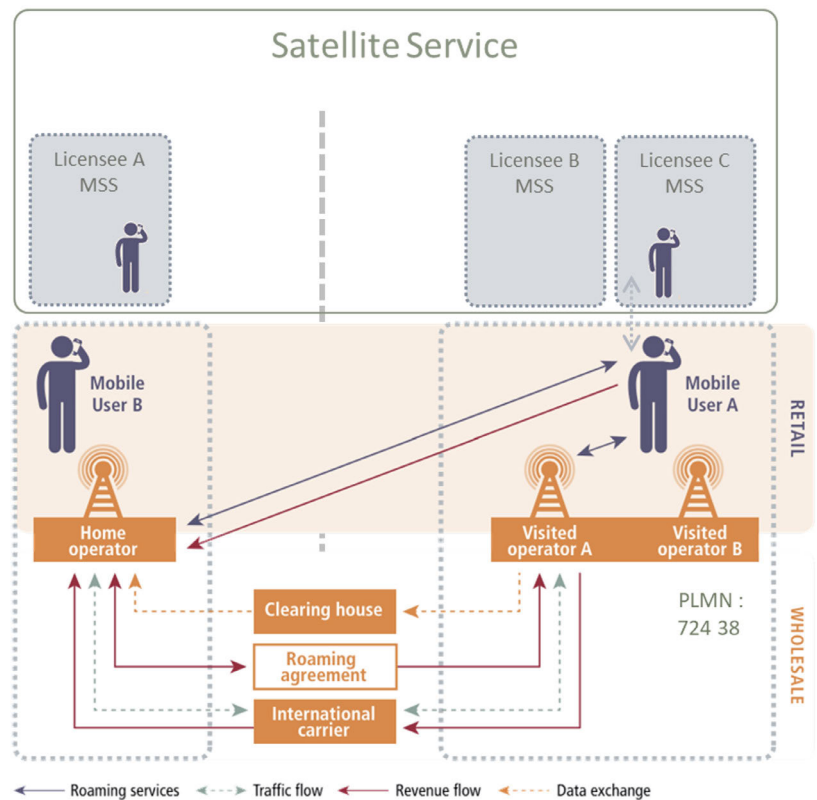


Figure 35 : Satellite Service, MSS Licensee and MNO Relationship



The following sections provide more details on key scenarios.

3.4.2.1.1 MSS to Roaming MSS

- MSS Licensees A & C use wholesale Satellite Service.
- MSS Licensee A in country A in a roaming agreement with MSS Licensee C in country B.
- MSS Licensees are interconnected by an IPX provider or using their own shared facilities.
- Licensee A can broadcast its own PLMN ID or a secondary PLMN ID from local MNO (e.g., 310 016).
- Licensee C can broadcast its own PLMN ID or a secondary PLMN ID from local MNO.

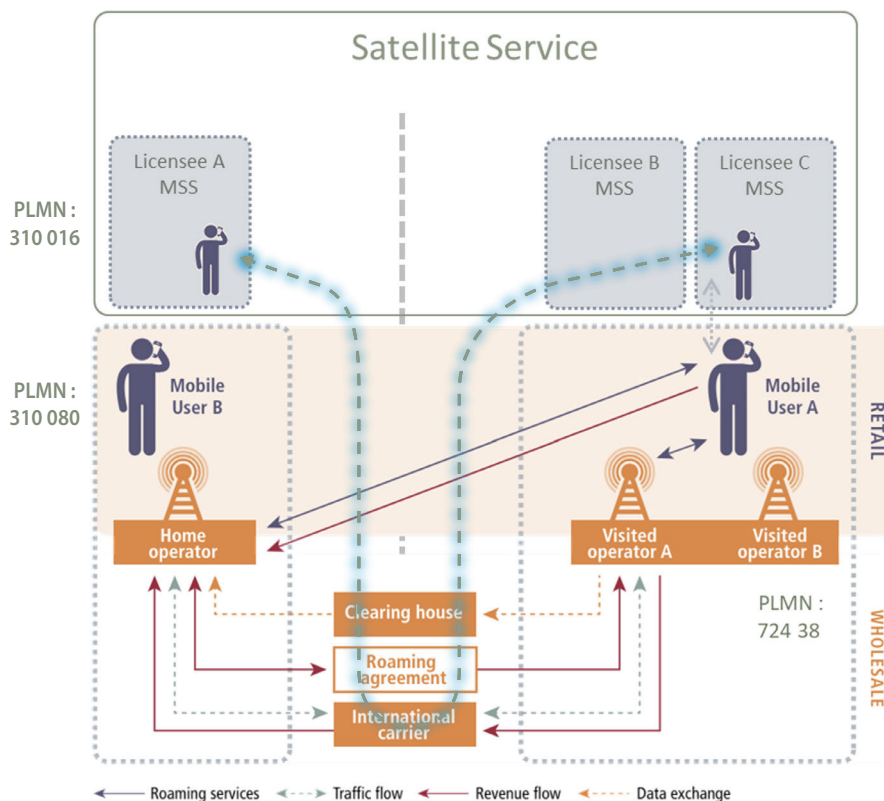


Figure 36 : MSS to Roaming MSS

3.4.2.1.2 MSS to Satellite Service Provider

- MSS Licensee A subscriber travels to another country where the Licensee is the Satellite Service Provider.
- MSS Licensee A in a Service Agreement with Satellite Service Operator for country B.
- Satellite Service Operator broadcasts its global PLMN ID (e.g., 9xx xxx) for MSS Licensee subscriber to attach to.
- Both Licensee A and Satellite Service providers use IPX provider services or are directly interconnected.

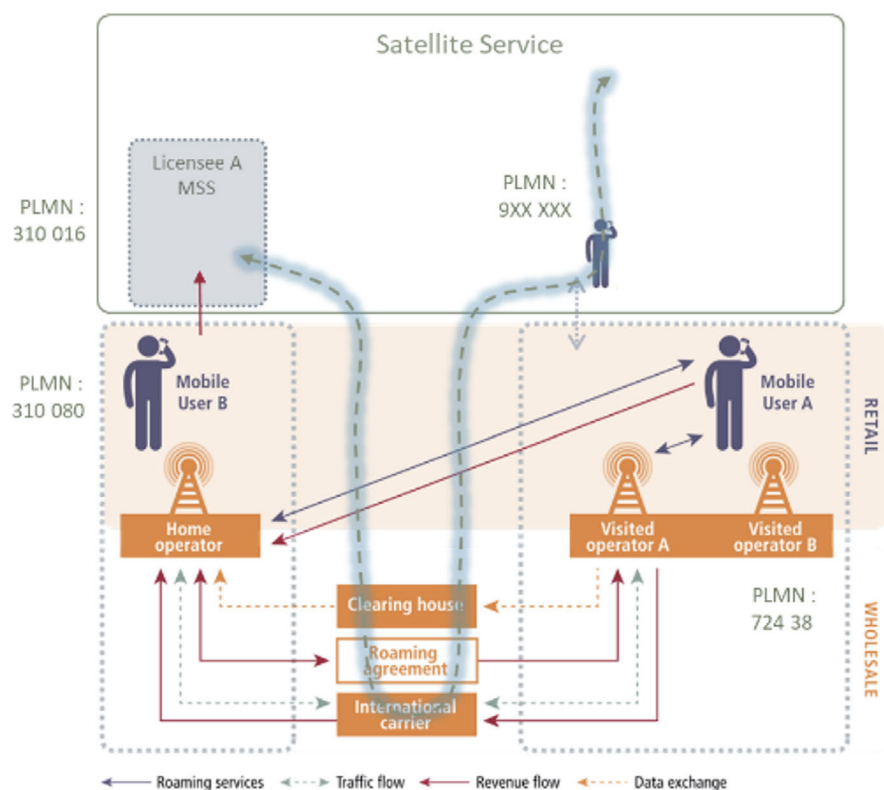


Figure 37 : MSS to Satellite Service Provider

3.4.2.1.3 Domestic MNO&MSS to Foreign MNO&MSS

- Licensee A has a network extension service with local MNO.
- Local MNO subscriber A travels to MSS Licensee C region, which has a network extension service with local MNO.
- Licensee C broadcasts local MNO's secondary PLMN ID to provide network extension services to roaming subscribers under the MNO's roaming agreement.
- International traffic is handled by the MNO's roaming agreement and their IPX providers. MSS licensees do not have to have a roaming agreement between them.

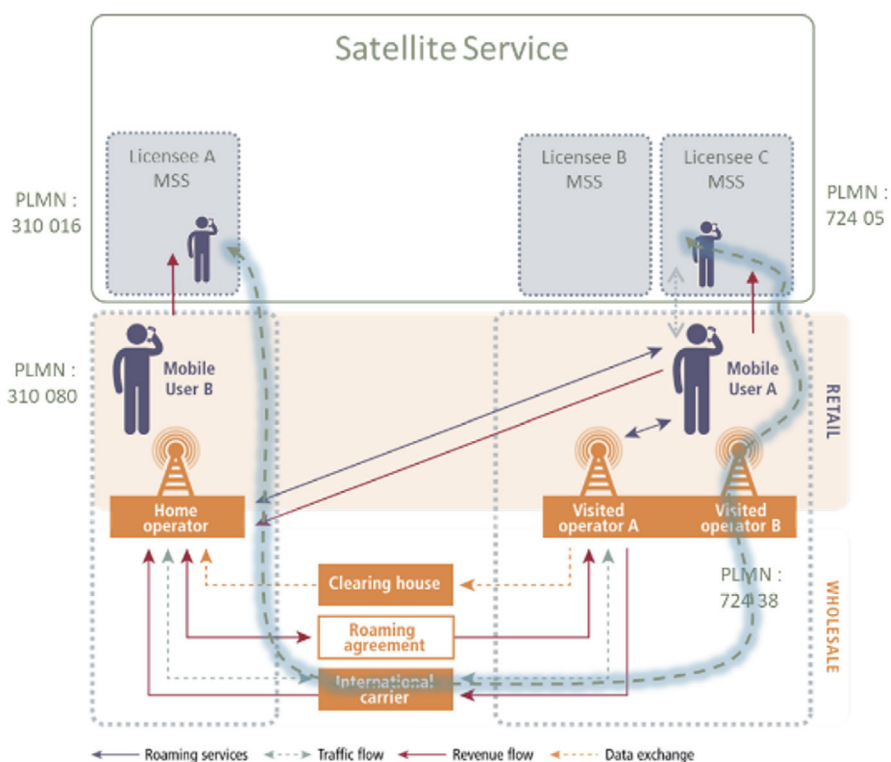


Figure 38 : Domestic MNO&MSS to foreign MNO&MSS

3.4.3 Recommendations

- The satellite system must support the broadcasting of multiple PLMN ID to ensure proper network selection and support the various Licensee to MNO agreement types as well as the MNO-to-MNO roaming agreements. This is essential also to comply with border conditions.
- The satellite system shall have its own IPX service provider agreement to support Licensee to Licensee and Licensee to MNO agreements.

3.5 Network Sharing

Network sharing has been an area of study within terrestrial networks for many years, with core capabilities being introduced into the 3GPP standards for 3G systems onwards, through to study of standardised ways of implementing those capabilities being studied by groups such as the Small Cell Forum (SCF). Most of this work is relevant to NTN but the space segment brings additional considerations.

There are multiple ways to share satellite system resources, ensuring efficient utilization and optimal performance. Sharing can be based on spectrum, time, or logically, each method offering unique advantages and challenges. Spectrum-based sharing involves allocating different frequency bands to various users or services, allowing simultaneous communication without interference. Time-based sharing, on the other hand, divides the satellite's operational time into slots, enabling different users to access the system at designated intervals. Logical sharing involves creating virtual partitions within the satellite system, allowing multiple users to operate independently within the same physical infrastructure.

For this document, three demarcation points from an SNO (Satellite Network Operator) viewpoint are examined. These demarcation points are crucial in understanding how resources are allocated and managed within the satellite system. Each demarcation point has a different impact on functionalities and business models. The first demarcation point focuses on the physical layer, affecting the hardware and transmission capabilities. The second point addresses the network layer, influencing data routing and connectivity. The third demarcation point pertains to the application layer, impacting service delivery and user experience. By examining these points, we can better understand the complexities and opportunities in satellite resource sharing.

References: GSMA 5G Network Co-construction and Sharing Guide; 3GPP TS 23.51 (MOCN); 3GPP TS 28.628 and TS 23.251 (MORAN)

3.5.1.1 Network Demarcation for Network Sharing

Figure 39 provides the main demarcation options – without the Beam Arbitrator function. For simplicity only a transparent architecture implementation is shown, however some of the options can be implemented in either a transparent or regenerative architecture.

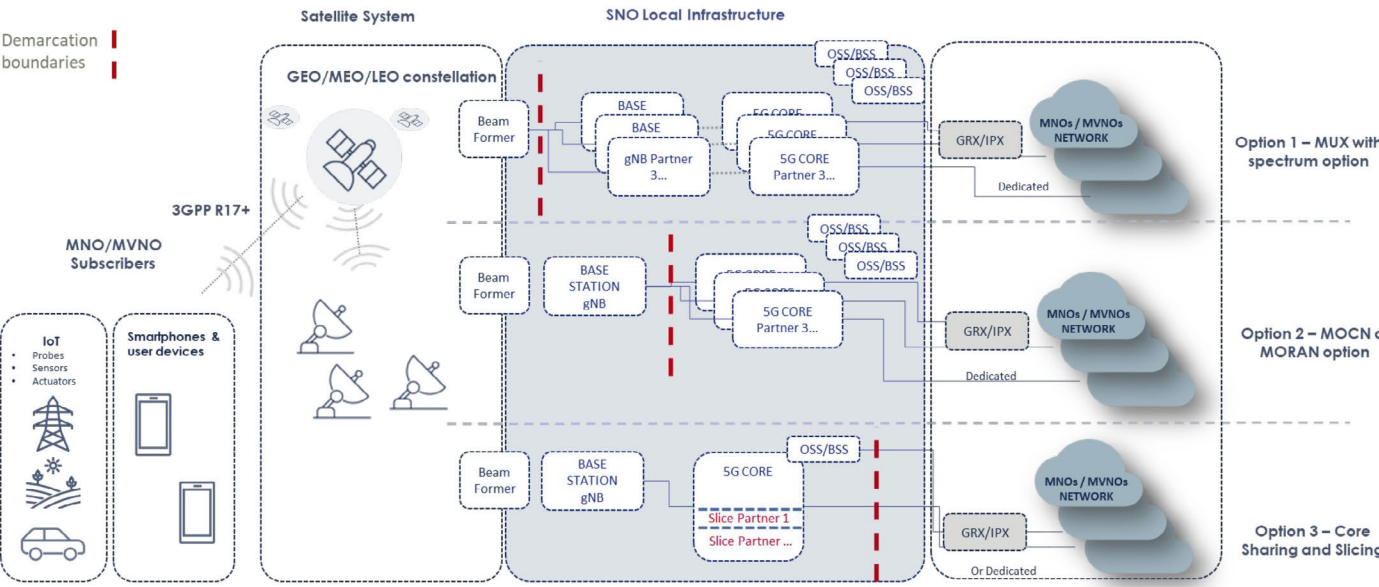


Figure 39 : Network Demarcation Options

Option 1 Multiplexing at the Gateway

- Multiple base stations (eNB, gNB) are connected to a multiplexing interface to the beam former.
- The multiplexing could be based on spectrum segmentation or a time/capacity division. Sharing proportion depends on agreement between the SNOs/Licensees.
- Each base station can broadcast its own PLMN ID group as a function of the commercial agreements between the Licensees and the MNOs.
- This option is typically only viable in a transparent architecture.

Option 2 Multiple Operator Core Network and Multiple Operator RAN

- MOCN and MORAN are 3GPP-supported sharing modes.
- The eNB or gNB functions are mutualized; multiple CNs from different Licensees connect logically to the mutualized functions.
- Traffic separation, for each Licensee or for each MNO, is typically based on IMSI identifiers' range.
- The Mutualized base station function can broadcast its own PLMN ID group as a function of the commercial agreements between the Licensees and the MNOs.
- This option can be implemented either as a transparent architecture as illustrated or a regenerative architecture with the shared gNB hosted on the satellite payload.

Option 3 Network Slicing and RAN Mutualization

- Both the RAN functions and CN functions are mutualized.
- The traffic separation between Licensee is handled by 3GPP specification (TR 28.530) for network slicing.
- The mutualized base station function can broadcast its own PLMN ID group as a function of commercial agreements between the Licensees and the MNOs.
- Traffic separation, for each Licensee or for each MNO, is typically based on IMSI identifiers' range.
- This option can be implemented either as a transparent architecture as illustrated or a regenerative architecture with the shared gNB hosted on the satellite payload. The 5G CN has to remain primarily on the ground; however, some elements of processing may be implemented as part of the payload.

3.5.1.2 Demarcation Options and Resources Management

This section examines the relationship between the three demarcation options and the Resources Management requirements, namely the interaction between the Beam Arbitrator function and the L2 schedulers.

Option 1 with Beam Arbitrator

- Requires the Beam Arbitrator to interact with multiple independent Base Stations/L2 Schedulers.
- Beam Arbitrator functions and the L2 Schedulers are no longer on the same processing plane.
- Supports both frequency domain and time domain sharing.

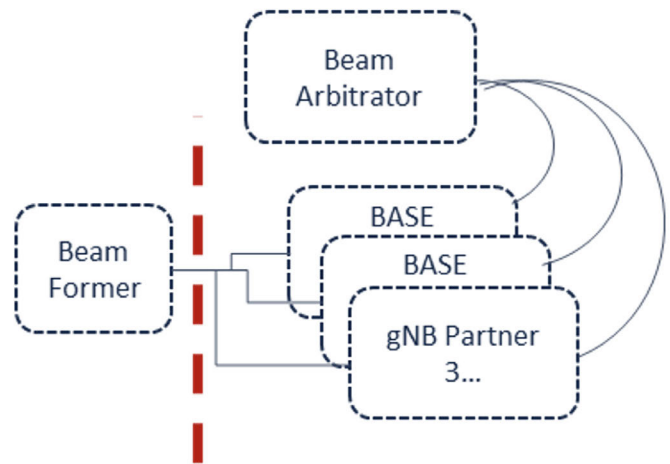


Figure 40 : Option 1 Multiplexing at the Gateway

Option 2 with Beam Arbitrator

- Beam Arbitrator and Base station functions can be collocated and on the same processing plane.
- Supports MOCN – Spectrum sharing, and MORAN – Separate spectrum.
- Supports both frequency domain and time domain sharing.

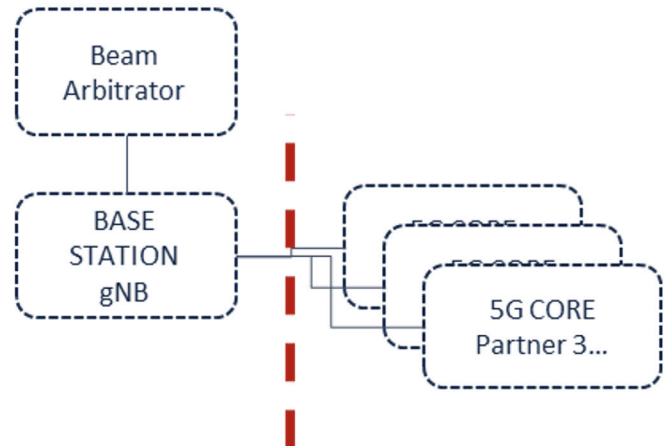


Figure 41 : Option 2 MOCN and MORAN

Option 3 with Beam Arbitrator

- Similar to Option 2.
- Logical separation by slices in a shared core network.
- 5G Core owned by operating SNO; slices provided to Licensees.
- Beam hopping from centralized Beam Arbitrator.
- Requires Slice awareness at both the UE and MNO core.

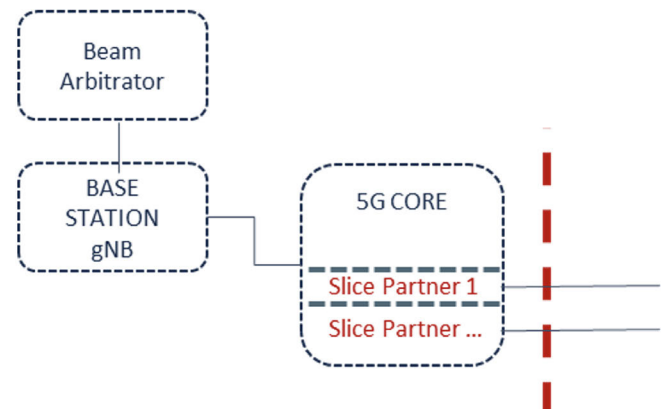


Figure 42 : Option 3 Network Slicing and RAN Mutualization

3.5.1.3 Recommendations

Given the time-sensitive nature and the close dependency between the Beam Arbitrator functions and the L2 scheduler functions, Demarcations Options 2 and 3 are the preferred options. These two options can be implemented with either a regenerative or transparent architecture and therefore do not dictate the choice of either one.

3.6 NB-IoT and NR Co-deployment

3.6.1 Operational considerations for co-deployment of NB-IoT with NR

3.6.1.1 Overview

This section discusses beam hopping and revisiting beam hopping architecture in the context of concurrent NB-IoT and NR NTN deployment and operations, including in-band and guard band NB-IoT deployments, with the corresponding constraints, feasibility assessment, and limitations.

This aligns with the wider topic of (Radio) Resources Management as it involves management of radio resources and optimization of the beam hopping. It may include guidelines for efficient use of spectrum and strategies for minimizing interference between different radio technologies, as well as strategies for optimizing the use of guard bands and ensuring that NB-IoT and NR can coexist without causing interference.

Furthermore, potential impacts to other aspects, such as mobility and load-balancing, UE features, potential UE backwards and forward compatibility issues, RAN feature and implementation requirements, as well as potential impacts to the Core Network capabilities and protocols will also be addressed in this and subsequent versions of the document.

3.6.1.2 Deployment Scenarios under consideration

The following deployment scenarios have been preliminarily identified as valid and likely, and are thus planned to be addressed in further detail in subsequent versions of this document:

- Case 1) LEO Intra-SAN (i.e., same satellite) deployment – i.e., LEO NR + LEO NB
- Case 2) GEO Intra-SAN deployment - i.e., GEO NR + GEO NB
- Case 3) Inter-SAN/inter-orbit deployment – i.e., LEO NR + GEO NB

In terms of deployment models, the following will be considered:

- Case A) Standalone NB-IoT deployment
- Case B) In-band NB-IoT deployment
- Case C) Guard band NB-IoT deployment

The following Satellite Access Node (SAN) architectures will be considered:

- SAN 1) Transparent
- SAN 2) Regenerative
- SAN 3) RAN Splits

3.6.1.3 Beam Hopping and Resource Management

As discussed in previous sections, which Beam Hopping or other general resource management strategies should be adopted and how is not explicitly specified in the 3GPP Standards, but rather left to network implementation, also due to the vast number of differences that might exist in each network, driving implementation choices.

However, as similarly described in previous sections (e.g., on 5G NR Mandatory Cell signals), air interface constraints must be taken into account, in line with the 3GPP specifications, when defining the appropriate beam handling mode of operations and techniques. The co-deployment of NB-IoT with 5G NR brings some additional complexities and constraints, due to the differences between the two RATs, which must be taken into account, not only in the context of radio resource management and beam hopping, but also in other operational contexts.

The following scenarios and approaches, with respect to resource management, and their corresponding potential impact are planned to be further examined in subsequent sections, as the document evolves:

- Basic resource grid alignment with NR and impact on “classic” beam hopping
- Considerations on Radio Unit (RU) sharing between NR and NB-IoT
- Use of same vs separate beamformers for NB-IoT vs NR, such as:
 - Same beamformer with duty-cycle alignment to NB-IoT
 - Wider Beams and/or different duty cycle beams for NB-IoT
- Lower bandwidth Beamformer with lower duty cycle for NB-IoT
- Considerations with different beamformer architectures (e.g., ASIC vs FPGA). NOTE: Transparent vs Regenerative architectures are expected to be impacted differently.
- Additional potential techniques with symbol and signal muting

3.6.1.4 Other areas requiring special considerations

In addition to resource management and beam hopping, introduction of NB-IoT in various modes of co-deployment with NR in an NTN network is expected to require additional considerations and cause impact to a number of other areas of the network and its architecture, design, planning, and management.

This section makes an attempt at providing an initial overview of what and where potential impacts might exist. Each aspect will then be tackled individually in subsequent sections as this document evolves.

The following therefore acts as an initial list, intended to be temporary and to be replaced by more detailed analysis in subsequent iterations:

- 1) Impact due to UE capabilities, generation, and compatibility (on both RAN and CN)
 - a) Existing UE with GEO-only NB-IoT support
 - b) UE with dual NB + NR capability
- 2) Mobility
 - a) Inter-satellite switching
 - b) Selection/reselection, including between NB-IoT and NR
 - c) Load balancing
 - d) Redirection between NR and NB-IoT
- 3) Inter-RAT Fallback Handling – e.g., in case NR Cells must be shut down for power saving, failure, or coverage prioritization reasons
- 4) RAN and Core Network impact and aspects
 - a) Separate vs combined eNB and gNB
 - b) Requirement for EPC vs 5GC
 - c) Single vs Dual Registration
 - d) Cellular IoT Core Network Functions and Features, e.g.,
 - i) IP vs Non-IP Data (NIDD)
 - ii) Core Network APIs (SCEF/NEF)
 - iii) Reachability and Coverage
 - e) Core Network Configuration and timers
 - f) Roaming (NTN-NTN, NTN-TN)
 - g) Discontinuous Coverage
 - h) Store & Forward (S&F) Operation

3.6.2 Radio Unit Sharing for NR and NB-IoT

This section discusses potential architectures of radio unit at the satellite to accommodate streams from different RATs, e.g., NR and NB-IoT with different bandwidth and duty cycle requirements. Each stream corresponds to one TxRU (Tx-Rx endpoint). While NR can tolerate a low duty cycle (see Table 4), NB-IoT beams typically require operation at a higher duty cycle (ideally close to 1). To support NB-IoT with a higher duty cycle, two main options are possible:

- 1) **Reserve more TxRUs for NB-IoT** (let K_{NB} and K_{NR} denote the number of TxRUs for NB-IoT and NR, respectively, so $K_{NB} > K_{NR}$).
- 2) **Use wide beams** and use a smaller number of TxRUs for NB-IoT.

A key question arising from this requirement is: **Is the capability-cost tradeoff of adding an NB-IoT TxRU more favorable than that of adding an NR TxRU?** As the following analysis shows, the answer depends heavily on the architecture of the radio unit.

3.6.2.1 Background and system model

Beam hopping (BH) is often used in satellite radio unit to share limited infrastructure to construct a set of simultaneously formed beams to cover a much larger set of coverage beams. Consider a radio unit with K TxRUs (Tx/Rx endpoints). This implies that, at any given time instant, the payload can form and radiate at most K simultaneous beams or equivalently generate at most K independent wideband RF waveforms. The overall coverage typically requires a much larger number of candidate beams K_{cov} , where $K_{\text{cov}} \gg K$. Under a round robin hopping strategy, a representative per beam duty cycle is

$$\delta \approx \frac{K}{K_{\text{cov}}}.$$

When multiple RATs must coexist on the same radio unit, the duty-cycle and scheduling constraints can differ significantly. NR typically includes mandatory signals and channel structures that constrain how sparse transmission can be over time, while NB-IoT services may push toward higher effective duty cycle. These scheduling considerations determine how often baseband processing is active; however, the per-active-time computational load depends strongly on where beamforming is placed relative to IFFT/CP generation and any sampling rate conversion.

In this section, we analyze the complexity comparison of three digital beamforming architectures, which differ in the arrangement and ordering of their signal processing steps (ADC/DAC, FFT/IFFT and Tx/Rx beamforming). Each of these architectures may be used to multiplex NR and NB-IoT for a given radio unit model in which K TxRUs drive a shared array of N antenna ports. Each antenna port corresponds to one DAC chain. Note that this analysis is Downlink (DL) centric, focusing on the transmission chain (DAC, IFFT, Tx Beamforming); however, the architectural trade-offs for the Uplink (UL) reception chain (ADC, FFT, Rx Beamforming) follow a symmetric logic.

Digital beamforming on a TxRU is represented by a complex weight vector $\mathbf{w} \in \mathbb{C}^{1 \times N}$. For a given baseband symbol (tone) or time-domain complex baseband sample x , the per-port outputs are:

$$\mathbf{y} = x \mathbf{w}, \quad \mathbf{y} \in \mathbb{C}^{1 \times N},$$

so, any beamforming stage that runs at a given sample (or tone) rate incurs N complex multiplications per processed sample (or tone).

The architectures, illustrated in Figure 43, differ mainly in where the summation step between NR and NB-IoT occurs relative to the IFFT and relative to beamforming, which in turn determines which blocks replicate per antenna port and which blocks occur once per TxRU (or equivalently, per stream).

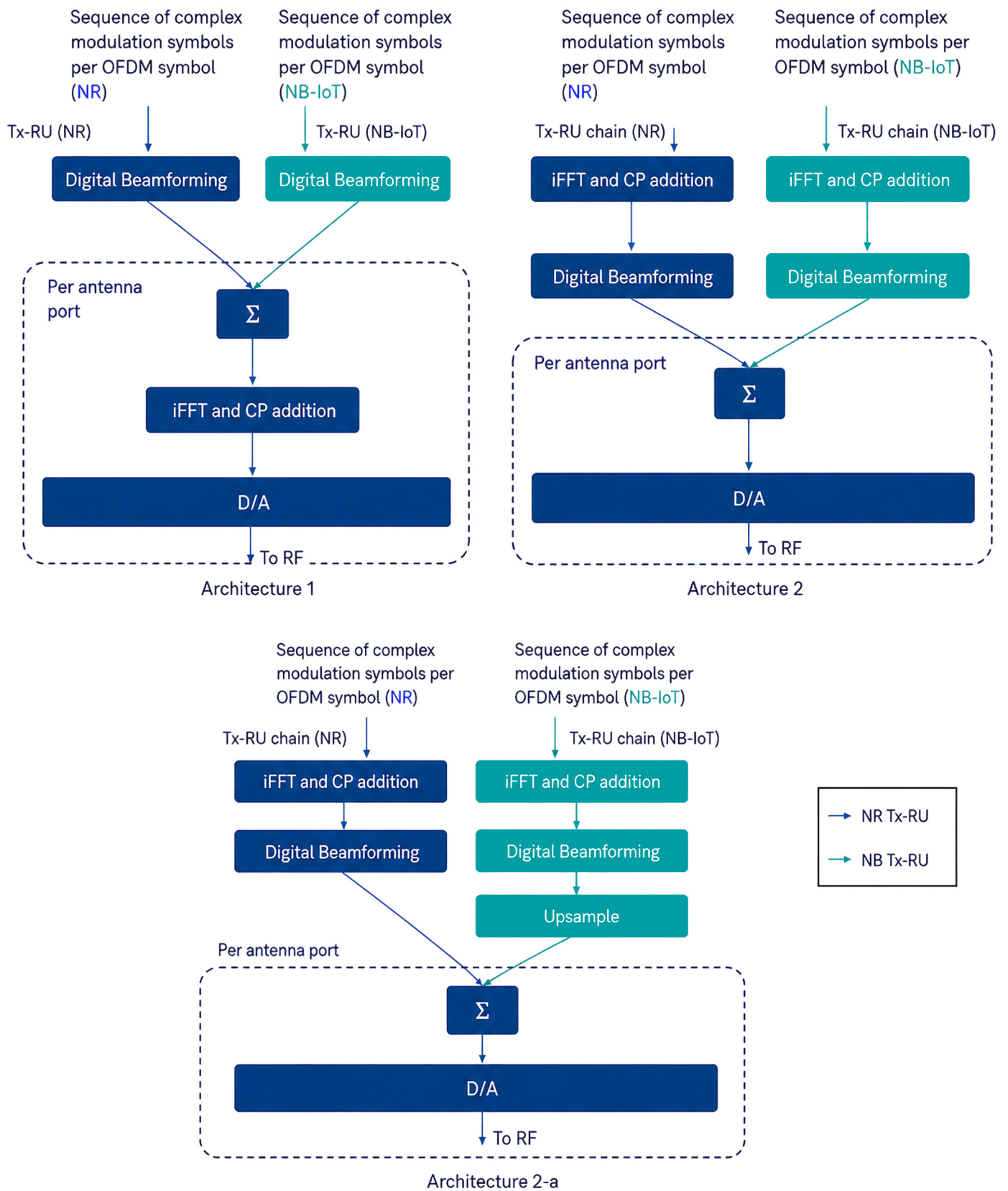


Figure 43: Digital Beamforming Architectures for Mixed NR + NB-IoT Payload

We now proceed to a detailed complexity analysis of the three candidate architectures. We evaluate the digital processing load in terms of complex multiplications and additions, distinguishing between operations that scale with the number of streams (K) and those that scale with the number of antenna ports (N).

3.6.2.2 Notation and scaling

Let B_{NR} denote the NR bandwidth and B_{NB} denote the NB-IoT bandwidth. The bandwidth ratio is:

$$R \triangleq \frac{B_{NR}}{B_{NB}}.$$

Let us define the abstract cost terms as follows.

$$C_{\text{IFFT}}(B) \approx \frac{B}{2} \log_2 \left(\frac{B}{\Delta f} \right) \quad \text{complex multiplications/s,}$$

where Δf is the sub-carrier spacing. This models the per-second complexity of one OFDM IFFT+CP chain at a rate corresponding to bandwidth B .

$C_{\Sigma}(B), C_{\text{DAC}}(B)$: per-second summation and DAC cost per antenna port at rate corresponding to bandwidth B .

$$C_{\text{BF,TD}}(B) \approx N B \quad \text{complex multiplications/s.}$$

This models the time-domain beamforming cost per stream at sample rate $\propto B$.

$$C_{\text{BF,FD}}(\kappa) \approx N \kappa \quad \text{complex multiplications/s.}$$

This models the frequency-domain beamforming cost per stream at “tone rate” κ (where $\kappa \approx B$). When the numerology is shared, and NB-IoT is mapped onto the NR OFDM grid, the FD tone rate for NB scales with occupied tones, so $\kappa_{NB} \approx B_{NB}$, even though the subsequent IFFT is still an NR-sized IFFT.

In the formulas below, C_{total} represents the total digital complexity for the payload (K TxRUs, N ports). We focus on complex multiplications per second.

3.6.2.3 Architecture 1: Frequency-domain beamforming

Architecture 1 applies digital beamforming in the frequency domain on the NR and NB branches, sums across the streams, and then converts the combined spectrum to time domain independently for each antenna port. The key implication of the block ordering is that IFFT+CP and D/A occur per antenna port and are shared by all streams. Under the present model, this means N copies of the IFFT+CP (and N DAC chains) run at the NR sampling rate, regardless of the number of TxRUs.

A compact total scaling for K TxRUs (comprising K_{NR} NR streams and K_{NB} NB streams) is:

$$C_{\text{total}}^{(1)} \approx (K_{NR} \cdot C_{\text{BF,FD}}^{(1)}(B_{NR}) + K_{NB} \cdot C_{\text{BF,FD}}^{(1)}(B_{NB})) + N \cdot C_{\text{IFFT}}^{(1)}(B_{NR}) + N \cdot (C_{\Sigma}^{(1)} + C_{\text{DAC}}^{(1)}).$$

Counting only *complex multiplications* per second and neglecting DAC/summation costs:

$$C_{\text{total}}^{(1)} \approx N \left(K_{NR} B_{NR} + K_{NB} B_{NB} + \frac{B_{NR}}{2} \log_2 \left(\frac{B_{NR}}{\Delta f} \right) \right).$$

This ordering preserves an occupancy advantage for NB-IoT inside the FD beamforming term (because only a small set of tones are processed) but requires a fixed baseline of N wideband IFFT+CP chains. Note that because the IFFTs are shared, the marginal cost of adding a TxRU is only the beamforming cost.

3.6.2.4 Architecture 2: Time-domain beamforming

Architecture 2 maps both NR and NB-IoT to the NR OFDM grid. NB-IoT reuse is realized by assigning only a narrow set of subcarriers to NB and setting the remaining subcarriers to zero, while still running the same NR-size IFFT and CP insertion. Under this assumption, the NB branch is created by reusing an NR-capable branch.

In this architecture, IFFT+CP is performed per stream (TxRU) before beamforming. Since both NR and NB streams utilize the NR grid, the cost per stream is that of an NR chain:

$$C_{\text{IFFT}}^{(2)} \approx C_{\text{IFFT}}(B_{NR}).$$

Time-domain beamforming occurs after IFFT+CP, so it operates on NR-rate time samples for both branches and replicates across N antenna ports:

$$C_{\text{BF,TD}}(B_{NR}) \approx N B_{NR}.$$

The total scaling for K streams ($K = K_{NR} + K_{NB}$) is:

$$C_{\text{total}}^{(2)} \approx K \cdot (C_{\text{IFFT}}(B_{NR}) + C_{\text{BF,TD}}(B_{NR})) + N (C_{\Sigma}(B_{NR}) + C_{\text{DAC}}(B_{NR})).$$

Counting only complex multiplications per second:

$$C_{\text{total}}^{(2)} \approx (K_{NR} + K_{NB}) \left(\frac{B_{NR}}{2} \log_2 \left(\frac{B_{NR}}{\Delta f} \right) + B_{NR} N \right).$$

In this form, the NB-IoT occupancy does not appear in the IFFT+CP or time-domain beamforming terms because both branches run at the NR rate. Unlike Architecture 1, the IFFT costs scale with the number of TxRUs, but there is no fixed IFFT cost per antenna port.

3.6.2.5 Architecture 2-a: Time-domain beamforming with NB-native narrowband chain

Architecture 2-a corresponds to a true NB-native waveform chain that remains narrowband through its own IFFT+CP and through NB beamforming, followed by an explicit up sampling stage to align the NB signal with the NR sample rate before summation.

The per-stream IFFT+CP costs differ for NR and NB:

$$C_{\text{IFFT}}^{(2a)} \approx \begin{cases} C_{\text{IFFT}}(B_{NR}) & \text{for NR chain} \\ C_{\text{IFFT}}(B_{NB}) & \text{for NB chain} \end{cases}$$

Beamforming also scales with the respective bandwidths:

$$C_{\text{BF,TD}}^{(2a)} \approx \begin{cases} NB_{NR} & \text{for NR chain} \\ NB_{NB} & \text{for NB chain} \end{cases}$$

However, up sampling is required for each NB stream. Since beamforming happens before summation (per stream), and summation happens per port, the up sampling must be done per port for each stream. The up sampling cost is replicated across N ports per NB stream,

$$C_{\text{US}}^{(2a)} \approx N C_{\text{US}}(B_{NB} \rightarrow B_{NR}),$$

where C_{US} is the complexity of sample rate conversion.

Counting only *complex multiplications* per second:

$$C_{\text{total}}^{(2a)} \approx K_{NR} \left(\frac{B_{NR}}{2} \log_2 \left(\frac{B_{NR}}{\Delta f} \right) + B_{NR} N \right) + K_{NB} \left(\frac{B_{NB}}{2} \log_2 \left(\frac{B_{NB}}{\Delta f} \right) + B_{NB} N + N \cdot O(B_{NR}) \right).$$

Architecture 2-a preserves narrowband scaling in NB IFFT and NB beamforming but introduces a per-port up sampling cost for each NB stream, which scales with the wideband rate B_{NR} .

Complexity of up sampling: For up sampling from B_{NB} to B_{NR} (interpolation factor $L = B_{NR}/B_{NB}$), a polyphase FIR interpolator is typically used. The conceptual FIR filter of length N_{filter} is decomposed into L polyphase branches, where each branch contains approximately N_{filter}/L taps. For every input sample (at rate B_{NB}), the interpolator computes L output samples (at rate B_{NR}). Each output sample requires computing one polyphase branch.

- Total work per input sample: $L \times (N_{\text{filter}}/L) = N_{\text{filter}}$ complex multiplications.
- Total complexity per second: $B_{NB} \times N_{\text{filter}}$ complex multiplications/s, which is equivalent to $B_{NR} \times (N_{\text{filter}}/L)$.

$$B_{NR} = 10 \text{ MHz}, \quad B_{NB} = 200 \text{ kHz}, \quad \Delta f = 15 \text{ kHz}.$$

3.6.2.6 Example

Let's Assume $N = 64$ antenna ports. We compare the complexity scaling for adding streams. First, let us estimate the component costs in complex multiplications per second (CM/s):

IFFT (NR): $C_{\text{IFFT}}(10\text{M}) \approx \frac{10^7}{2} \log_2\left(\frac{10^7}{15 \times 10^3}\right) \approx 5 \times 10^6 \times 9.4 \approx 47 \text{ M CM/s}.$

IFFT (NB): $C_{\text{IFFT}}(200\text{k}) \approx \frac{2 \times 10^5}{2} \log_2\left(\frac{2 \times 10^5}{15 \times 10^3}\right) \approx 10^5 \times 3.7 \approx 0.37 \text{ M CM/s}.$

Beamforming (NR): $NB_{\text{NR}} = 64 \times 10^7 = 640 \text{ M CM/s}.$

Beamforming (NB): $NB_{\text{NB}} = 64 \times 2 \times 10^5 = 12.8 \text{ M CM/s}.$

Up sampling (NB): Assume a polyphase filter with length $N_{\text{filter}} = 64$ (modest quality).

- Cost per port: $C_{\text{US}} = B_{\text{NB}} \times N_{\text{filter}} = 200 \text{ kHz} \times 64 = 12.8 \text{ M CM/s}.$
- Total Up sampling Cost (N ports): $64 \times 12.8 \text{ M} \approx 819 \text{ M CM/s} \approx 0.82 \text{ G CM/s}.$

Architecture 1

- **Fixed Cost (Shared IFFTs):** $N \times C_{\text{IFFT}}(10\text{M}) = 64 \times 47 \text{ M} \approx 3008 \text{ M CM/s}.$
- **Marginal Cost per NR Stream:** $NB_{\text{NR}} = 640 \text{ M CM/s}.$
- **Marginal Cost per NB Stream:** $NB_{\text{NB}} = 12.8 \text{ M CM/s}.$

Architecture 1 has a high fixed cost ($\sim 3 \text{ G CM/s}$), but adding NB streams is extremely cheap (12.8 M CM/s).

Architecture 2 (NB mapped onto NR grid)

- **Marginal Cost per NR Stream:** $C_{\text{IFFT}}(10\text{M}) + NB_{\text{NR}} = 47 + 640 = 687 \text{ M CM/s}.$
- **Marginal Cost per NB Stream:** Same as NR = 687 M CM/s (since it uses a wideband chain).

Architecture 2 has no fixed cost, but adding an NB stream is much more expensive than in Architecture 1 (687 M vs 12.8 M).

Architecture 2-a (NB-native narrowband chain)

- **Marginal Cost per NR Stream:** Same as Architecture 2 = $687 \text{ M CM/s}.$
- **Marginal Cost per NB Stream:** $C_{\text{IFFT}}(\text{NB}) + C_{\text{BF}}(\text{NB}) + C_{\text{US}}(\text{Total}) \approx 0.37 + 12.8 + 819 \approx 832 \text{ M CM/s}.$

Adding an NB-IoT stream in Architecture 2-a ($\sim 0.83 \text{ G CM/s}$) is approximately 1.2 times more expensive than in Architecture 2 ($\sim 0.69 \text{ G CM/s}$). While the beamforming itself is cheaper, the requirement to perform up sampling on sixty-four antenna ports outweighs the savings, even with a modest 64-tap filter. Higher quality filters ($N_{\text{filter}} > 100$) would widen this gap further.

Architecture	Fixed Cost (M CM/S)	Variable Cost Per Nr Stream (M CM/S)	Variable Cost Per Nb Stream (M CM/S)
Arch 1 (FD Beamforming)	~3008	640	12.8
Arch 2 (TD Beamforming, Shared Grid)	0	~687	~687
Arch 2-a (TD Beamforming, Native NB)	0	~687	~832

Table 5: Fixed and Marginal Costs per Stream (in Terms of # of Complex Multiplications per Second) for the Case of $N=64$ Antenna Ports, $BNR=10$ MHz, $BNB=200$ kHz, and $SCS=15$ kHz.

The above discussion illustrates that depending on the radio unit architecture, for sharing NR and NB-IoT streams, there is a trade-off between fixed per-port complexity and variable per-stream complexity. For example,

- **Architecture 1** places the wideband generation (IFFT) at the ports. This incurs a high fixed complexity cost (N IFFTs) regardless of traffic load but offers the lowest marginal cost for adding new beams (streams). This is the most efficient architecture for adding many NB-IoT streams. Furthermore, the beamformer in Architecture 1 depends on the knowledge of the structure of the OFDM waveform (e.g., changing numerology will result in changing the beamformer).
- **Architecture 2** places the wideband generation (IFFT) at the streams. This avoids the fixed per-port cost, making it efficient for sparse scenarios ($K \ll N$). However, for NB-IoT, it forces the use of a wideband chain for a narrowband signal, resulting in inefficient scaling.
- **Architecture 2-a** attempts to reduce the stream-side cost by keeping NB-IoT narrowband. However, with digital beamforming (pre-summation), up sampling must be performed for every antenna port for every stream. For a large number of ports ($N = 64$), the up sampling complexity ($N \cdot C_{UG}$) per NB-IoT stream may become prohibitive.

3.7 Use of Existing Space Segment

By far the biggest challenge for the deployment of NTN direct-to-device mobile satellite services, in particular targeting mass-scale consumer-grade or otherwise commercially available smartphones, wearable and IoT devices, is the cost associated with Space Segment infrastructure, including launch costs.

On the other hand, many satellite operators have existing space segment assets, both transparent and regenerative, that have been used to deliver dedicated mobile satellite services via a mix of different technologies.

We can therefore anticipate (and witness already) an evolution path that starts with existing space segment assets, enabling capitalization on past capital investment:

- **GEO Satellites:** Commercially available handsets together with 5G NTN link budgets present a challenge from GEO for 5G NR services. Thus, the starting point for GEO-based space segment has so far been to realize NTN services by leveraging the NB-IoT protocol and a 4G LTE-based architecture for the RAN and CN.

- **LEO Satellites:** Even in LEO, 5G NR NTN services to consumer-grade smartphones require large antennas on the user link of the space segment. Many legacy or small-sat MSS LEO constellations are not capable enough of serving consumer smartphones, so deployment of NB-IoT is also a valid approach.

For GEO satellites and those LEO satellites with limited capabilities, a 5G NR-based architecture for the RAN and Core Network, coupled with dedicated terminals featuring high gain antennas and specific modifications to the radio frequency (RF) frontend capabilities (e.g., to increase receiver sensitivity and transmitter output power) could provide an attractive solution and serve as the starting point.

These classes of devices have been a staple of mobile satellite systems for decades, and their form factor and enhanced performance characteristics could be of strong relevance also as the NTN MSS networks evolve to higher space segment capabilities.

As of Release 19, the 3GPP NTN RAN4 specifications do not include codified specification for User Equipment (UE) with such characteristics, but their out-of-spec realization is also “not precluded.” Therefore, whilst it is important that support for such types of devices is ultimately introduced in the 3GPP specifications, their implementation can already be considered as a solution that is complementary to the existing normative specification and can enable interoperability and network sharing on existing infrastructure, resulting in structural cost benefits.

3.8 MSS User Services

3.8.1 Background

5G NTN requires MSS spectrum, e.g. band n252, n253, n254, n255, n256 in the L- and S-bands. Today, this spectrum is allocated and in use in both in NTNs and TNs.

5G NTN exists in two different modes:

- **NR NTN:** Based on 5G NR RAN protocols and 5GC protocols
- **NB NTN:** Based on NB-IoT RAN protocols and EPC protocols

3.8.2 Messaging

The messaging services identified in [1] are:

- One-way emergency messaging
- Two-way messaging services – text only
- Two-way messaging services – IoT
- Two-way messaging services – multimedia messages

The realizations options are:

- SMS over NAS
- IMS- (SMS over IP and RCS) based messaging
- NIDD-based messaging
- Over-the-top application

3.8.2.1 SMS over NAS

Description:

- SMS over NAS is text-only and thus applicable for emergency, text, and IoT messaging services.
- SMS-based messaging exists in two flavors: one (a) using SGs interface from the MME via MSC to SMS-C, and one (b) using the SGd interface from the MME direct to SMS-C.

Device requirements:

- Support of SMS over NAS.

Network (RAN/CN) requirements.

- In case of SGs-based service: MSC implemented in visited network (SNO) and MAP protocol from MSC to SMS-C in home network (MNO). MAP-based protocol between HLR and SMS-C in home network (MNO).
- In case of SGd-based service: Diameter interface between MME in visited network (SNO) and SMS-C in home network (MNO). Diameter-based S6c between SMS-C and HSS in home network (MNO).

Note: SGd-based SMS is not commonplace in the market. Most MNOs are running SGs-based service (legacy reasons). This may impact the SNO, which may need to implement an MSC/VLR rather than pushing requirements to the MNOs to change their SMS-C solutions to a Diameter-based solution.

3.8.2.2 IMS- (SMS over IP and RCS) Based Messaging

Description:

- IMS-based messaging can be used as an IP-SM-GW for plain text SMS or an RCS AS for multimedia messaging. IMS-based messaging is applicable for all services.
- The roaming architecture used is S8HR, meaning that the service is fully executed in the home network (the MNO).

Device requirements:

- Support of SMS over IP and/or RCS
- Able to use the IMS APN

Network (RAN/CN) requirements.

- Home network support of IMS with SMS over IP or RCS
- S8HR roaming to visited network (SNO) in place
- Voice and data restricted in home HSS.
- Support of the IMS APN

Comments:

- NB-IoT using DoNAS and IMS APN + S8HR is not commonplace and not widely verified.

3.8.2.3 NIDD-Based Messaging (NB NTN only)

Description:

- NIDD allows a device to send data to the network without using an IP stack.
- This is applicable for emergency, text, and IoT messaging services.
- A Network Abstraction Layer (NAL) in the home network implements an API tailored for the service that the Application Server (AS) accesses.

Device requirements:

- NIDD is an integrated service in the modem/UE.

Network (RAN/CN) requirements:

- Data over NAS
- SCEF terminates the data transfer in CN
- Implementation of API towards the AS

Comments:

- Roaming is possible through T7, but this is not established in the market.
- NIDD is thus not recommended for roaming services but works for SNO subscriber services or device OEM services that have a direct relationship with MNO.

3.8.2.4 Over-the-top Application

Description:

- Messaging as an Over-the-top IP service is not possible to distinguish from other data services and is not further discussed here.

3.8.3 Voice/Telephony

The voice/telephony services identified in [1] are:

- Non-regulated emergency voice calls
- Non-regulated two-way voice calls
- Fully regulated two-way voice calls (telephony service)

The realizations options are:

- Over-the-top (not applicable for fully regulated two-way voice calls)
- Standard S8HR model for IMS
- Dual-SIM model for IMS

3.8.3.1 Over-the-Top

Description:

- Applicable for non-regulated emergency voice calls and non-regulated two-way voice calls
- Voice through OTT app, overlaid on data
- No (direct) connection to PSTN; App specific identifiers used
- Voice service and codec implemented in OTT-SW, no dependency on IMS client in device and IMS network

Device requirements:

- OS support for 3P OTT voice app
- New codec running in OS
- URSP allows QoS and RoHC to be used with 3P voice app in OS (NR NTN only)

Network (RAN/CN/IMS) requirements:

- RoHC in RAN (NR TNT) or CN (NB NTN) to lower overhead
- NW slicing/URSP support. (NR NTN only)

Comments:

- Fast way to implement a first voice service
- Ideal for user originated services such as non-regulated emergency voice calls

3.8.3.2 Standard S8HR Model for IMS

Description:

- Applicable for all voice service types
- Voice using IMS using the same non-roaming and roaming architecture as for TN services
- Same phone numbers for TN and NTN calls (connected to PSTN)

Device and Network (RAN/CN/IMS) requirements:

- Follows the GSMA PRD NG.114 minimum requirements, but with adaptations
- Same phone numbers for TN and NTN calls favor the use of the native IMS app for both TN and NTN connectivity
- If new codec is needed (NB-NTN), it is implemented in native IMS app and IMS MGW
- If signaling enhancements are needed, e.g., Sigcomp, (NB-NTN), it is implemented in the native IMS app and IMS P-CSC
- New codec and signaling enhancements need to be implemented in IMS MGW and P-CSCF by all MNOs with NTN-enabled subscribers

Comments:

- When on NTN connectivity, the system may or may not be compliant with sections 4 Radio and Packet Core Feature Set and 5 Common Functionalities of GSMA PRD NG.114 Support for, e.g., QoS flow/radio bearer services, IMS APN/DNN, IMS emergency services, etc. may be omitted. Especially if NB-NTN is used, adaptations are foreseen.

3.8.3.3 Dual SIM Model for IMS Voice

Description:

- Applicable for all voice service types
- All voice users are provisioned by the SNO
- Not the same phone numbers for TN and NTN calls, but both systems are connected to PSTN
- Dual SIM is a model used in TN (but it is less usable in an NTN context)

Device and Network (RAN/CN/IMS) requirements:

- Follows the GSMA PRD NG.114 minimum requirements, but with adaptations
- Different phone numbers for voice calls when on TN or NTN; open for usage of OTT IMS app for when on the NTN connection
- New codec in IMS app may need to be implemented (NB-NTN)
- Signaling enhancements, e.g., Sigcomp, may need to be implemented (NB-NTN)
- Implementation of new codec and signaling enhancements in IMS needs to be done only by the SNO

Comments:

- When on NTN connectivity, the system may or may not be compliant with sections 4 Radio and Packet Core Feature Set and 5 Common Functionalities of GSMA PRD NG.114 Support for e.g. QoS flow/radio bearer services, IMS APN/DNN, IMS emergency services, etc. may be omitted. Especially if NB-NTN is used, adaptations are foreseen.
- Dual-SIM is used in terrestrial networks, and one can choose to have the primary voice service from one MNO and the data services from another. This works mainly due to the overlapping coverage of TNs. There is a different situation for a TN/NTN set-up where you would expect to have your calls routed to the NTN connection outside of TN coverage.

3.8.4 Data Services

The data applications identified in [1] are:

- Data applications for specific usage
- Data applications for general usage

The realization options are:

- Data over NAS (DoNAS) (NB-NTN)
- IP over User Plane (NB-NTN)
- Applications for 5GC-based best-effort data service (NR-NTN)
- Applications for 5GC-based data services (NR-NTN) using NW slicing mechanisms

3.8.4.1 Data over NAS (DoNAS) (NB-NTN)

Description:

- Data over NAS (DoNAS) is a control plane for cellular IoT optimization that allows for transport of user data within signaling messages through the MME.

Comments:

- Security is a benefit since user data is encrypted with the same mechanism as network signaling.
- Supports NIDD via SCEF.
- This is the market dominating data transport model for NB-IoT.
- Bearer separation (e.g., using a dedicated voice bearer) is not possible.
- No handover mechanisms are supported (RRC Connection Reestablishment is used instead).

3.8.4.2 IP Over User Plane (NB-NTN)

Description:

- Traditional data path through the gateways (SGW and PGW)

Comments:

- Support bearer separation and handovers – hence, QoS mechanisms may be used, but in the EPC architecture, dynamic QoS is not easily used for 3rd party apps.
- NIDD is supported but is more complicated than in DoNAS.
- This is not widely adopted on the market.

3.8.4.3 Best-effort 5GC-based Data Service (NR-NTN)

Description:

- Best effort data service without any specific NW or Device requirements

Comments:

- May be provided in combination with NW slicing/URSP; then this best-effort service would be provided by the default slice.
- In case of combination with NW slicing/URSP, some data applications may be blocked and only allowed on other slices.

3.8.4.4 5GC-based Data Services (NR-NTN) using NW Slicing

Description:

- In a rather narrow channel via a satellite, it could be so that access to specific popular or demanding data applications is sold via a higher tier subscription/extra cost.
- These data applications may be associated with a network slice via URSP.

Device and Network (RAN/CN/IMS) requirements:

- Support for URSP
- Support for network slicing
- Ability to block applications

Comments:

- To make this work end-to-end, it is likely you need to make the data applications NTN-aware, so they choose (e.g., codecs to be optimized for an NTN channel).

References:

- [1] 5G NTN Service evolution, presented to MSSA in October 2024

4 3GPP MANDATORY FEATURES FOR NTN

4.1 General configuration

- The NR FDD bands needed for interoperability are (in groups):
 - n256, n252 (S-bands)
 - n255, n253, n251, n250 (L-bands)
 - n254 (L/S-band)
- The UE should be capable of supporting an NGSO scenario.
- The UE should be capable of supporting both TN and NTNs.
- The ephemeris information sent to the UE could both be in Position Velocity and Orbital format.

4.2 Test cases the UE must pass to be interoperable

4.2.1 Test cases covered by 3GPP RAN5

The 3GPP TSG RAN WG5 (RAN5) is the Conformance Testing Group for User Equipment (UE). The main objective of RAN5 is the development of UE conformance test specifications, and RAN 5 is responsible for the following areas related to the MSSA solution outlined in this document.

RRM conformance tests

- Protocol test cases for 5G NR NTN, inter RAT mobility, NAS, 5GC and IMS

RRM conformance test cases are outlined in TS38.523.



Test Case Family	Test Case Group
Arch 1 (FD Beamforming)	Cell Selection (38.523)
	Cell Reselection (38.523)
	Cell Reselection (38.533)
CONNECTED Mode Mobility Tests	Handover for SAN
	Conditional Handover (38.523)
	RACH-less Handover
	Random Access for SAN
	Coverage Enhancements for Random Access Procedure (38.523)
	RRC Reestablishment for SAN
	RRC Re Connection Release with Redirection for SAN
Timing for Satellite Access Tests	Timing Advance for Satellite Access
	SIB19 T430 Timer Test (38.523)
MAC Tests	DRX Operation (38.523)
RLC Tests	RLC UM (38.523)
	RLC AM (38.523)
PDCCP Tests	RLC UM (38.523)
PDCCP Tests	PDCCP Discard (38.523)
Signaling Characteristics Tests	Radio Link Monitoring for Satellite Access
	Beam Failure Detection and Link recovery procedures for satellite access
Measurement Procedure Tests	Measurement configuration Event D1 (38.523)
	TN measurement skipping (38.523)
	Intra-frequency Measurements for SAN (38.533)
	Inter-frequency Measurements for SAN (38.533)
	L1-RSRP measurement for beam reporting for SAN (38.533)
Measurement Performance Tests	SS-RSRP for SAN
	SS-RSRQ for SAN
	SS-SINR Requirements for SAN
	L1-RSRP Beam Reporting Measurement Requirements for SAN
5GS Mobility Management Tests	GNSS position reporting
	Mobility registration update
	TAI(s) not allowed / forbidden TAI(s)
	Disabling and re-enabling of UE's Satellite NG-RAN capability

Table 6 : Test Cases the UE Must Pass to be Interoperable

4.3 Minimum RAN features

- Operating bands for NTN access
- SIB19 set-up and configuration
- Idle mode operations
 - Cell selection options
 - Cell re-selection options
- Connected mode operations
 - Handover options incl conditional HO (inc. T1/D1 CHO)
 - Reestablishment, Random Access, Connection Release
 - Unchange PCI Satellite switch with Resync
 - Coverage enhancement
 - Data transfer (RLC, MAC, PDCP)
 - DRX operations
- Timing for satellite access
 - UE transmit timing, timing advance
- Signaling characteristics
 - SSB, CSI-RS
 - Beam failure detection and link recovery
 - CBW/BWP changes
- Measurement procedure
 - Inter/Intra frequency measurements

4.4 Minimum CN features

The core network features/configurations to be tested, discussed in this section are spread over the packet core (5GC) domain and the IMS domain.

- Mobility management
 - GNSS reporting
 - Earth fixed cells/moving cells
 - Tracking area/cell barring
 - Enhanced support for discontinuous coverage
- QoS settings
 - 5QI characteristics for telephony/voice
- IMS settings
 - Feature set and services to be supported
 - Media considerations
 - SIP timer considerations
 - Emergency service settings
 - Data off and service availability

5 ANNEXES

5.1 Annex A: Glossary

TERM	DEFINITION
3GPP	3rd Generation Partnership Project
5GC	5G Core
AS	Application Server
BH	Beam Hopping
CN	Core Network
CU	Central Unit
D2D	Direct-to-Device
Diameter	Diameter Protocol
DoNAS	Data over NAS
DU	Distributed Unit
eCPRI	Enhanced Common Public Radio Interface
EFA	Earth-Fixed Area
EIRP	Effective Isotropic Radiated Power
EPC	Evolved Packet Core
FR	Frequency Reuse
GEO	Geostationary Earth Orbit
gNB	Next Generation Node B
GW	Gateway
HARQ	Hybrid Automatic Repeat Request
HLR	Home Location Register
IMS	IP Multimedia Subsystem
IMSI	International Mobile Subscriber Identity
IPX	Internetwork Packet Exchange
L2	Layer 2
LEO	Low Earth Orbit

5.1 Annex A: Glossary Continued

TERM	DEFINITION
MAP	Mobile Application Part
MEO	Medium Earth Orbit
MGw	Media Gateway
MME	Mobility Management Entity
MNO	Mobile Network Operator
MSC	Mobile Switching Center
MSS	Mobile Satellite Services
NAL	Network Abstraction Layer
NAS	Non-Access Stratum
NB-IoT	Narrowband Internet of Things
NIDD	Non-IP Data Delivery
NOC	Network Operations Center
NR	New Radio
NTN	Non-Terrestrial Networks
NW	Network
OISL	Optical Inter-Satellite Link
ORAN-LLS	Open Radio Access Network Lower Layer Split
OSS	Operations Support System
OTT	Over-the-Top
PCI	Physical Cell Identity
P-CSCF	Proxy Call Session Control Function
PFD	Power Flux Density
PGW	Packet Gateway
PLMN	Public Land Mobile Network
QoS	Quality of Service
RA	Routing Area
RAN	Radio Access Network
rApps	Non-Real-Time Applications Running on RIC

5.1 Annex A: Glossary Continued

TERM	DEFINITION
RCS	Rich Communication Services
RG	Reuse Group
RIC	Radio Intelligent Controller
RoHC	Robust Header Compression
RTT	Round Trip Time
RU	Radio Unit
S6c	Diameter-based Interface
SCEF	Service Capability Exposure Function
SCS	Satellite Communication System
SGd	Signaling Gateway Diameter
SGs	Signaling Gateway Service
SGW	Serving Gateway
SIB	System Information Block
SIB19	System Information Block 19
Sigcomp	Signaling Compression
SMO	Service Management and Orchestration
SMS	Short Message Service
SMS-C	Short Message Service Center
SNO	Satellite Network Operator
SOC	Satellite Operation Center
TA	Tracking Area
TR	Time Reuse
URSP	User Equipment Route Selection Policy
VLR	Visitor Location Register
xApps	Near-Real-Time Applications Running on RIC